Complex Analysis in Several Variables

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1 Calculus on Complex Functions

1.1 Preliminaries

Notation 1.1. *Let $z \in \mathbb{C}^n$, we define the max norm and the Euclidean norm as*

$$|\underline{z}| = \max_{i \leq j \leq n} |z_j|,$$

$$\|\underline{z}\| = \sqrt{\sum_{j=1}^{n} |z_j|^2},$$

respectively.

Definition 1.1. *Let $\underline{a} \in \mathbb{C}^n$ and $\underline{r} \in (0, \infty)^n$. We define a polydisk around $\underline{a}$ to be*

$$D_{\underline{r}}(\underline{a}) = \{\underline{z} \in \mathbb{C}^n \mid \forall 1 \leq j \leq n, |z_j - a_j| < r_j\}.$$

We also define for $r \in (0, \infty)$,

$$D_r(\underline{a}) = \{\underline{z} \in \mathbb{C}^n \mid \forall 1 \leq j \leq n, |z_j - a_j| < r\}.$$

In particular, when $r = 1$ and $\underline{a} = \underline{o}$, we define the unit polydisk

$$\mathbb{D} = D_1(\underline{o}).$$

Definition 1.2. *The unit ball around the origin is*

$$\mathbb{B} = \{\underline{z} \in \mathbb{C}^n \mid \|z\| < 1\}.$$

Remark 1.1. *Observe that $\mathbb{C}^n$ is $2n$-dimensional $\mathbb{R}$-vector space. Therefore, we have*

$$\partial\mathbb{B} \cong S^{2n-1}.$$

Remark 1.2. *While $\partial\mathbb{B}$ is smooth, $\partial\mathbb{D}$ is not.*

Definition 1.3. *The n-dimensional torus is*

$$\partial_0 \mathbb{D} := \{\underline{z} \in \mathbb{C}^n \mid \forall 1 \leq j \leq n, |z_j| = 1\}.$$

Indeed we see

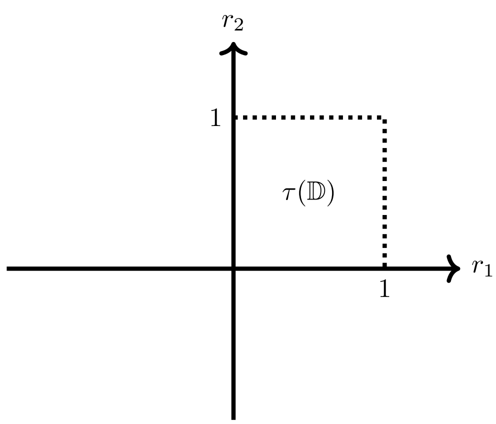
$$\partial_0 \mathbb{D} = (S^1)^n.$$

Proposition 1.1. *Let $\tau : \mathbb{C}^n \to (0, \infty)^n$ be such that*

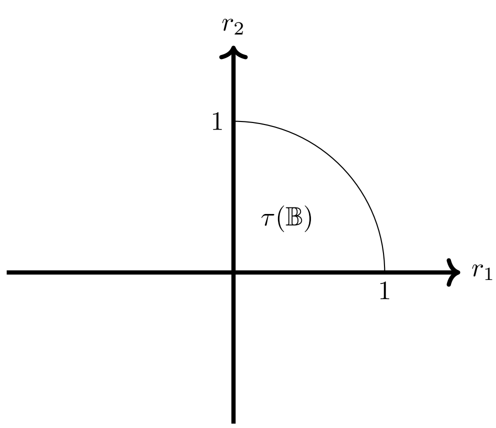
$$\tau(z_1, \cdots, z_n) = (|z_1|, \cdots, |z_n|).$$

Furthermore, let $\underline{r} \in [0, \infty)^n$ be such that k-many coordinates are non-zero. Then $\tau^{-1}(\underline{r})$ is a k-dimensional torus.

Example 1.1. *For $n = 2$, we have*



and



Definition 1.4. *Let $M \subseteq \mathbb{C}^n$. A function $F : M \to \mathbb{C}^m$ is continuous if for $F = (f_1, \cdots, f_m)$, where for each $1 \leq i \leq m$, $f_i : M \to \mathbb{C}$, f_i is continuous.*

Definition 1.5. *$f : U \to \mathbb{C}$ is differentiable at $\underline{z}_0 \in U$ if there exists $\Delta_i, E_i : U \to \mathbb{C}, (1 \le i \le n)$ such that they are continuous at $\underline{z}_0$ and*

$$f(\underline{z}) - f(\underline{z}_0) = \sum_{i=1}^{n}(z_i - z_{0,i})\Delta_i(\underline{z}) + \sum_{i=1}^{n}(\overline{z_i} - \overline{z_{0,i}})E_i(\underline{z}).$$

Δ_i, E_i are called Wirtinger derivatives. They are usually denoted as

$$\Delta_i(\underline{z}_0) = \frac{\partial f}{\partial z_i}(\underline{z}_0), \quad E_i(\underline{z}_0) = \frac{\partial f}{\partial \overline{z_i}}(\underline{z}_0).$$

Proposition 1.2. *Let $f : U \to \mathbb{C}$ be differentiable. Then we have*

$$\frac{\partial f}{\partial z_i} = \frac{1}{2}\left(\frac{\partial f}{\partial x_i} - i\frac{\partial f}{\partial y_i}\right),$$
$$\frac{\partial f}{\partial \overline{z_i}} = \frac{1}{2}\left(\frac{\partial f}{\partial x_i} + i\frac{\partial f}{\partial y_i}\right).$$

Corollary 1.1.

$$\frac{\partial z_i}{\partial z_j} = \delta_{ij}, \quad \frac{\partial z_i}{\partial \overline{z_j}} = 0,$$
$$\frac{\partial \overline{z_i}}{\partial z_j} = 0, \quad \frac{\partial \overline{z_i}}{\partial \overline{z_j}} = \delta_{ij}.$$

In particular,

$$\overline{\left(\frac{\partial f}{\partial z_i}\right)} = \frac{\partial \overline{f}}{\partial \overline{z_i}}.$$

Definition 1.6. *Let $U \subseteq \mathbb{C}^n$. A function $F : U \to \mathbb{C}^m$ is differentiable if for $F = (f_1, \cdots, f_m)$, where for each $1 \le i \le m$, $f_i : M \to \mathbb{C}$, f_i is differentiable.*

Proposition 1.3. *Let $U \subseteq \mathbb{C}^n$ be open. A function $f : U \to \mathbb{C}$ with $f = g + ih$ for some $g, h : U \to \mathbb{R}$ is differentiable if g, h are differentiable.*

Proposition 1.4 (Chain Rule)**.** *Let $U \subseteq \mathbb{C}^n, V \subseteq \mathbb{C}^m$ be open sets and consider differentiable functions $F : U \to V$ and $f : V \to \mathbb{C}$ where $F = (f_1, \cdots, f_m)$. We have*

$$\frac{\partial (f \circ F)}{\partial z_i}(\underline{z}) = \sum_{j=1}^{m}\frac{\partial f}{\partial w_j}(F(\underline{z}))\frac{\partial f_j}{\partial z_i}(\underline{z}) + \sum_{j=1}^{m}\frac{\partial f}{\partial \overline{w_j}}(F(\underline{z}))\frac{\partial \overline{f_j}}{\partial z_i}(\underline{z}).$$

1.2 Holomorphic Functions

Definition 1.7. *Let $U \subseteq \mathbb{C}^n$ be open and $f : U \to \mathbb{C}$ be a function. f is complex differentiable at $\underline{z}_0 \in U$ if there exists functions $\Delta_1, \cdots, \Delta_n : U \to \mathbb{C}$ which are continuous at $\underline{z}_0$ with*

$$f(\underline{z}) - f(\underline{z}_0) = \sum_{i=1}^{n} (z_i - z_{0,i})\Delta_i(z).$$

f is holomorphic at $\underline{z}_0$ if it is complex differentiable in a neighborhood of $\underline{z}_0$. The values

$$\Delta_i(\underline{z}_0) = \frac{\partial f}{\partial z_i}(\underline{z}_0)$$

are called the complex partial derivatives.

Remark 1.3. *f is holomorphic at $\underline{z}_0$ if and only if it is differentiable at $\underline{z}_0$ with $E_i \equiv 0$ for all $1 \leq i \leq n$.*

Example 1.2. *z_i are holomorphic but $\overline{z_i}$ are not. Polynomials in $z_1, \cdots, z_n$ are holomorphic.*

Definition 1.8. *A function $F : U \to \mathbb{C}^m$ given by $F = (f_1, \cdots, f_m)$ is holomorphic if all f_i are holomorphic.*

Proposition 1.5. *$F : U \to \mathbb{C}^m$ is holomorphic if and only if for any open subset $V \subset \operatorname{Im} F$ and a holomorphic function $f : V \to \mathbb{C}$, $f \circ F$ is holomorphic.*

Theorem 1.1. *$f : U \to \mathbb{C}$ is holomorphic if and only if f is real-differentiable (ie. with respect to x_i and y_i for $z_i = x_i + iy_i$), and for all $1 \leq i \leq n$, $\frac{\partial f}{\partial \overline{z}_i} = 0$.*

Remark 1.4. *$f : U \to \mathbb{C}$ is holomorphic if and only if for each $1 \leq i \leq n$ and fixed $z_1, \cdots, z_n$, $z_i \mapsto f(z_1, \cdots, z_n)$ is holomorphic.*

Definition 1.9. *A complex line is a function $l : \mathbb{C} \to \mathbb{C}^n$ for some $\underline{a}, \underline{w} \in \mathbb{C}^n, w \neq \underline{o}$, we have*

$$l(t) = \underline{a} + t\underline{w}.$$

The image of l is denoted by $l(\mathbb{C}) = L$.

Notation 1.2. *For a function $f : U \to \mathbb{C}$, we denote*

$$f|_L(t) := f \circ l(t) = f(\underline{a} + t\underline{w}).$$

Remark 1.5. *Using the chain rule, we see*

$$\frac{d}{dt} f|_L(t) = \sum_{i=1}^{n} \frac{\partial f}{\partial z_i}(\underline{a} + t\underline{w})w_i.$$

1.3 Cauchy Integrals

We will consider a holomorphic function $f : U \to \mathbb{C}$ where U is a neighborhood of the closure of $D = D_{\underline{r}}(\underline{a})$. Without loss of generality we assume $\underline{a} = \underline{o}$. Let $\underline{z} \in D$, applying the Cauchy integral to the first variable, we see that

$$f(z_1, \cdots, z_n) = \frac{1}{2\pi i} \int_{|\zeta_1|=r_1} \frac{f(\zeta_1, z_2, \cdots, z_n)}{\zeta_1 - z_1} d\zeta_1.$$

Repeating this process, we get

$$f(z_1, \cdots, z_n) = \left(\frac{1}{2\pi i}\right)^n \int_{|\zeta_n|=r_n} \cdots \int_{|\zeta_1|=r_1} \frac{f(\zeta_1, \cdots, \zeta_n)}{\prod_{i=1}^n (\zeta_i - z_i)} d\zeta_1 \cdots d\zeta_n.$$

This is valid on the disk D. Motivated by the above equation, we have,

Definition 1.10. *The Cauchy kernel in $\mathbb{C}^n$ is*

$$C(\underline{\zeta}, \underline{z}) = \left(\frac{1}{2\pi i}\right)^n \frac{d\zeta_1 \wedge \cdots \wedge d\zeta_n}{(\zeta_1 - z_1) \cdots (\zeta_n - z_n)}.$$

Theorem 1.2. *Let $f : \overline{D} \to \mathbb{C}$ be a holomorphic function and $z \in D$, we have*

$$f(z) = \int_T f(\underline{\zeta}) C(\underline{\zeta}, \underline{z}),$$

where $T = \partial_0 D$, the distinguished boundary of the poly disk D.

Proposition 1.6. *Suppose h is a continuous function on $T = \partial_0 D$ for some disk D, then the function*

$$F(z) = \int_T h(\underline{\zeta}) C(\underline{\zeta}, z),$$

is holomorphic on D

Remark 1.6. *The theorem tells us that a holomorphic function f on a disk D is completely determined by its values on $T = \partial_0 D$.*

Remark 1.7. *The theorem also holds if the function f is continuous on $\overline{D}$ and holomorphic on D.*

This theorem tells us moreover all the local properties about the holomorphic functions. The strategies are to work on the one dimensional cases or restrict functions to complex lines. All the proofs are listed in [Fi/Li, II section 1] of A Course in Complex Analysis.

Theorem 1.3. *f is holomorphic on an open set U that is smooth. Then all the derivatives are again holomorphic.*

Sketch of the Proof. Since an open set contains a disk, we can use the Cauchy integral on that disk. Since holomorphicness is a local property, we obtain the statement. □

Corollary 1.2. *If f is as in Theorem 1.2, then for $\underline{\alpha} \in \mathbb{N}_0^n$,*

$$D^{\underline{\alpha}} f(\underline{z}) = \frac{\alpha_1! \cdots \alpha_n!}{(2\pi i)^n} \int_T \frac{f(\underline{z})}{(\zeta_1 - z_1)^{\alpha_1+1} \cdots (\zeta_n - z_n)^{\alpha_n+1}} d\zeta_1 \cdots d\zeta_n.$$

We will denote the above formula as,

$$D^{\underline{\alpha}} f(\underline{z}) = \frac{\underline{\alpha}!}{(2\pi i)^n} \int_T \frac{f(\underline{z})}{(\underline{\zeta} - \underline{z})^{\underline{\alpha}+n}} d\underline{\zeta}.$$

Corollary 1.3 (Cauchy's estimate)**.** *Motivated by Corollary 1.2. If M is the supremum of $|f|$ in some ball $B(a, r)$, then for any $n \in \mathbb{N}$, we have,*

$$|f^{(n)}(a)| \leq \frac{n!}{r^n} M.$$

Theorem 1.4 (Weierstrass)**.** *Let $(f_n : D \to \mathbb{C})_{n \in \mathbb{N}}$ be a sequence of holomorphic functions uniformly converging to $f : D \to \mathbb{C}$ where D is compact. Then f is holomorphic and all derivatives of f_i are also uniformly converging to the ones of f.*

Definition 1.11. *A domain is an open connected subset of $\mathbb{C}^n$.*

Theorem 1.5 (Maximal Principle)**.** *Let f be holomorphic on a domain U. If $|f|$ attains a local maximal then $f \equiv c$ for some constant $c \in \mathbb{C}$.*

Corollary 1.4. *Let f be a non-constant holomorphic function on a domain U. If U is bounded then $|f|$ attains the maximum in $\overline{U}$. More specifically, the maximum must be attained at boundary point of U.*

Theorem 1.6. *Suppose f is holomorphic on a closure $\overline{D}$ of some polydisk $D = D_r(\underline{z}_0)$. For $\underline{z} \in D$,*

$$\left(\frac{1}{2\pi i}\right)^n \int \cdots \int_T \frac{f(\underline{z})}{(\underline{\zeta} - \underline{z})} d\underline{\zeta}.$$

We develop this into a geometric series, and interchange the summation, we get

$$f(\underline{z}) = \sum_{\alpha \in \mathbb{N}_0^n} a_{\underline{\alpha}} (\underline{z} - \underline{z}_0)^{\underline{\alpha}}$$

where for each $\underline{\alpha} \in \mathbb{N}_0^n$,

$$a_{\underline{\alpha}} = \frac{D^{\underline{\alpha}} f(\underline{z}_0)}{\underline{\alpha}!}.$$

The series is moreover uniformly convergent on D.

Theorem 1.7. *Let f be holomorphic on a domain U. Then for each $\underline{z}_0 \in U$, there exists a convergent series*

$$f(z) = \sum a_{\underline{\alpha}}(\underline{z} - \underline{z}_0)^{\underline{\alpha}}$$

where for each $\underline{\alpha} \in \mathbb{N}_0^n$,

$$a_{\underline{\alpha}} = \frac{D^{\underline{\alpha}} f(\underline{z}_0)}{\underline{\alpha}!}.$$

Series converges at least in the largest polydisk about $D_r(\underline{z}_0) \subset\subset U$ (the boundary is contained in U).

Theorem 1.8. *Holomorphic functions are analytic, in other words, it can be locally developable into a power series.*

Theorem 1.9 (Identity Theorem)**.** *If f is holomorphic on a domain U, and it is identically 0 in a non-empty open neighborhood V in U, then f is identically 0 on U.*

Theorem 1.10. *Let f be a continuous and holomorphic in each variable separately. Then f is holomorphic.*

Proof. Can be shown using Cauchy's integral formula. □

Definition 1.12. *Let $\mathcal{G}$ be a domain. We define $\mathcal{O}(\mathcal{G})$ to be the set of all holomorphic functions on $\mathcal{G}$.*

Proposition 1.7. *$\mathcal{O}(\mathcal{G})$ is a $\mathbb{C}$-algebra. Furthermore, $\mathcal{O}(\mathcal{G})$ has no zero divisors. (Follows from Theorem 1.9).*

Proposition 1.8. *$\mathcal{O}(\mathcal{G})$ can be introduced a topology by the following way. Given a compact subset $K \subset \mathcal{G}$, we define*

$$V(K, \varepsilon) = \{f \in \mathcal{O}(\mathcal{G}) \mid |f|_K < \varepsilon\}$$

where

$$|f|_K = \max_{\underline{z} \in K} |f(\underline{z})|.$$

Such $V(K, \varepsilon)$ form a neighborhood basis for $0 \in \mathcal{O}(\mathcal{G})$.

We observe that $\mathcal{O}(\mathcal{G})$ is a Fréchet algebra, (ie. convergence automatically means locally uniform convergence) and

$$\frac{\partial}{\partial z_i} : \mathcal{O}(\mathcal{G}) \to \mathcal{O}(\mathcal{G})$$

are continuous linear maps.

We may say that $\mathcal{O}(\mathcal{G})$ is a topological $\mathbb{C}$-algebra.

Theorem 1.11 (Liouville)**.** *Let f be a bounded holomorphic function on $\mathbb{C}^n$. Then it is identically constant.*

Theorem 1.12 (Hartogs)**.** *Let f be a function which is holomorphic in each variable separately then f is holomorphic.*

Proof. See [3]. $\square$

Theorem 1.13 (Hartogs's Kugelsatz)**.** *Suppose $n \geq 2$. Let $K \subset U$ be a compact subset of an open set such that $U \backslash K$ is connected. If f is holomorphic on $U \backslash K$, then f extends holomorphically to the whole U.*

Proof. Will be shown later. $\square$

Remark 1.8. *Using the theorem above, we conclude that there are no isolated singularities and isolated zeroes. (ie. a holomorphic function on a punctured domain can be extended to that point). The latter follows that if $f(\underline{z}) = 0$, then $\frac{1}{f}(\underline{z})$ has an isolated singularity.*

Proposition 1.9. *Let $f \not\equiv 0$ be a holomorphic function on a domain $\mathcal{G}$. Then the set*

$$V(f) := \{z \in \mathcal{G} \mid f(z) = 0\}$$

has no interior points and has Lebesgue measure 0.

1.4 Holomorphic Maps and Differential Forms

For this subsection, we follow [4].

Recall that for $f : \mathbb{C} \supseteq U \to \mathbb{C}$, we have,

$$f(z) = f\left(\frac{1}{2}(z + \overline{z}), -i\frac{1}{2}(z - \overline{z})\right) =: f(x + iy)$$

Note that $\frac{\partial f}{\partial z}$ and $\frac{\partial f}{\partial \overline{z}}$ are defined in the symbolic sense. That is we have,

$$\frac{\partial f}{\partial z} = \frac{1}{2}\left(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\right), \frac{\partial f}{\partial z} = \frac{1}{2}\left(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\right).$$

Remark 1.9.

$$\begin{aligned}
\frac{\partial \overline{f}}{\partial z} &= \frac{1}{2}\left(\frac{\partial g}{\partial x} - i\frac{\partial h}{\partial x} - i\left(\frac{\partial g}{\partial y} - i\frac{\partial h}{\partial y}\right)\right), \\
&= \frac{1}{2}\overline{\left(\frac{\partial g}{\partial x} + i\frac{\partial h}{\partial x} + i\left(\frac{\partial g}{\partial y} + i\frac{\partial h}{\partial y}\right)\right)}, \\
&= \frac{1}{2}\overline{\left(\frac{\partial f}{\partial \overline{z}}\right)}.
\end{aligned}$$

Recall Proposition 1.5. Using the notation we introduced, we can rewrite it as

Proposition 1.10. *Let $F : \mathbb{C}^n \supset U \to V \subset \mathbb{C}^m$ be a function given by $F = (f_1, \cdots, f_m)$. It is holomorphic if f_i is holomorphic for each i. Or equivalently for any $f \in \mathcal{O}(V)$, the pullback $f \circ F$ is holomorphic on U.*

Corollary 1.5. *Given $U \xrightarrow{F} V \xrightarrow{G} W$ holomorphic functions, the composition $G \circ F$ is holomorphic. Obviously $\mathrm{id} : U \to U$ is holomorphic.*

Definition 1.13. *Let $U, V \subseteq \mathbb{C}^n$ be open and $F : U \to V$ be differentiable given by $F = (f_1, \cdots, f_n)$ such that for each f_i, it decomposes into*

$$f_i = g_i + ih_i.$$

We define the real Jacobian by

$$J_F^{\mathbb{R}} = \begin{pmatrix} \frac{\partial g_i}{\partial x_j} & \frac{\partial h_i}{\partial x_j} \\ \frac{\partial g_i}{\partial y_j} & \frac{\partial h_i}{\partial y_j} \end{pmatrix}_{1 \le i,j \le n} \in \mathbb{R}^{2n \times 2n},$$

and the complex Jacobian by

$$J_F^{\mathbb{C}} = \begin{pmatrix} \frac{\partial f_i}{\partial z_j} & \frac{\partial \overline{f}_i}{\partial z_j} \\ \frac{\partial f_i}{\partial \overline{z}_j} & \frac{\partial \overline{f}_i}{\partial \overline{z}_j} \end{pmatrix}_{1 \le i,j \le n} \in \mathbb{C}^{2n \times 2n}.$$

Furthermore, if f is holomorphic we have the holomorphic Jacobian

$$J_F^h = \left(\frac{\partial f_i}{\partial z_j} \right)_{1 \le i,j \le n} \in \mathbb{C}^{n \times n}.$$

Proposition 1.11. *We have the following formulas.*

$$\mathrm{rk}\, J_F^{\mathbb{R}} = \mathrm{rk}\, J_F^{\mathbb{C}}, \quad \det J_F^{\mathbb{R}} = \det J_F^{\mathbb{C}}.$$

In particular $\det J_F^{\mathbb{C}}$ is a real number. Furthermore, if F is holomorphic,

$$J_F^{\mathbb{C}} = \begin{pmatrix} J_F^h & O \\ O & \overline{J_F^h}. \end{pmatrix}$$

and therefore,

$$\det J_F^{\mathbb{C}} = |\det J_F^h|^2 \ge 0.$$

Proof.

$$J_F^{\mathbb{C}} = \begin{pmatrix} \frac{1}{2}\left(\frac{\partial g_i}{\partial x_j} - i\frac{\partial g_i}{\partial y_j}\right) + \frac{i}{2}\left(\frac{\partial h_i}{\partial x_j} - i\frac{\partial h_i}{\partial y_j}\right) & \frac{1}{2}\left(\frac{\partial g_i}{\partial x_j} - i\frac{\partial g_i}{\partial y_j}\right) - \frac{i}{2}\left(\frac{\partial h_i}{\partial x_j} - i\frac{\partial h_i}{\partial y_j}\right) \\ \frac{1}{2}\left(\frac{\partial g_i}{\partial x_j} + i\frac{\partial g_i}{\partial y_j}\right) + \frac{i}{2}\left(\frac{\partial h_i}{\partial x_j} + i\frac{\partial h_i}{\partial y_j}\right) & \frac{1}{2}\left(\frac{\partial g_i}{\partial x_j} + i\frac{\partial g_i}{\partial y_j}\right) - \frac{i}{2}\left(\frac{\partial h_i}{\partial x_j} + i\frac{\partial h_i}{\partial y_j}\right) \end{pmatrix}$$

By elementary row operations we see this has the same RREF as $J_F^{\mathbb{R}}$. Thus the first assertion about the rank is proven, furthermore this also shows that their determinants coincide as the operations do not involve multiplying any rows.

The second assertion is clear from Remark 1.9. $\square$

Definition 1.14. *A function $F : U \to V$ is biholomorphic if it is bijective and F^{-1} is also holomorphic.*

Proposition 1.12. *Let F be a biholomorphic function, then we have*

$$\det J_F^{\mathbb{C}} > 0.$$

Corollary 1.6. *Biholomorphic functions preserve orientation.*

Recall that if $f : U \to \mathbb{C}$ where $U \subset \mathbb{C}$ is a holomorphic and injective function. Then necessary we have $f'(z) \neq 0$.

Proposition 1.13. *Let $F : U \to V$ be a holomorphic bijective such that J_F^h is everywhere regular (ie. its determinant is nowhere 0). Then F is biholomorphic.*

Proof. We know that F^{-1} is differentiable. We get

$$J_F^{\mathbb{C}} \circ J_{F^{-1}}^{\mathbb{C}} = I_{2n}.$$

By the assumption, we have,

$$J_F^{\mathbb{C}} = \begin{pmatrix} M & O \\ O & \overline{M} \end{pmatrix}.$$

Thus we rewrite the first equation to be

$$\begin{pmatrix} M & O \\ O & \overline{M} \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} I_n & O \\ O & I_n \end{pmatrix}$$

We have

$$MA = E, MB = 0.$$

Thus we conclude $A = M^{-1}, B = O$(similarly $C = O, D = \overline{M^{-1}}$) which gives us the Cauchy-Riemann equations. □

Theorem 1.14 (Osgood's Theorem)**.** *Suppose $F : U \to V$ be bijective holomorphic function where $U, V \subset \mathbb{C}^n$. Then J_F^h is everywhere regular and F is therefore biholomorphic.*

Proof. [4] chapter 2.5. □

1.5 Differential Forms

Definition 1.15. *Let $\underline{x_0} \in \mathbb{R}^N$. We set*

$\mathcal{S}(\underline{x_0}) = \{f : U_f \to \mathbb{R}|$ *a continuous function defined on some neighborhood of* $\underline{x_0}\}$

We can introduce an algebraic structure by for all $f, g \in \mathcal{S}(\underline{x_0})$ and $c \in \mathbb{R}$,

$$f + g : U_f \cap U_g \to \mathbb{R}, \quad fg : U_f \cap U_g \to \mathbb{R}, cf : U_f \to \mathbb{R}.$$

Also we define

$$\mathcal{D}(\underline{x}_0) = \{f \in \mathcal{S}(\underline{x}_0) \mid \textit{differentiable at } \underline{x}_0\}.$$

A mapping $L : \mathcal{S}(\underline{x}_0) \to \mathbb{R}$ *is linear if for any* $a, b \in \mathbb{R}$ *and* $f, g \in \mathcal{S}(\underline{x}_0)$, *we have*

$$L(af + bg) = aLf + bLg.$$

Remark 1.10. $\mathcal{S}(\underline{x}_0)$ *is not a vector space since the additive identity is not unique.*

Lemma 1.1. *Let* $f \in \mathcal{D}(\underline{x}_0), g \in \mathcal{S}(\underline{x}_0)$ *such that* $f(\underline{x}_0) = 0$, *then* $f \cdot g \in \mathcal{D}(\underline{x}_0)$.

Proof. Without loss of generality we assume $\underline{x}_0 = x_0 \in \mathbb{R}$. Then

$$\begin{aligned} f(x)g(x) - f(x_0)g(x_0) &= f(x)g(x) - f(x_0)g(x) + f(x_0)g(x) - f(x_0)g(x_0), \\ &= (f(x) - f(x_0))(g(x)). \end{aligned}$$

□

Definition 1.16. *A linear map* $D : \mathcal{D}(\underline{x}_0) \to \mathbb{R}$ *is called a tangent vector to* $\underline{x}_0 \in \mathbb{R}^N$, *if*

1). $D(1) = 0$

2). For any $f, g \in \mathcal{D}(\underline{x}_0)$, *such that* $f(\underline{x}_0) = g(\underline{x}_0) = 0$, *then* $D(fg) = 0$.

Proposition 1.14. D *satisfies the following properties.*

1. *D satisfies the Leipniz rule.*
2. $\frac{\partial}{\partial x_i}\Big|_{\underline{x}_0}$ *is a tangent vector at* $\underline{x}_0$.
3. *The sets of tangent vectors at D form a vector space and it has a basis consists* $\frac{\partial}{\partial x_i}\Big|_{\underline{x}_0}$ *where* $1 \leq i \leq N$.

Proof.

$$D(fg - f(\underline{x}_0)g(\underline{x}_0)) = D((f - f(\underline{x}_0))(g - g(\underline{x}_0)) + fg(\underline{x}_0) + f(\underline{x}_0)g).$$

This proves the first assertion. The second is trivial. For the third, observe that

$$f = \sum_{i=1}^{N} \frac{\partial f}{\partial x_i}(\underline{x}_0)(x_i - \underline{x}_{0,i}) + f(\underline{x}_0).$$

Therefore, applying D, we get,

$$Df = \sum_{i=1}^{N} Dx_i \frac{\partial f}{\partial x_i}(\underline{x}_0).$$

□

Definition 1.17. *The vector space of tangent vectors at $\underline{x}_0$ is called the tangent space at $\underline{x}_0$ and denoted by $T_{\underline{x}_0}$. Its dual space is called the cotangent space and denoted by $T^*_{\underline{x}_0}$.*

Example 1.3. *Let $f \in \mathcal{D}(\underline{x_0})$, we define*

$$df : T_{\underline{x}_0} \to \mathbb{R}, \quad df(D) = Df,$$

is an element of the contangent space called a total differentiation.

Proposition 1.15. *The dx_i where $i = 1, \cdots, N$ is a basis for the cotangent space $T^*_{\underline{x}_0}$. Such basis is a dual basis to $\{\frac{\partial}{\partial x_i}\Big|_{\underline{x}_0}\}_{i=1,\cdots,N}$.*

Definition 1.18. *Let $M \subseteq \mathbb{R}^N$. A Pfaffian form or (1-form) on M associates each point $\underline{x} \in M$ to a cotangent vector $a(x) \in T^*_{\underline{x}}$. From the basis of the cotangent spaces, we know*

$$a(x) = \sum_{i=1}^{N} a_i(\underline{x}) dx_i$$

where $a_i : M \to \mathbb{R}$.

We call a is continuous, differentiable, or measurable, if all a_i are so.

Remark 1.11. *The set of 1-forms on M which is denoted by $E^1(M)$ is a module over $E^0(M)$, where $E^0(M) = \mathcal{C}^\infty(M)$ is a ring of functions.*

Definition 1.19 (Exterior Products)**.** *$E^p(M) = \bigwedge^p E^1(M)$ module of p-forms, $p = 0, \cdots, N$. We call an element a of $E^p(M)$, a p-dimensional differential form, which is of the form*

$$a = \sum_{\substack{I \subseteq \{1,\cdots,N\} \\ |I|=p}} a_I dx^I.$$

where dx^I denotes that

$$dx^I = dx_{i_1} \wedge \cdots \wedge dx_{i_N}, i_j \in I, k < l \Rightarrow i_k \leq i_l.$$

The rank of E^p is $\binom{N}{p}$.

Definition 1.20. *We define*

$$E = \bigoplus_{1 \leq p \leq N} E^p.$$

The rank of E is $\sum_{p=0}^{N} \binom{N}{p} = 2^N$.

Remark 1.12. *Let f be a differential function on an open set $M \subset \mathbb{R}^N$. We get a 1-form by*

$$M \ni x \mapsto [df(x) \in T^*_x],$$

called the total differential and this is of the form

$$df = \sum_{i=1}^{N} \frac{\partial f}{\partial x_i} dx_i.$$

Definition 1.21. *Let $a \in E^p$. a is said to be sufficiently differentiable if*

$$a = \sum_{I} a_I dx^I,$$

then

$$da = \sum_{I} da_I \wedge dx^I \in E^{p+1},$$

where a_I is a differentiable function since a is differentiable. Thus we can consider the total differentiable of a_I.

Let a be a p-form. At each $x \in M$, we have a p-covector

$$a(x) \in \bigwedge^p T_x^*.$$

Definition 1.22. *For any $p, q \in \mathbb{N}$, we define an exterior product as a bilinear map $E^p \times E^q \to E^{p+q}$ denoted by*

$$(a, b) \mapsto a \wedge b.$$

such that we have,

$$a \wedge b = (-1)^{p+q} b \wedge a.$$

In particular, if p is odd, we have,

$$a \wedge a = 0.$$

Therefore,

$$dx_i \wedge dx_i = 0.$$

More details will be found in [1]Chapter 3.

Theorem 1.15.

$$d \wedge d = 0$$

and

$$d(a \wedge b) = da \wedge b + (-1)^p a \wedge db$$

where $a \in E^p(M)$, and whenever these expression make sense.

Definition 1.23 (Transformation of Forms)**.** *Let $F : R^N \supseteq U \to V \subseteq \mathbb{R}^M$, $F = (f_1, \cdots, f_n)$ and $a \in E^p(V)$ where*

$$a = \sum_{I} a_I dy^I.$$

We define $a \circ F$ to be

$$a \circ F = \sum_I (a_I \circ F) df^I,$$

where $df^I = df_{i_1} \wedge \cdots \wedge df_{i_{|I|}}$. Then this yields a p-form on U. We denote this by $F^* a$.

Theorem 1.16.

$$d(a \circ F) = da \circ F.$$

In other words, total differential commutes with pullbacks.

1.6 Integration

Definition 1.24. *Let $M \subseteq \mathbb{R}^N$ and $a \in E^N(M)$ such that*

$$a = \tilde{a} \cdot d_1 \wedge \cdots \wedge d_N.$$

We define

$$\int_M a = \int_M \tilde{a}(x) dx_1 \wedge \cdots \wedge dx_n.$$

Definition 1.25. *Let a be a k-form in $\mathbb{R}^N$ and M be a k-dimensional surface on $\mathbb{R}^N$ such that*

$$F : Q \to M$$

parametrizes M where Q is some cube.

$$\int_M a = \int_Q F^* a := \int_Q a \circ F.$$

Theorem 1.17 (Stokes)**.** *Let $\mathcal{G} \subset\subset \mathbb{R}^N$ be a relatively compact domain such that $\partial \mathcal{G}$ is $N-1$ dimensional sufficiently smooth (ie. a union of piecewise smooth, simple(no-crossing), and closed(the initial and the terminal points agree) curves). Let a be a $N-1$-form defined and continuously differentiable on $\overline{\mathcal{G}}$, then*

$$\int_{\partial \mathcal{G}} a = \int_{\mathcal{G}} da.$$

[1] volume 3 chapter 3 integration

Now we define these notions in terms of complex analysis.

Definition 1.26. *Let $M \subset \mathbb{C}^n$ then $E^0_{\mathbb{C}}(M)$ denotes the $\mathbb{C}$-valued functions on M, and*

$$E^p_{\mathbb{C}}(M) = E^p(M) \underset{E^0(M)}{\otimes} E^0_{\mathbb{C}}(M).$$

A p-form on M is defined as

$$a = \sum_{j=1}^n a_j dx_j + i \sum_{j=1}^n b_j dy_j$$

where $z_j = x_j + iy_j$ and a_j, b_j are complex-valued functions.

If we have $f = g + ih$, then we have $df = dg + idh$. In particular ,

$$dz_j = dx_j + idy_j, \quad d\overline{z}_j = dx_j - idy_j.$$

Thus we could write instead,

$$a = \sum_{j=1}^{n} \tilde{a}_j dz_j + \sum_{j=1}^{n} \tilde{b}_j d\overline{z}_j.$$

Remark 1.13. *Let $a \in E_{\mathbb{C}}(M) := E(M) \otimes_{E^0(M)} E^0_{\mathbb{C}}(M)$ then*

$$a = \sum_{r=p+q, 0 \leq r \leq 2n} \sum_{\substack{1 \leq i_1 \leq \cdots \leq i_p \leq N \\ 1 \leq j_1 \leq \cdots \leq j_q \leq N}} a_{i_1, \cdots, ip, j_1, \cdots, j_q} \cdot dz_{i_1} \wedge \cdots \wedge dz_{i_p} \wedge d\overline{z}_{j_1} \wedge \cdots \wedge d\overline{z}_{j_q}.$$

This can also be written as

$$a = \sum_{\substack{I,J \subseteq \{1, \cdots, n\}, \\ |I| = p, \\ |J| = q}} a_{I,J} dz^I d\overline{z}^J.$$

Thus

$$E^r(M) = \bigoplus_{p+q=r} E^{p,q}(M),$$

and

$$E_{\mathbb{C}}(M) = \bigoplus_{r=0,\cdots,2n} E^r(M) = \bigoplus_r \bigoplus_{p,q} E^{p,q}(M).$$

The rank of $E_{\mathbb{C}}(M) = \sum \binom{n}{p}\binom{n}{q} = 2^{2n}$.

Consider

$$da = \sum_{I,J} da_{I,J} \wedge dz^I d\overline{z}^J,$$

In particular,

$$da = \sum a_j dz_j + \sum b_k d\overline{z}_k,$$

for

$$a_j = \frac{\partial a}{\partial z_j}, \quad b_k = \frac{\partial a}{\partial \overline{z}_k}.$$

Therefore

$$df = \partial f + \overline{\partial} f = \sum \frac{\partial f}{\partial z_j} dz_j + \sum \frac{\partial f}{\partial \overline{z}_k} d\overline{z}_k.$$

Thus we get

$$\partial : E^{p,q} \to E^{p+1,q}$$

and

$$\overline{\partial} : E^{p,q} \to E^{p,q+1}$$

Proposition 1.16. *From the remark above, we have the following statements.*

1). $d = \partial + \overline{\partial}$,

2). $d \circ d = 0$, $\partial\overline{\partial} + \overline{\partial}\partial = \partial \circ \partial = \overline{\partial} \circ \overline{\partial} = 0$.

3). Let f *be a function,* f *is holomorphic if and only if* $\overline{\partial} f = 0$

4). Graded-commutativity for $\wedge$ *holds.*

5). Leipniz rules hold for $\partial(f \wedge g)$.

Example 1.4.

$$dx_1 \wedge dy_1 \wedge \cdots \wedge dx_n \wedge dy_n = \left(\frac{i}{2}\right)^n dz_1 \wedge d\overline{z}_1 \wedge \cdots \wedge dz_n \wedge d\overline{z}_n.$$

This is called the volume form dV.

$$\int_M dV = \int_M d\lambda(z) = \lambda(M),$$

where λ *is the Lebesgue measure.*

Example 1.5. *Recall the Cauchy kernel.*

$$C(\underline{\zeta}, \underline{z}) = \left(\frac{1}{2\pi i}\right)^n \frac{d\underline{\zeta}}{(\underline{\zeta} - \underline{z})}.$$

This is a form of type $(n, 0)$ *in* $\underline{\zeta}$ *defined on the open set*

$$\mathbb{C}^n \times \mathbb{C}^n \backslash \Delta,$$

where $\Delta = \{(\underline{\zeta}, \underline{z}) \mid \forall j = 1, \cdots, n, \zeta_j = z_j\}$.

Proposition 1.17 (Transformation Rules)**.** *Let* $F : U \to V$ *be holomorphic where* $U \subset \mathbb{C}^n$ *and* $V \subset \mathbb{C}^m$. *For* $a \in E^{p,q}(V)$,

$$F^*a := a \circ F = \left(\sum_{I,J} a_{I,J} dw^I \wedge d\overline{w}^J\right) \circ F = \sum_{I,J} (a_{I,J} \circ F)(dw^I \circ F) \wedge (d\overline{w}^J \circ F),$$

where $F = (f_1, \cdots f_n)$ *and*

$$dw_j \circ F = d(w_j \circ F) = df_j,$$

for each j *and similarly for* $d\overline{w}_j$.

In particular, we have $a \in E^{p,q}(V)$, $F^*a \in E^{p,q}(U)$. *If* F *is just differentiable we get* $a \circ F \in E^{p+q}(U)$.

Proposition 1.18. *Let Let $F : U \to V$ be differentiable and $a \in E^{p,q}(V)$, we have*

$$d(a \circ F) = da \circ F.$$

Furthermore, if F is holomorphic, we have,

$$\partial(a \circ F) = \partial a \circ F, \overline{\partial}(a \circ F) = \overline{\partial} a \circ F.$$

Definition 1.27 (Poincaré Equation)**.** *Let $\mathcal{G} \subseteq \mathbb{R}^N$ be a domain and $f \in E^r$ such that $df = 0$. The equation*

$$du = f,$$

is called the Poincaré equation.

Definition 1.28. *Let $a \in E^r$. a is called*

1). a closed form if $da = 0$.

2). an exact form if there exists $u \in E^{r-1}$ such that $du = a$.

Theorem 1.18. *If the domain is convex in the above then the Poincaré equation is solvable. (ie. all the de Rham cohomologies are 0 except the $r = 0$ term.)*

Furthermore,

$$Z^r := \{f \in E^r \mid df = 0\}, B^r := \{du \mid u \in E^{r-1}\}.$$

We have $B^r \subset Z^r$, $H^r_{dR}(\mathcal{G}) := Z^r/B^r$, gives the de Rham cohomology and

$$H^0_{dR}(\mathcal{G}) = \mathbb{C}.$$

If $r \neq 0$, then

$$H^r_{dR}(\mathcal{G}) = 0.$$

Note that B^0 is empty as there is no such thing as -1-form.

Definition 1.29. *Let $\mathcal{G} \subset \mathbb{C}^n$, $f \in E^{p,q}$. Consider the equation,*

$$\overline{\partial} u = f \quad u \in E^{p,q-1}.$$

If such equation holds, then we necessarily have $\overline{\partial} f = 0$. We get

$$\overline{\partial} u = f, \overline{\partial} f = 0$$

in $\mathcal{G}$. We define similarly,

$$\begin{aligned} Z^{p,q} &= \{f \in E^{p,q} \mid \overline{\partial} f = 0\}, \\ B^{p,q} &= \{\overline{\partial} u \mid u \in E^{p,q-1}\}, \\ H^{p,q}_{\mathrm{Dol}}(\mathcal{G}) = Z^{p,q}/B^{p,q} & \end{aligned}$$

This cohomology is called Dolbeault cohomology. This gives

$$H^{0,0}_{\mathrm{Dol}}(\mathcal{G}) = \mathcal{O}(\mathcal{G})$$

the sheaf of holomorphic functions.

1.7 Bochner-Martinelli Integral Formulas

Definition 1.30. *We define the form*

$$\beta = \beta(\underline{\zeta}, \underline{z}) = \sum_{j=1}^{n} (\overline{\zeta}_j - \overline{z}_j) d\zeta_j.$$

We also define

$$B(\underline{\zeta}, \underline{z}) = \left(\frac{1}{2\pi i}\right)^n \frac{1}{\|\underline{\zeta} - \underline{z}\|^{2n}} \beta \wedge (\overline{\partial}_\zeta \beta)^{n-1},$$

where

$$\overline{\partial}_{\underline{\zeta}} \beta = \sum_{j=1}^{n} d\overline{\zeta}_j \wedge d\zeta_j = \sum_{j=1}^{n} \omega_j,$$

such that $\omega_j = d\overline{\zeta}_j \wedge d\zeta_j$ and the power means the exterior product. This B above is called the Bochner-Martinelli formel which is defined on $\mathbb{C}^n \times \mathbb{C}^n$ except the diagonal.

Remark 1.14. *For $n = 1$, we have*

$$B = \frac{1}{2\pi i} \frac{1}{(\zeta - z)(\overline{\zeta} - \overline{z})} (\overline{\zeta} - \overline{z}) d\zeta = \frac{1}{2\pi i} \frac{d\zeta}{\zeta - z}.$$

Remark 1.15. *Given a $(n, n-1)$ form in ζ which is a function in z.*

Lemma 1.2.

$$d_\zeta B(\underline{\zeta}, \underline{z}) = \overline{\partial}_\zeta B(\underline{\zeta}, \underline{z}) = 0.$$

Proof. For $\underline{z} = 0$, we get

$$\beta = \sum_j \overline{\zeta}_j d\zeta_j, \quad \omega_j = d\overline{\zeta}_j \wedge d\zeta_j.$$

If $K \subset \{1, \cdots, n\} = N$, then

$$\omega^K = \bigwedge_{j \in K} \omega_j.$$

We already know,

$$\overline{\partial}\beta = \sum \omega_j,$$

and

$$(\overline{\partial}\beta)^k = \left(\sum \omega_j\right)^k = k! \sum_{|K|=k} \omega^K.$$

Furthermore,

$$\overline{\partial}_\zeta \|\underline{\zeta}\|^2 = \overline{\partial} \sum \overline{\zeta}_j \zeta = \sum \zeta_j d\overline{\zeta}_j = \overline{\beta}.$$

$$\begin{aligned}(2\pi i)^n\overline{\partial}B(\underline{\zeta},\underline{z}) &= \overline{\partial}\left(\beta\wedge(\overline{\partial}\beta)^{n-1}\frac{1}{\|\zeta\|^{2n}}\right)\\ &= \frac{-n\overline{\beta}}{\|\underline{\zeta}\|^{2n+2}}\wedge\beta\wedge(\overline{\partial_\zeta}\beta)^{n-1}+\frac{1}{\|\zeta\|^{2n}}(\overline{\partial}\beta)^n\\ &= \frac{1}{\|\zeta\|^{2n}}\left[\frac{-n\overline{\beta}\wedge\beta}{\|\zeta\|^2}+\overline{\partial}\beta\right]\wedge(\overline{\partial}\beta)^{n-1}.\end{aligned}$$

In particular,

$$\overline{\beta}\wedge\beta=\sum_{j,k}\zeta_j\overline{\zeta}_k d\overline{\zeta}_j\wedge d\zeta_k=\sum_{j\neq k}\zeta_j\overline{\zeta}_k d\overline{\zeta}_j\wedge d\zeta_k+\sum|\zeta_j|^2\omega_j.$$

Thus we get

$$\overline{\beta}\wedge\beta\wedge(\overline{\partial}_\zeta\beta)^{n-1}=0+\sum|\zeta_j|^2\omega_j(n-1)!\sum_{k=1}^n\omega^{N\setminus\{k\}}=(n-1)!\sum_{j=1}^n|\zeta_j|^2\omega^N=(n-1)!\|\underline{\zeta}\|^2\omega^N.$$

$$(2\pi i)^n\overline{\partial}B(\underline{\zeta},0)=\frac{1}{\|\underline{\zeta}\|^{2n}}\left[-n!\omega^N+n!\omega^N\right]=0.$$

$\square$

Definition 1.31. *Let $\mathcal{G}\subseteq\mathbb{C}^n$ be a domain. $\mathcal{G}$ is said to have a reasonable boundary if the Stoke's theorem applies to $\mathcal{G}$.*

Theorem 1.19. *Let $\mathcal{G}\subset\subset\mathbb{C}^n$ with $\partial\mathcal{G}$ is reasonable Let $f\in\mathcal{C}^1(\overline{\mathcal{G}})$, and $z\in\mathcal{G}$. We have*

$$f(z)=\int_{\partial\mathcal{G}}f(\underline{\zeta})B(\underline{\zeta},\underline{z})-\int_{\mathcal{G}}\overline{\partial}f(\underline{\zeta})\wedge B(\underline{\zeta},\underline{z}).$$

The special case is that for $n=1$, we have,

$$f(z)=\frac{1}{2\pi i}\int_{\partial\mathcal{G}}\frac{f(\zeta)}{\zeta-z}d\zeta+\frac{1}{2\pi i}\int_{\mathcal{G}}\frac{\partial f(\zeta)}{\partial\overline{\zeta}}\frac{1}{\zeta-z}d\zeta\wedge d\overline{\zeta}.$$

Proof. Observe that B has a singularity of order $\frac{1}{\|\zeta-z\|^{2n-1}}$ in $\mathbb{C}^n\cong\mathbb{R}^{2n}$ since β consists of linear terms. Denote $\mathcal{G}_\varepsilon=\mathcal{G}\setminus\overline{B(z,\varepsilon)}$ for sufficiently small $\varepsilon>0$. Using the lemma, we get,

$$\begin{aligned}\int_{\mathcal{G}_\varepsilon}\overline{\partial}f(\underline{\zeta})\wedge B(\underline{\zeta},\underline{z}) &= \int_{\mathcal{G}_\varepsilon}\overline{\partial}(f(\underline{\zeta})B(\underline{\zeta},\underline{z})),\\ &= \int_{\mathcal{G}_\varepsilon}d_\zeta(f(\underline{\zeta})B(\underline{\zeta},\underline{z})),\\ &= \int_{\partial\mathcal{G}_\varepsilon}f(\underline{\zeta})B(\underline{\zeta},\underline{z}),\\ &= \int_{\partial\mathcal{G}}f(\underline{\zeta})B(\underline{\zeta},\underline{z})-\int_{\partial B(z,\varepsilon)}f(\underline{\zeta})B(\underline{\zeta},\underline{z}).\end{aligned}$$

Taking $\varepsilon \to 0$, we get

$$\int_{\mathcal{G}} \overline{\partial} f \wedge B = \int_{\partial\mathcal{G}} f(\underline{\zeta}) B(\underline{\zeta}, \underline{z}) - \lim_{\varepsilon \to 0} \int_{\partial B(z,\varepsilon)} f(\underline{\zeta}) B(\underline{\zeta}, \underline{z}).$$

□

Remark 1.16. *In $\mathbb{R}^n$ and $\alpha < n$, we have*

$$\int_{B(0,R)} \frac{dV}{r^\alpha} < \infty,$$

and if $\alpha > n$, we have

$$\int_{\mathbb{R}^n \setminus B(0,R)} \frac{dV}{r^\alpha} < \infty,$$

but if $\alpha = n$, neither of them exsits.

Lemma 1.3. *Let us define,*

$$B^1(\underline{\zeta}, \underline{z}) := (n-1) \left(\frac{1}{2\pi i} \right)^n \frac{1}{(\|\underline{\zeta} - \underline{z}\|)^{2n}} \beta \wedge (\overline{\partial_\zeta} \beta)^{n-2} \overline{\partial_z} \beta.$$

Then we have

$$\overline{\partial}_z B(\underline{\zeta}, \underline{z}) = \overline{\partial}_\zeta B^1(\underline{\zeta}, \underline{z}).$$

Let $\mathcal{G}$ be a domain and consider a ball $B(z, \varepsilon)$ inside $\mathcal{G}$. We know that

$$\int_{\mathcal{G}} \overline{\partial} f(\zeta) \wedge B(\zeta, z) = \int_{\partial\mathcal{G}} f(\zeta) B(\zeta, z) - \lim_{\varepsilon \to 0} \int_{S_\varepsilon} f(\zeta) B(\zeta, z).$$

Examine the last term,

$$\int_{S_\varepsilon} f(\zeta) B(\zeta, z) = \left(\frac{1}{2\pi i} \right)^n \frac{1}{\varepsilon^{2n}} \int_{\partial\mathcal{G}} f(\zeta) \tilde{B}(\zeta, z),$$

where

$$\tilde{B}(\zeta, z) = \beta \wedge (\overline{\partial} \beta)^{n-1}.$$

That is

$$\int_{S_\varepsilon} f(\zeta) \tilde{B}(\zeta, z) = \underbrace{\int_{S_\varepsilon} (f(\zeta) - f(z)) \tilde{B}(\zeta, z)}_{=I_1} + \underbrace{f(z) \int_{S_\varepsilon} \tilde{B}(\zeta, z)}_{=I_2}.$$

Call the integrals on the right hand side I_1, I_2, respectively.

Clearly we have,

$$|I_1| \leq \max_{\zeta \in B_\varepsilon} |f(\zeta) - f(z)| c_1 \cdot \varepsilon \cdot c_2 \cdot \varepsilon^{2n-1},$$

where the constants c_1 depends on β and c_2 depends on .Furthermore,

$$\leq c_1 c_2 \varepsilon^{2n} \max |f(\zeta) - f(z)|.$$

Therefore, the upper bound can be

$$\left| \frac{1}{2\pi i} \frac{1}{\varepsilon^{2n}} I_1 \right| \leq c \max |f(\zeta) - f(z)|.$$

By the continuity, we have $\varepsilon \to 0$, then $\zeta \to z$ thus I_1 vanishes as we take the limit.

For the another term, by applying Stoke's theorem,

$$I_2 = f(z) \int_{S_\varepsilon} \tilde{B}(\zeta, z) = f(z) \int_{B_\varepsilon} d_\zeta \tilde{B}(\zeta, z).$$

Substituting the definition of $\tilde{B}$, we get,

$$= f(z) \int_{B_\varepsilon} (\overline{\partial}\beta)^n.$$

Furthermore,

$$\overline{\partial}\beta = \sum_j \overline{\zeta}_j \wedge d\zeta_j = \sum_j \omega_j.$$

Thus we obtain,

$$f(z) n! \int_{B_\varepsilon} \omega_1 \wedge \cdots \wedge \omega_n = n! \int_{B_\varepsilon} d \operatorname{Vol} (2\pi i)^n = n! \frac{\pi^n}{n!} \varepsilon^{2n} (2i)^n = (2\pi i)^n \varepsilon^{2n}.$$

Thus taking $\varepsilon \to 0$, we get

$$I_2 \to f(z) \int_{S_\varepsilon} B(\zeta, z) = f(z).$$

Theorem 1.20. *Let $\mathcal{G}$ be a domain with $\partial\mathcal{G}$ resonable.*

If f is holomorphic on $\overline{\mathcal{G}}$, then for any $z \in \mathcal{G}$, we have

$$f(z) = \int_{\partial\mathcal{G}} f(\zeta) B(\zeta, z).$$

If f is holomorphic on a neighborhood U of $\mathcal{G}$, then we have

$$F(z) = \int_{\partial\mathcal{G}} f(\zeta) B(\zeta, z),$$

is holomorphic on $\mathcal{G}$.

Proof. We compute the Following

$$\overline{\partial}_z F(z) = \int_{\partial \mathcal{G}} f(\zeta) \overline{\partial}_z B(\zeta, z).$$

By Lemma 1.3, we get

$$\begin{aligned}
&= \int_{\partial \mathcal{G}} f(\zeta) \overline{\partial}_\zeta B^1(\zeta, z), \\
&= \int_{\partial \mathcal{G}} \overline{\partial}_\zeta (f(\zeta) B^1(\zeta, z)), \\
&= \int_{\partial \mathcal{G}} d_\zeta (f(\zeta) B^1(\zeta, z)), \\
&= \int_{\partial \partial \mathcal{G}} \cdots = 0.
\end{aligned}$$

The last equality follows from the exactness. □

Theorem 1.21 (Hartogos' Kugelsatz or Hartogos' Ball Theorem)**.** *For $n > 1$, let $U \subset \mathbb{C}^n$ be open and $K \subset U$ be compact. Assume U, K are both connected. If f is holomorphic on $U \backslash K$, there is F on U such that*

$$F|_{U \backslash K} = f.$$

In other words, f extends holomorphically to the whole U.

Proof. Let $\mathcal{G}_0, \mathcal{G}$ be domains such that $K \subset \mathcal{G}_0 \subset \mathcal{G}$ and $\mathcal{G}_0, \mathcal{G}$ have reasonable boudaries. Let $z \in \mathcal{G} \backslash \overline{\mathcal{G}_0}$. Using Bochner-Martinelli, we get

$$f(z) = \int_{\partial \mathcal{G}} f(\zeta) B(\zeta, z) - \int_{\partial \mathcal{G}_0} f(\zeta) B(\zeta, z) = F_1(z) - F_2(z).$$

Let us denote the two integral terms on the right hand sides $F_1(z), F_2(z)$, respectively. F_1 is holomorphic in z on $\mathcal{G}$, and F_2 is holomorphic in z on $\mathbb{C}^n \backslash \mathcal{G}_0$.

Let L be a complex line such that $L \cap \mathcal{G}_0 = \overline{\emptyset}$. By construction,

$$F_2(z) = \int \frac{\text{something bounded}}{\|\zeta - z\|^{2n}} \|\zeta - z\| \to 0,$$

as $z \to \infty$ under the assumption $n > 1$. Restating $z \to \infty \Rightarrow F_2(z) \to 0$.

Observe that $F_2 | L \equiv 0$, by Liouville's theorem using the single variable complex analysis. In general, $F_2 \equiv 0$ on any open subsets $U \backslash K$. By the connectedness, we conclude $F_2 \equiv 0$ on $U \backslash K$.

Thus we conclude that

$$F_1 \equiv f \text{ on } \mathcal{G} \backslash K$$

yields the holomorphic extension. □

Corollary 1.7. *There are no isolated singularities nor isolated zeros for $n > 1$.*

Proof. Suppose a function f has a singularity at z_0. Suppose f is holomorphic on $U\backslash\{z_0\}$. By the theorem above and $\{z_0\}$ is compact, we see f can be holomorphically extended to U. We derived a contradiction.

Similarly, if f has 0 at z_0 and $\frac{1}{f}$ is holomorphic on $U\backslash\{z_0\}$, then by the same argument, we derived a contradiction. □

Remark 1.17 (Caution!!)**.** *Suppose we are given a holomorphic function f on variable at least 2, such that*

$$f = g + ih.$$

where $g, h : \mathbb{R}^2 n \to \mathbb{R}$. Then

$$V(f) = V(g) \cap V(h),$$

where $V(f)$ denotes the set of zeros of f. Observe that $V(g), V(h)$ are $2n-1$ dimensional sets thus the intersection is $2n-2$ unless they do not coincide.

Theorem 1.22 (Cauchy/Pmperi)**.** *Let $\mathcal{G}$ be a domain with a resonable boundary $\partial\mathcal{G}$ such that $\mathcal{G} \subset\subset \mathbb{C}$. Let $f \in \mathcal{C}^1(\overline{G})$. For $z \in \mathcal{G}$, we have*

$$f(z) = \frac{1}{2\pi i}\int_{\partial\mathcal{G}} \frac{f(\zeta)}{\zeta - z} d\zeta + \frac{1}{2\pi i}\int_{\mathcal{G}} \frac{\frac{\partial f}{\partial \overline{\zeta}}(\zeta)}{\zeta - z} d\zeta \wedge d\overline{\zeta}.$$

Apply $f \in \mathcal{C}^1(\overline{G})$, and solve,

$$\frac{\partial u}{\partial \overline{z}} = f.$$

Proof. Assume u is a solution $\in \mathcal{C}(\overline{\mathcal{G}})$,

$$u(z) = \frac{1}{2\pi i}\int_{\partial\mathcal{G}} \frac{u(\zeta)}{\zeta - z} d\zeta + \frac{1}{2\pi i}\int_{\mathcal{G}} \frac{f(\zeta)}{\zeta - z} d\zeta \wedge d\overline{\zeta}.$$

By the assumption, we get

$$f(z) = \frac{\partial u}{\partial \overline{z}} = \frac{\partial}{\partial \overline{z}} \frac{1}{2\pi i}\int_{\mathcal{G}} \frac{f(\zeta)}{\zeta - z} d\zeta \wedge d\overline{\zeta}.$$

□

Theorem 1.23. *Let $\mathcal{G}, \partial\mathcal{G}$ as before in $\mathbb{C}$. Suppose $f \in \mathcal{C}^1(\mathcal{G}) \cap L^1(\mathbb{C})$. Then there is a solution $u \in \mathcal{C}^1(\mathcal{G})$ such that u solves,*

$$\frac{\partial u}{\partial \overline{z}} = f.$$

For $n > 1$, and $\mathcal{G}, \partial\mathcal{G}$ a domain with a reasonable boundary. We see $f \in \mathcal{C}^1_{0,1}(\overline{\mathcal{G}})$, $\overline{\partial} f = 0$. We get

$$\overline{\partial} u = f.$$

Suppose such u exists. By Bochner-Martinelli, we get

$$u(z) = \int_{\partial \mathcal{G}} u(\zeta) B(\zeta, z) - \int_{\mathcal{G}} f(\zeta) \wedge B(\zeta, z).$$

Then we get

$$f = \overline{\partial} u = \overline{\partial} \int_{\partial \mathcal{G}} \cdots - \partial \int_{\mathcal{G}} \cdots .$$

However, the first integral term on the right hand side does not equal 0.

1.8 Hartogos Levi problems.

The first problem is, given $n > 1$, there exists a domain $\mathcal{G}$ such that all holomorphic functions on $\mathcal{G}$ extends holomorphically to a larger domain. We will characterize the domains where this does not happen.

Definition 1.32. *A domain of holomorphy is a domain such that there exists a holomorphic function on it which cannnot be extended holomorphically to a larger domain.*

The second problem is that if $\mathcal{G}$ is a domain of holomorphy, can we have a function theory there?

Example 1.6. *$\mathbb{C}$ is a domain of holomorphy as it cannnot be extended to a larger domain. Let f_1, f_2 be entire function which are holomorphic in $\mathbb{C}$ which do not have common zeros.*

Solve the Diophantine equation,

$$g_1 f_1 + g_2 f_2 \equiv 1.$$

The solutions indeed exists. Assume that the zeros $\{a_1, \cdots\}$ of f_2 are simple and set

$$c_i = f_1(a_i).$$

Find g_1 with

$$g_1(a_i) = \frac{1}{c_i}.$$

Set g_2 to be

$$g_2 = \frac{1 - g_1 f_1}{f_2}.$$

Example 1.7. *Let $\mathcal{G} = \mathbb{C}^2 \backslash \{0\}$, Let us define*

$$f_1 = z_1, f_2 = z_2.$$

If we have

$$g_1 z_1 + g_2 z_2 = 1$$

on $\mathcal{G}$.

The third problem is Study the zero set of holomorphic function. This leads to analytic sets.

2 Power Series

2.1 Generalities

Notation 2.1. *We will use the following conventions.*

1). $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$

2). $\mathbb{N}_0^n = \{\alpha = (\alpha_1, \cdots, \alpha_n) \mid 1 \leq i \leq n, \alpha_i \in \mathbb{N}_0\}$

3). $z = (z_1, \cdots, z_n)$

4). For $\alpha \in \mathbb{N}_0^n$, $z^\alpha = \prod_{i=1}^n z_i^{\alpha_i}$

5). $|\alpha| = \sum_{i=1}^n \alpha_i$.

6). $f = f(z) = \sum_{\alpha \in \mathbb{N}_0^n} a_\alpha z^\alpha$

Definition 2.1. *We define the formal power series as*

$$f = \sum_{j \in \mathbb{N}_0} f_j,$$

where f_j is a homogeneous polynomial of degree j. The sums and products are given by

$$f + g = \sum_{i \in \mathbb{N}_0} f_i + g_i, \quad fg = \sum_{i=0}^{n} \sum_{j+k=i} f_j g_k.$$

The ring of formal power series with the above operation over a commutative ring $\mathbb{C}$ is denoted by

$$\mathbb{C}[\![z]\!].$$

Definition 2.2. *Let $f = \sum_{i \in \mathbb{N}_0} f_i$, then we define the order of f to be the minimum index i such that $f_i \neq 0$. We will denote this by $o(f)$.*

Proposition 2.1. *Let $f, g \in \mathbb{C}[\![z]\!]$, we have*

1). $o(f) \geq 0$ for any non-zero element. Moreover $o(f) = 0$ if and only if f is a unit.

2). $o(f) = \infty \Leftrightarrow f \equiv 0$

3). $o(fg) = o(f) + o(g)$

4). $o(f + g) \geq \min\{o(f), o(g)\}$

It follows that $\mathbb{C}[\![z]\!]$ is an integral domain.

Corollary 2.1. *$\mathbb{C}[\![z]\!]$ has a unique maximal ideal.*

$$\mathfrak{m} = \{f \in \mathbb{C}[\![z]\!] \mid o(f) > 0\}.$$

Definition 2.3 (Convergent Power Series). *For any $\underline{r} \in \mathbb{N}_0^n$, we set $c_{\underline{r}} \in \mathbb{C}$, then*

$$\sum_{\underline{r}\in\mathbb{N}_0^n} c_{\underline{r}}$$

converges to $c \in \mathbb{C}$ if for any $\varepsilon > 0$, there is a finite subset $J_0 \subset \mathbb{N}_0^n$ such that for all finite $J \subset \mathbb{N}_0^n$, with $J \supset J_0$ we have,

$$\left|\sum_{\underline{r}\in J} c_{\underline{r}} - c\right| < \varepsilon.$$

Proposition 2.2. *The convergence of the series implies the absolute, unconditional convergence, and uniform convergence.*

Definition 2.4. *Let $c_{\underline{r}}$ be a function on M then*

$$\sum_{\underline{r}\in\mathbb{N}_0^n} c_{\underline{r}} \overset{M}{\Rightarrow} c$$

if and only if for any $\varepsilon > 0$, there exists a finite set $J_0 \subset \mathbb{N}_0^n$ such that for all finite set with $J \supset J_0$, and all $x \in M$, we have

$$\left|\sum_{\underline{r}\in J} c_{\underline{r}}(x) - c(x)\right| < \varepsilon.$$

Example 2.1 (Geometric Series). *Let $0 \leq q_i \leq 1$ for $i = 1, \cdots, n$, then*

$$\sum_{\underline{r}\in\mathbb{N}_0^n} \prod_{i=1}^{n} q_i^{r_i} = \prod_{i=1}^{n} \frac{1}{1-q_i}.$$

Proposition 2.3. *Let*

$$f = \sum_{\underline{r}\in\mathbb{N}_0^n} a_{\underline{r}} z^{\underline{r}} \in \mathbb{C}[\![z]\!].$$

Take

$$\underline{z}_0 = (z_1^0, \cdots, z_n^0) \in \mathbb{C}^n, z_j^0 \neq 0.$$

If there exists $S > 0$ such that

$$|a_{\underline{r}} z_0^{\underline{r}}| \leq S,$$

then $f(z)$ converges for all $\underline{z}$ with

$$\forall j, |z_j| \leq z_{0,j}.$$

Converges uniformly for all $\underline{z}$ such that there exists $r_j > 0$ with

$$|z_j| \leq r_j < |z_{0,j}|.$$

Corollary 2.2. *A sum of power series is holomorphic function.*

Remark 2.1 (Warning)**.** *If $f \in \mathbb{C}[\![z]\!]$, the domain of covergence is equal to the interior of the set of convergence.*

Example 2.2. *Let*

$$f(z_1, z_2) = z_2 \sum_{i=1}^{\infty} z_1^i.$$

Then the set of convergence is

$$\{(z_1, z_2) \mid z_2 = 0\} \cup \{(z_1, z_2) \mid |z_1| < 1\}.$$

The domain of convergense is

$$\{|z_1| < 1\}.$$

Remark 2.2. *The convergence set is logarithmetically convex complete Reinhardt domain. For details, see Chapter 1 of [2].*

Definition 2.5. *We define the ring of convergent power sereis to be*

$$\mathbb{C}\{z_1, \cdots, z_n\} \subset \mathbb{C}[\![z_1, \cdots, z_n]\!],$$

which consists of those formal power series whose domain of convergence contains some neighborhood around the origin.

Remark 2.3. *This inherits the properties from $\mathbb{C}[\![z_1, \cdots, z_n]\!]$, namely, it is an integral domain, which is local with unique maximal ideals*

$$\mathfrak{m} = \{f \mid f(0) = 0\},$$

and units are the one such that $o(f) = 0$.

Remark 2.4. *It is clear that if f is a unit then there exists $g \in \mathbb{C}\{z_1, \cdots, z_n\}$ such that $fg = 1$. Therefore,*

$$o(f) + o(g) = o(1) = 0.$$

Therefore $o(f) = 0$. if f is holomorphic at 0 and $f(0) \neq 0$. Then $\frac{1}{f}$ is also holomorphic at 0, therefore taking the Taylor series of $\frac{1}{f}$ to be g, we obtain the desired unit.

Remark 2.5. *For $n = 1$, we get,*

$$(a_0 + a_1 z + \cdots)(b_0 + b_1 z + \cdots) = 1,$$

where $a_0 \neq 0$.

Definition 2.6. *Let $z \in \mathbb{C}^n$, f, g be holomorphic functions defined in z. We denote $f \sim_z g$ if f coincides with g on some neighborhood of z. Such equivalence classes are called the germs of holomorphic functions. The set of those equivalence classes is denoted by $\mathcal{O}_z$ and called the stalk at z.*

Definition 2.7. *The ring of germs at* $z_0 \in \mathbb{C}^n$ *which is denoted by* $\mathcal{O}_{z_0}$ *is isomorphic to* $\mathbb{C}\{z - z_0\} \cong \mathbb{C}\{z\}$.

Definition 2.8. *If* f *is a holomorphic function on* U *and* $z \in U$*. The germ of* f *at* z *is denoted by* f_z *or* $\gamma_z f$ *(ie. an equivalence class which is represented by the taylor series of* f *around* z*).*

Remark 2.6. *Let* $\mathcal{D}_z$ *be the ring of smooth funtions defined arond* z *and* $\mathcal{C}_z$ *be the continuous function defined around* z*, then we have*

$$\mathcal{O}_z \subset \mathcal{D}_z \subset \mathcal{C}_z.$$

Definition 2.9. *The sheaf of holomorphic function is defined to be*

$$\mathcal{O} = \bigsqcup_{z \in \mathbb{C}^n} \mathcal{O}_z,$$

together with a canonical projection $p : \mathcal{O} \to \mathbb{C}^n$.

Definition 2.10. *A (continuous) section in* $\mathcal{O}$ *corresponding to an open set* $U \subset \mathbb{C}^n$ *is such that*

$$\sigma : U \to \mathbb{C}, p \circ \sigma = \mathrm{id}, \sigma(z) \in \mathcal{O}_z.$$

There exists a holomorphic function f *on* U *such that*

$$\sigma(z) = f_z = \gamma_z f.$$

We define the topology over it by setting the base of the topology to be a collection consists of those $\sigma(U)$ *where* σ *is a section and* U *is open in* $\mathbb{C}^n$*,*

Definition 2.11. *Let* X, Y *be topological space and* $f : X \to Y$*.* f *is said to be locally topological/homeomorphic if any point* $x_0 \in X$*, there exits open neighborhood* U *around* x_0 *and* V *around* $f(x_0)$ *such that*

$$f|_U : U \to V$$

is a homeomorphism.

Proposition 2.4. $\mathcal{O}$ *is a Hausdorff space.* $p : \mathcal{O} \to \mathbb{C}^n$ *is continuous and locally topological(ie. locally homeomorphic). The fiber is*

$$\mathcal{O}_z = p^{-1}(z)$$

The addition and multiplication are continuous.

We have the fiber product,

$$\mathcal{O} \times_p \mathcal{O} \xrightarrow{+,\cdot} \mathcal{O}$$

with

$$\begin{array}{ccc} \mathcal{O} \times_p \mathcal{O} & \xrightarrow{\pi_1} & \mathcal{O} \\ {\scriptstyle \pi_2}\downarrow & & \downarrow{\scriptstyle p} \\ \mathcal{O} & \xrightarrow[p]{} & \mathbb{C}^n \end{array}$$

We have a one-to-one correspondence between (continuous) sections and holomorphic functions.

We observe that $\mathcal{O}$ is a structure sheaf of $\mathbb{C}^n$. This can also be generalized for $\mathcal{D}, \mathcal{C}$.

Remark 2.7. *The above proposition almost holds for $\mathcal{D}, \mathcal{C}$ except that they are not Hausdorff.*

2.2 The Weierstrass Theorem

Notation 2.2.

$$H_n := \mathbb{C}\{z_1, \cdots, z_n\} = \mathbb{C}\{z\}.$$

Definition 2.12. *$f \in H_n$, then*

$$f = \sum_{i=0}^{\infty} f_i z_n^i,$$

where $f_i \in H_{n-1}$.

Definition 2.13. *$f \in H_n$ is called z_n-regular of order k if $f_i(0, \cdots, 0, z_n) = 0$ for $i = 0, \cdots, k-1$ and $f_k(0, \cdots, 0, z_n) \neq 0$. That is to say*

$$f(0, \cdots, 0, z_n) = \sum_{i \geq k}^{\infty} c_i z_n^i \neq 0,$$

and $c_k \neq 0$ is a constant.

Theorem 2.1 (Vorbereitungs Satz)**.** *Let $f \in H_n$ be z_n-regular of k. Then there are elements $e \in H_n, \omega \in H_{n-1}[z_n]$ with*

1. *e is a unit in H_n,*
2. *$\omega = z_n^k + c_1(z_1, \cdots, z_{n-1}) z_n^{k-1} + \cdots + c_n(z_1, \cdots, z_{n-1})$.*
3. *$f = e \cdot \omega$.*

Furthermore, e, ω are uniquely determined and

$$\omega(0, \cdots, 0, z_n) = z_n^k.$$

ie $H_{n-1} \ni c_r(0, \cdots, 0) = 0$. We call ω the Weierstrass polynomial of f.

Proof. Without loss of generality, we may assume that the coefficient of z_n^k is 1. Thus we can write,

$$f(z) = \sum_{i=0}^{k-1} c_i(z_1, \cdots, z_{n-1}) z_n^i + z_n^k + \sum_{i>k} c_i(z_1, \cdots, z_{n-1}) z_n^i.$$

where for all $i < k$,

$$c_i(\underbrace{0, \cdots, 0}_{n-1}) = 0.$$

$$f(z) = z_n^k \left(\sum_{i=0}^{k-1} c_i(0, \cdots, 0) z_n^{n-i} + 1 + \sum_{i>k} c_i(0, \cdots, 0) z_n^{i-k} \right) = z_n^k (A + 1 + B).$$

where

$$A = \sum_{i=0}^{k-1} c_i(z_1, \cdots, z_{n-1}) z_n^{i-k}, \quad B = \sum_{i=k+1}^{\infty} c_i(z_1, \cdots, z_{n-1}) z_n^{r-k}.$$

Let $W = A + B$ and R be the radius of convergence. Pick $R > \rho > 0$ such that

$$|z_n| \leq \rho \Rightarrow |B| < \frac{1}{2}$$

For A, take $\frac{\rho}{2} \leq |z_n| \leq \rho$, and find $\delta > 0$ such that

$$|(z_1, \cdots, z_{n-1})| \leq \delta \Rightarrow |A| < \frac{1}{2}.$$

Together we have, $|W| < 1$. Now consider,

$$\log(1 + W) = W - \frac{W^2}{2} + \frac{W^3}{3} - \cdots = C + D.$$

where C is the Taylor expansion part and D is the principal part.

$$f(z) = z_n^k \exp(\log(1 + W)) = z_m^k \exp(C + D) \Rightarrow e^{-C} f(z) = z_n^k \exp D.$$

On the left hand side, it is powers in z_n, so is the right hand side. That is, D consists of only regular powers of z_n.

$$z_n^k \left(1 + D + \frac{D^2}{2} + \cdots \right) = z_n^k + a_1(z_1, \cdots, c_{n-1}) z_n^{k-1} + \cdots + a_n(z_1, \cdots, c_{n-1}) + R,$$

where R is the remainder. Since the Laurent development is unique we conclude R to be 0.

We will set

$$\omega(z_1, \cdots, z_n) = z_n^k + a_1(z_1, \cdots, z_{n-1}) z_n^{k-1} + \cdots + a_n(z_1, \cdots, z_{n-1}).$$

We now have the decomposition,

$$f(z) = e^C w(z', z_n).$$

Note that e^C is a unit. Thus we both proved the existence and the uniqueness. $\square$

Theorem 2.2 (Weierstrass Formal Division Theorem)**.** *Let $g \in H_n$ be z_n-regular of order k. Then for each $f \in H_n$, there are elements $q \in H_{n-1}, r \in H_{n-1}[z_n]$ where $\deg_{z_n} r < k$ which are uniquely determined with*

$$f = qg + r.$$

Proof. By Weierstrass Vorbereitungs Satz, we may assume that $g = e\omega$ where ω is the Weierstrass polynomial of order k. Without loss of genearlity, we may assume $e = 1$. Thus ,we have,

$$g = z_n^k \exp(D),$$

where D is regular polynomial of z_n. Similarly as in the previous statement, we get

$$\frac{f}{g} = q + H,$$

where q is the Taylor part, and H is the principle part. We have,

$$f = qg + gH \Rightarrow f - qg = gH.$$

Let $r = gH$, we see $f - qg$ has only non-negative powers of z_n. Therefore gH also contains no negative powers. The power in H is a polynomial of degree at most $k - 1$.

Thus we obtained,

$$f = qg + r.$$

For the uniqueness, it is enough to consider,

$$0 = qg + r.$$

However, observe that

$$-qg = r,$$

on the left hand side, qg is of order k, but $\deg_{z_N} r < k$. We conclude $q, r = 0$. $\square$

Corollary 2.3. *Consider the formula,*

$$f = qg + r.$$

If $g = \omega$ a Weierstrass polynomial and f is a polynomial in $H_{n-1}[z_n]$ of degree m ,then q is a polynomial of degree $m - k$ or it is identically 0.

If f is a polynomial of degree n in the Weierstrass Vorbereitungs Satz, then e is a polynomial of degree $n - k$.

Proof. For the first claim, it follows from Euclidean division. For the second claim, $f = e\omega$, then this is nothing but a division of f by ω. Therefore, e must be a polynomial. $\square$

Corollary 2.4. *Suppose $f_1, \cdots, f_m \in H_n \backslash \{0\}$. Then there is a linear transformation, $\sigma : \mathbb{C}^n \to \mathbb{C}^n$ such that $f_j \circ \sigma$ are z_n-regular of order k_j.*

Proof. Find a line L in H_n such that $f_j|_L \neq 0$. Change the coordinate so that L appears as an axis. $\square$

Remark 2.8. *Weierstrass Vorbereitungs Satz and Weierstrass Formal Division Theorem are equivalent and true for convergent power series over completely valued field.*

Remark 2.9. *A natural proof for these theorems are,*

$$f(z) = e(z)\omega(z_1, \cdots, z_n).$$

leads to equivalence system for the coefficients monomials.

Example 2.3.

$$z_1 z_2 \in H_2$$

is neither z_1 nor z_2 regular. But take $z_1 = u_1 + u_2$, and $z_2 = u_2$, then

$$z_1 z_2 = (u_1 + u_2)u_2 = u_2^2 + u_1 u_2,$$

which is u_2 regular of order 2.

3 Some Algebra

3.1 Properties of Rings

In this section, we assume that R is a ring and $A, B, M_1, \cdots, M_i, \cdots$, are R-modules.

Definition 3.1. *A R-module M is called Noetherian if one of the following equivalent conditions holds.*

i). Each submodule $N \subseteq M$ is finite (ie. finitely generated).

ii). Each ascending sequence $N_0 \subsetneq N_1 \subsetneq \cdots \subsetneq M$ stabilizes.

iii). Each non-empty set of submodules contains a maximal element.

The ring R itself is Noetherian if R is Noetherian as a moduler over itself. (ie. ideals are finitely generated and so are other conditions hold for ideals).

Example 3.1. *Fields are Noetherian. Principle ideal domains are Noetherian, in particular, $\mathbb{Z}$ or $k[X]$ where k is a field are all Noetherian.*

Theorem 3.1 (Hilbert Nullstellensatz)**.** *If R is Noetherian then so is $R[x]$. In particular for any $R[x_1, \cdots, x_n]$ is Noetherian.*

Theorem 3.2. *Given an exact sequence in R-modules,*

$$0 \longrightarrow M \longrightarrow N \longrightarrow P \longrightarrow 0$$

If any two of them are Noetherian so is the other one.

Theorem 3.3. *Suppose R is Noetherian and M is a R-module. M is Noetherian if and only if M is finite.*

Definition 3.2. *A ring R is local if it contains a unique maximal ideal $\mathfrak{m}$.*

Example 3.2. *H_n-s are local.*

Theorem 3.4 (Krull's lemma)**.** *Let $(R, \mathfrak{m})$ is a local Noetherian ring. Let M be a finite R-module with two submodules $M_1, M_2 \subseteq M$. We have,*

$$\forall l \in \mathbb{N}, M_1 \subseteq M_2 + \mathfrak{m}^l M \Rightarrow M_1 \subseteq M_2.$$

In particular, $M = R$, we set $\mathfrak{a} = \bigcap_{k \geq 0} \mathfrak{m}^k$. We have,

$$\mathfrak{a} \subset (0) + \mathfrak{m}^l.$$

Thus $\mathfrak{a} = (0)$.

Corollary 3.1. *$\mathcal{C}_0^\infty$ be the ring of germs of smooth function at 0 and $\mathfrak{m} = \{f \in \mathcal{C}_0^\infty \mid f(0) = 0\}$ be the ideal of germs at 0. Then*

$$\bigcap \mathfrak{m}^l \neq (0).$$

Therefore $\mathcal{C}_0^\infty$ is not Noetherian.

We assume R to be an integral domain.

Definition 3.3. *An element $x \in R$ is irrducible if x is neither 0 nor a unit, and*

$$x = uv \Rightarrow u \text{ or } v \text{ is a unit..}$$

An element $p \in R$ is prime if

$$p | ab \Rightarrow p | a \vee p | b.$$

Remark 3.1. *Primes are irreducible.*

Definition 3.4. *A ring R is called factorial if for any $x \in R \backslash \{0\}$ is a product of primes.*

Proposition 3.1. *The following conditions are equivalent for a ring R.*

i). R is factorial.

ii). Each nonzero element $x \in R$ is a product of irreducibles which is unique up to ordering.

iii). Each nonzero element is a product of irreducibles and all irreducibles are primes.

Example 3.3. $\mathbb{Z}, k[x]$ *are factorial. The ring of Gauss integers* $\mathbb{Z}[i]$ *is factorial, it is in particular, a principal ideal domain.*

Theorem 3.5 (Gauss)**.** *If R is factorial then $R[x]$ is also factorial. The units of $R[x]$ are of those in R. The primes in $R[x]$ are*

1). the primes of R,

2). the primes which are also primitive polynomials which are in $Q = \mathrm{Frac}(R)$ (ie. the polynomial with coefficients in R and all of them are coprime.).

Definition 3.5. *Let k be a field and $(A, \mathfrak{m})$ be a local ring. Consider*

$$k \xrightarrow{a \mapsto a \cdot 1_A} A \xrightarrow{a \cdot 1_A \to \overline{a \cdot 1_A}} A/\mathfrak{m}$$

Let us denote the whole map $\sigma : k \to A/\mathfrak{m}$. If σ is an isomorphism, we call $(A, \mathfrak{m})$ a local k-algebra. This induces $\rho : A \to k$ by sending $f \mapsto \sigma(\overline{f})^{-1}$.

Definition 3.6. *A is called hensel/hensilian if it satisfies the following conditions.*

Let $f(t)$ be a monic polynomial in Let $\omega(t) = \rho(f(t))$ where $\rho : A \to k$ which induces $\rho : A[t] \to k[t]$ by $t \mapsto t$ and $\omega(t) = \omega_1(t)\omega_2(t)$ where both of them are monic polynomial coprime to one another. Then there are monic polynoial $f_1(t), f_2(t) \in A[t]$ with $f(t) = f_1(t)f_2(t)$, with

$$\rho(f_i(t)) = \omega_i(t).$$

Example 3.4. *Valuation rings are Hensel.*

3.2 Algebras of H_n

The typical case, we have,

$$H_1 = \mathbb{C}\{x\},$$

which is a principal ideal domain and also is local with the unique maximal ring (z) and all non-trivial ideals are of the form. (z^k) for $k \in \mathbb{N}$.

Definition 3.7. *Let*

$$f = \sum_{j=0}^{\infty} f_j,$$

where f_j is a homogeneous polynomial of degree j. The we define

$$o(f) = \min\{j \in \mathbb{N} \mid f_j \neq 0\}$$

Remark 3.2. *We have the following properties.*

i). $o(fg) = o(f) + o(g)$.

ii). $o(f) = 0 \Leftrightarrow f$ *is a unit.*

Theorem 3.6. *Let $\omega \in H_n[t]$ be monic of degree s. Let*

$$\Omega(t) = \omega(0, t) \in \mathbb{C}[t].$$

Thus by the fundamental theorem of algebra, we have,

$$\Omega(t) = \prod_{i=1}^{l} (t - c_i)^{s_i},$$

where $s = \sum_{i=1}^{l} s_i$ and c_i-s are distinct.

Then there are monic polynomials $\omega_i(z, t) \in H_n[t]$ such that

$$\omega(0, t) = (t - c_i)^{s_i}, \omega(z, t) = \prod_{i=1}^{l} \omega_i(z, t).$$

Furthermore, ω_i-s are unique.

Proof. We prove this by induction on l. For $l = 1$ is trivial as take $\omega_1(z, t) = \omega(z, t)$.

Suppose the statement holds for $l - 1$. If $\omega(0, 0) = 0$, then

$$\Omega(t) = t_1^{s_1} \tilde{\Omega(t)}.$$

Then this decomposition tells us that $\omega(z, t) \in H_n[t] \subset H_{n+1}$ which is t-regular of order s_1 by Weierstrass Vorbereitungs Satz. We get,

$$\omega = \omega_1 e.$$

We have,

$$e \in H_n[t], \quad o(e) = s - s_1.$$

$$\omega(z, t) = \omega_1(z, t) e(z, t).$$

Observe that,

$$e(0, t) = \tilde{\Omega}(t) = \prod_{i=2}^{l} (t - c_i)^{s_i}.$$

apply the induction hypothesis we get the dcomposition.

For the general case, let c_1 be such that $\omega(c_1, 0) = 0$. Then consider $\omega'(z, t) = \omega(z + c_1, t)$.

$$\prod_{i=1}^{l} \omega_i'(z, t) = \omega'(z, t).$$

Set $\omega_i(z, t) = \omega_i'(z - c_1, t)$. $\square$

Theorem 3.7. *H_n is Noetherian, factorial and henselian.*

Proof. Let $\mathfrak{a} \subseteq H_n$ be an ideal and $f \in \mathfrak{a}, f \neq 0$. Consider $\sigma : H_n \xrightarrow{\sim} H_n$ an automorphism such that σf is z_n-regular of order k. Observe that $\sigma\mathfrak{a}$ is finite so is $\mathfrak{a}$.

Let $h \in H_n$, by Weierstrass division formula, we have,

$$h = qf + r,$$

where $r \in H_{n-1}[z_n]$ and $\deg r < k$. Write

$$r(z) = a_0 z_n^{k-1} + a_1 z_n^{k-1} + \cdots + a_{k-1},$$

where $a_k \in H_{n-1}$. Associate,

$$h \mapsto (a_0, \cdots, a_{k-1}) \in H_{n-1}^k.$$

Then such map $H_n \to H_{n-1}^k$ is surjective with kernel (f).

$$\begin{aligned} H_n \to A := H_n/(f) &\xrightarrow{\sim} H_{n-1}^k \\ \mathfrak{a} \mapsto \overline{\mathfrak{a}} &\overset{\sim}{\mapsto} \{\text{finite submodule of } H_{n-1}\}. \end{aligned}$$

Suppose H_{n-1} is Noetherian then $\overline{\mathfrak{a}}$ is finite. So there are $f_1, \cdots, f_l \in \mathfrak{a}$ such that

$$(\overline{f}_1, \cdots, \overline{f}_l) = \overline{\mathfrak{a}}.$$

Therefore, we see

$$(f, f_1, \cdots, f_l) = \mathfrak{a}.$$

For the factorial part, let $f \in H_n$.

If f is irreducible there is nothing to show. If f is not irreducible then $f = f_1 f_2$ and by the defnition, we can assume f_1, f_2 are not units. Therefore,

$$o(f_1), o(f_2) < o(f).$$

Thus by induction, we see f is a product of irreducible.

We have H_1 is factorial. Suppose the statement holds for $n-1$, then for an irreducible $f \in H_n$, consider,

$$f | f_1 f_2,$$

we may assume they are z_n-regular. By Weierstrass Vorbereitungs Satz, we have

$$f = e\omega, f_1 = e_1\omega_1, f_2 = e_2\omega_2.$$

Since e, e_1, e_2 are all unit, in particular ω is irreducible. Thus if

$$\omega | \omega_1\omega_2 \Leftrightarrow \omega_1\omega_2 = q\omega,$$

for some $q \in H_n$. By the assumption we have $q \in H_{n-1}[z_n]$. This means that

$$\omega | \omega_1\omega_2$$

in $H_{n-1}[z_n]$. We have ω is prime in $H_{n-1}[z_n]$. Thus either $\omega|\omega_1$ or $\omega|\omega_2$. Using Theorem 3.5, we conclude $\omega|\omega_1$ or $\omega|\omega_2$ in H_n.

Finally for the Henselian part, let $\omega \in H_n[t]$, and consider

$$\rho : H_n \to \mathbb{C}, \rho(f) = f(0).$$

This induces

$$\rho(\omega(z,t)) = \omega(0,t) \in \mathbb{C}[t].$$

□

Definition 3.8. *Let $\mathcal{K}$ be the sheaf of meromorphic functions on $\mathbb{C}^n$. We define*

$$\mathcal{K}_0 = K_n = \mathrm{Frac}(H_n).$$

Remark 3.3. *As we have seen H_n is factorial, any element $m \in H_n$ can be expressed as $m = \frac{f}{g}$ where f, g are coprime.*

Theorem 3.8. *Let $f, g \in \mathcal{O}(U)$. Assume that at $z \in U$, f_z and g_z are coprime. Then there is a neighborhood V of z such that for all $\zeta \in V$ the germs f_ζ, g_ζ are coprime. Moreover let $a \in \hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ and let*

$$m(z) = \frac{f(z)}{g(z)} \text{ on } V.$$

Then there is, for any neighborhood V' of z, a point $\zeta \in V'$ with $m(\zeta) = a$.

Proof. Without loss of generality, we may assume that $z = 0$, and $f_0, g_0 \in H_n = \mathcal{O}_0$ which are z_n-regular. By Weierstrass Vorbereitungs Satz, we have,

$$f_0 = e_0\omega_0, g_0 = \tilde{e_0}\tilde{\omega_0}.$$

By assumption $\omega_0, \tilde{\omega_0}$ are relatively prime in $\mathcal{O}_0$. Sor f, g are monic polynomial in z_n which is a Weierstrass polynomial at 0.

Again using f_0, g_0 are relatively prime in $H_{n-1}[z_n]$. But this implies that f_0, g_0 are also relatively prime in $\mathcal{K}_{n-1}[z_n]$. Formally, there exists $A, B \in \mathcal{K}_{n-1}$ such that

$$Af + Bg = 1 \quad (\text{in } H_{n-1}[z_n]),$$

and multiply this by the common denominator, we get,

$$af + bg = c.$$

Note that a, b are holomorphic in 0 in $\mathbb{C}^n$, and $c = c(z_1, \cdots, z_{n-1}) \in H_{n-1}$.

In a neighborhood of the origin,

$$U = U_{(z_1,\cdots,z_{n-1})} \times U_{z_n},$$

for $z \in U$.

$$a(z_1, \cdots, z_n)f(z_1, \cdots, z_n) + b(z_1, \cdots, z_n)g(z_1, \cdots, z_n) = c(z_1, \cdots, z_{n-1}).$$

Let $\zeta = (\zeta_1, \cdots, \zeta_n)$. Then f, g are $(z_n - \zeta_n)$-regular of some order. And the same is true for all the common divisors of f and g. Thus by applying Weierstrass Vorbereitungs Satz to $z_n - \zeta_n$, we obtained a Weierstrass polynomial in $z_n - \zeta_n$ which divide f, g simultaneously. On the other hand, we have the equation,

$$af + bg = c.$$

Consequently, the common divisor of f, g divides c. That is to say for a common divisor ω in (ζ', ζ) we have $\omega_\zeta | c_\zeta$.

But $\deg_{z_n, \zeta_n} c_\zeta = 0$, thus,

$$\deg_{z_n, \zeta_n} \omega_\zeta = 0.$$

In $z_n - \zeta_n$, we have $\omega = 1$, thus f_ζ, g_ζ are with no common divisors.

For the second part, we assume $h = 0$ and replace f by $f - hg$. Suppose f, g are Weierstrass polynomial at 0 such that

$$af + bg = c.$$

f has zeros for $|z_n|$ arbitrary small. Assume zeros of f are also zeros of g. Then for all z' we get

$$c(z') = 0.$$

This is a contradiction. Thus there must be a point in the neighborhood of the origin such that f is 0 but g is not. $\square$

Remark 3.4. *f_z, g_z above are not units. We have,*

$$m(z) = \frac{z_1}{z_2}.$$

Several cases to consider that the value $\frac{f_z}{g_z}$ takes poles, zeros, or indeterminants. The above case is a trivial example of it.

4 Analytic Sets

4.1 Basics

Definition 4.1. *Let $U \subseteq \mathbb{C}^n$ be a domain. A subset $M \subseteq U$ is said to be locally analytic if for each point $x \in M$, there is a neighborhood $V = V(x)$ and finitely many functions $f_1, \cdots, f_k \in \mathcal{O}(V)$ such that*

$$M \cap V = \{z \in M \mid i = 1, \cdots, k, f_i(z) = 0\}.$$

We will denote,

$$V(f_1, \cdots, f_k) := \{z \in M \mid i = 1, \cdots, k, f_i(z) = 0\}.$$

$\mathbb{M}$ is called analytic if it is locally analytic and closed in U.

Definition 4.2. *Let $U \subseteq \mathbb{C}^n$ be a domain. A subset $M \subseteq U$ is called a locally analytic submanifold if for any $z \in M$ there exists a neighborhood $W = W(z)$ of z and finitely many holomorphic functions $f_{k+1}, \cdots, f_n \in \mathcal{O}(W)$ such that*

$$M \cap W = V(f_{k+1}, \cdots, f_n),$$

and the matrix

$$\left(\frac{\partial f_i}{\partial z_j}\right)_{\substack{i=k+1,\cdots,n \\ j=1,\cdots,n}},$$

has the highest rank possible.

M is called an analytic submanifold if it is locally analytic submanifold and closed.

Remark 4.1. *Clearly, analytic submanifolds are also analytic sets. Also there are coordinates $w_1, \cdots w_n$ such that*

$$M \cap W = \{w \mid w_{k+1} = \cdots w_n = 0\}.$$

The above assertion follows from implicit function theorem.

Definition 4.3. *Let M be an analytic submanifold. The dimension of M is k such that*

$$M \cap W = \{w \mid w_{k+1} = \cdots w_n = 0\}.$$

Definition 4.4. *Let M be an analytic set. A point $z \in M$ is called regular(or simple) if in a neighborhoodof $W = W(z)$ of z, $M \cap W$ is also an alaytic submanifold of W. The other points are called singular. The set of singular points is denoted by*

$$S(M) = \{z \mid z \text{ is singular}\}.$$

Remark 4.2. *$S(M)$ is closed.*

Remark 4.3. *Suppose $F : U \to U'$ is biholomorphic and $M \subset U$ is an analytic set then $F(M)$ is also analytic in U'.*

Remark 4.4. *In $\mathbb{C} = \mathbb{C}^1$, all the locally analytic sets are discrete points.*

Definition 4.5. *Let U be a domain, consider holomorphic functions $\mathcal{O}(U)$ over U. Take any subset $\mathcal{M} \subset \mathcal{O}(U)$ and let us define,*

$$M = V(\mathcal{M}) := \{z \in \mathbb{C}^n \mid \forall f \in \mathcal{M}, f(z) = 0\}.$$

Remark 4.5. *$V(\mathcal{M})$ above is analytic.*

Example 4.1. *Let $V(z_1 z_2) \in \mathbb{C}^2$. Then we get,*

$$V(z_1 z_2) = \{z_1 = 0\} \cup \{z_2 = 0\}.$$

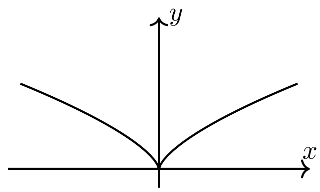
All the points are regular except for the origin.

Example 4.2. *In $\mathbb{C}^2$, the set $V(z_1^2 - z_2^3)$ contains only regular points except for the origin.*

Consider the map,

$$\mathbb{C} \ni t \mapsto (t^3, t^2) \in \mathbb{C}^2.$$

This is a continuous bijection, but not biholomorphic.



Proposition 4.1. *Let $U \subset \mathbb{C}^n$ be domain. We have,*

1). $\emptyset, U$ are both analytic in U.

2). Arbitrary intersection and finite unions of analytic sets are analytics.

Proof. Clearly, $\emptyset = V(1), U = V(0)$. For the second part, suppose we have,

$$A = V(f_1, \cdots, f_n), B = V(g_1, \cdots, g_m).$$

Then we have,

$$A \cap B = V(f_1, \cdots, f_n, g_1, \cdots, g_m), A \cup B = V\left(\{f_i g_j\}_{\substack{i=1,\cdots,n \\ j=1,\cdots,m}}\right).$$

□

Remark 4.6. *For a proper analytic subset A of a domain U, we have $A^\circ = \emptyset$.*

Proposition 4.2 (Riemann First Extension Theorem)**.** *Let U be a domain and $A \subsetneqq U$ be analytic. Suppose $f \in \mathcal{O}(U \backslash A)$ is locally bounded on U, then f extends to U.*

Proof. We may assume that for some $z_0 \in A$, A is given by $A = V(g)$ for a single holomorphic function g on $W(z_0)$. Assume after change of coordinates by Weierstrass Vorbereitungs Satz, we assume g is a Weierstrass polynomial in $z_0 = 0$.

Write $f = f(z', z)$ where z' is fixed, then f is holomorphic in z_n The extension is given by

$$f(z', z_n) = \frac{1}{2\pi} \int_{\text{boundary of the ball around the critical points}} \frac{f(z', \zeta_n)}{(\zeta_n - z_n)} d\zeta_n.$$

This is holomorphic in (z', z_n). This tells us that f can be extended. □

Remark 4.7. *Suppose U is a domain and $A \subsetneq U$ is an analytic set such that $U \backslash A$ is connected. If $U \backslash A$ is not connected, that is there exists two open sets U_1, U_2 such that $U_1 \cap U_2 = \emptyset$ and $U_1 \cup U_2 = U \backslash A$. Set,*

$$f(z) = \begin{cases} 1 & (on\ U_1), \\ 0 & (on\ U_2). \end{cases}$$

Then f can be extended.

Remark 4.8. *There are no isolated singularities in $\mathbb{C}^n$ for $n > 1$. Suppose f has an isolated 0 and f is holomorphic on $\overline{U} \backslash \{0\}$ and consider a line l, passing through some point z in U but does not cotain 0. Then f is bounded on $l \cap \partial U$. In particular, f is bounded by z in its maximum $|f|_{\partial U}$.*

4.2 Some More Algebras and Topologies

Definition 4.6. *Let K be a field of characteristic 0 and $\overline{K}$ be its algebraic closure. Consider a monic polynomial $f(t) \in K[t]$ such that*

$$f(t) = t^k + a_1 t^{k-1} + \cdots a_k.$$

Then f has k-many zeros $t_1, \cdots, t_k$ allowing multiplicity.

We then define the discriminant of f to be

$$D_f := \prod_{j<k} (t_j - t_k)^2.$$

Remark 4.9. *$D_f \in \mathbb{Z}[a_1, \cdots, a_k] \subset K$ and $D_f = 0$ if and only if it contains a multiple roots.*

Definition 4.7. *Let*

$$f(t) = \sum_{i=0}^{n} a_i t^{n-i}, g(t) = \sum_{i=0}^{m} b_i t^{m-i}.$$

Then we define

$$R(f,g) = \det \begin{pmatrix} a_0 & a_1 & \cdots & a_n & 0 & \cdots & 0 \\ 0 & a_0 & \cdots & a_{n-1} & a_n & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & \cdots & \cdots & \cdots & \cdots & a_n \\ b_0 & b_1 & \cdots & b_m & 0 & \cdot & 0 \\ 0 & b_0 & \cdots & b_{m-1} & b_m & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & \cdots & \cdots & \cdots & \cdots & b_m \end{pmatrix}$$

which is a determinant of $(n+m)(n+m)$ times matrix where we placed $a_0, \cdots, a_n$ m-many times and $b_0, \cdots, b_n$ n-many times. We call this the resultant of f, g. and this belongs to $\mathbb{Z}[a_0, \cdots, a_n, b_0, \cdots, b_m]$.

Proposition 4.3. *Let*

$$f(t) = \sum_{i=0}^{n} a_i t^{n-i}, g(t) = \sum_{i=0}^{m} b_i t^{m-i}.$$

such that $a_0 b_0 \neq 0$. Then $R(f, g) = 0$ if and only if f, g have common divisors.

Proposition 4.4. *Let $f(t) = t^m + a_1 t^{m-1} + \cdots$, then we have,*

$$D_f = R(f, f').$$

Proposition 4.5. *Suppose R is a factorial ring and $\omega \in R[t]$. Let*

$$\omega = \omega_1 \cdots \omega_l,$$

be its prime decomposition. We have $D_\omega = 0$ if and only if there are prime appearing multiple times in ω.

Proof. Let $D_\omega = 0$ for $\omega \in R[t]$. Then ω, ω' have a common prime. We have,

$$\omega' = \sum_{i=1}^{n} \omega_i' \prod_{j \neq i} \omega_j.$$

Thus there is ω_i appearing at least twice. $\square$

Definition 4.8. *Let X, Y be connected, locally path connected Hausdorff spaces. Consider a continuous map $p : X \to Y$. It is called an unlimited unbranched covering map if for all $y \in Y$ there exists a neighborhood $V = V(y)$ of g such that*

$$p^{-1}(V) = \coprod_{i \in \mathbb{N}} U_i,$$

where U_i-s are pairwise disjoint and $p|_{U_i} \cong V$.

A continuous map $p : X \to Y$ is called a covering if it is discrete (ie. each fiber is discrete) and open in the discrete topology, $p^{-1}(\{y\})$ is an isolated points. We call the cardinality of $p^{-1}(\{y\})$ the number of sheets.

Example 4.3.

$$\mathbb{C} \to \mathbb{C}, z \mapsto z^2,$$

as well as

$$\mathbb{C}^\times \to \mathbb{C}^\times, z \mapsto z^2,$$

are unlimited unbranched coverings.

Definition 4.9. *Let $f : X \to Y$ be a continuous function of locally compact spaces.*

1. *f is proper if for any compact set $K \subseteq Y$, $f^{-1}(K)$ is compact.*
2. *f is finite if it is proper and discrete.*

4.3 Analytic Hypersurfaces

Lemma 4.1. *Let $U \subseteq \mathbb{C}^n$ be a domain. Consider a polynomial*

$$f(z,t) = t^k + a_1(z)t^{k-1} + \cdots + a_k(z).$$

where a_i is holomorphic in U for each i. Suppose there is c such that $f(c,t)$ has k-distinct zeros $\{\alpha_1, \cdots, \alpha_k\}$ in $\mathbb{C}$. Then there is a neighborhood W of c, and holomorphic functions $\varphi_1, \cdots, \varphi_k$ on W such that

$$f(z,t) = \prod_{i=1,\cdots,k} (t - \varphi_i(z)).$$

In particular $\varphi_i(c) = \alpha_i$.

Proof. Suppose $c = 0$ and $f_0 \in \mathcal{O}_0 = H_{n+1}$, (ie the coefficients of f are in H_n). We have,

$$f_0 = \prod_{i=1}^{k}(t - \alpha_i).$$

Since H_n is hensilian, there exists germs $f_1, \cdots, f_k \in H_n[t]$ such that $f_i(0,t) = t - \alpha_i$ thus set $f_i = t - \varphi_i$ for $\varphi_i \in H_n$. Choose a neighborhood $0 \in W$, such that all the germs are given by holomorphic functions. Then $\varphi_i : W \to \mathbb{C}$ with

$$f(z,t) = \prod_i (t - \varphi_i(z)).$$

□

Notation 4.1. *For a function $f : D \to \mathbb{C}$ defined on some domain D. For a subset $S \subseteq D$, we denote*

$$\mathcal{N}(f,S) = |\{s \in S \mid f(s) = 0\}.$$

Theorem 4.1 (Argument Principle)**.** *Let $D \subseteq \mathbb{C}$ be a domain and $f : D \to \mathbb{C}$ be a holomorphic function. Then we have,*

$$\mathcal{N}(f,D) = \frac{1}{2\pi i} \oint_{\partial D} \frac{f'(z)}{f(z)} dz.$$

Lemma 4.2. *Let $\underline{a} = (a_1, \cdots, a_k) \in \mathbb{C}^k$ and set $f(t) = t^k + a_1 t^{k-1} + \cdots \in \mathbb{C}[t]$. Suppose f has zeros at $\alpha_1, \cdots, \alpha_l \in \mathbb{C}$ with multiplicities $r_1, \cdots, r_l$. Then for any $\varepsilon > 0$, there is $\delta > 0$ such that if $b = (b_1, \cdots, b_k) \in D_\delta := D_\delta(\underline{a})$ then for each $\lambda = 1, \cdots, l$, the polynomial*

$$g(t) = t^k + b_1 t^{k-1} + \cdots + b_k,$$

has exactly r_λ zeros in $D_\lambda := D_\varepsilon(\alpha_\lambda)$.

Proof. Let

$$F(\underline{z}, t) = t^k + z_1 t^{k-1} + \cdots,$$

be holomorphic on $\mathbb{C}^k \times \mathbb{C}$. Consider,

$$X := V(F) \subseteq \mathbb{C}^{k+1} = \{F(z,t) = 0\}.$$

For $\underline{a} \in \mathbb{C}^k$, we have,

$$\left(\bigcup_{i=1}^{l} \{\underline{b}\} \times \partial D_\lambda\right) \cap X = \emptyset,$$

then there exists $\delta > 0$ such that

$$\left(\bigcup_{i=1}^{l} D_\delta(\underline{a}) \times \partial D_\lambda\right) \cap X = 0.$$

Using F we can write $g(t) = F(b,t)$ for $\underline{b} \in D_\delta := D_\delta(\underline{a})$. Using Theorem 4.1, we have for each $\lambda = 1, \cdots, l$,

$$\mathcal{N}(g, D_\lambda) = \frac{1}{2\pi i}\int_{\partial D_\lambda} \frac{F'(\underline{b},t)}{F(\underline{b},t)} dt,$$

which is an integer but the expression in the right hand side is continuous in $\underline{b}$. Thus this is constant and we can take $\underline{b} = \underline{a}$ and the above quantity equals to r_λ. $\square$

Lemma 4.3. *Let $f(z,t) = t^k + a_1(z)k^{k-1} + \cdots$ on $U \times \mathbb{C}$ where $U \subseteq \mathbb{C}^k$ open. Let $X = V(f) = \{f = 0\}$. Suppose $p : U \times \mathbb{C} \to U$ is a natural projection and $\pi = p|_X : X \to U$. We have,*

1). π is discrete (i.e. the fibers are discrete).

2). π is proper (i.e. the preimages of compacts sets are again comapct)

Proof. The first assertion is clear. For the second part, consider a compact subset $K \subseteq U$. Then each $a_1, \cdots, a_k$ are bounded on K. Lemma 4.2 says that for $z \in K$, the zeros of $f(z,t)$ is bounded by C which is independent of z. We have,

$$\pi^{-1}(K) = V(f) \cap (K \times \{t \mid |t| \leq C\}),$$

which is clearly a compact set in $X = V(f)$. $\square$

Definition 4.10. *Let X be an analytic set of a domain $U \subseteq \mathbb{C}^n$. X is called an analytic hypersurface if for all $x \in X$, there is a neighborhood W of x and $f \in \mathcal{O}(W)$ such that $X \cap W = V(f)$.*

Remark 4.10. *In Definition 4.10 above, if $0 \in W$ and $f \in \mathcal{O}(W)$ which is z_n-regular at 0 then by Weierstrass Vorbereitungs Satz, we have,*

$$f = e\omega,$$

where ω is a Weierstrass polynomial at 0. So choose W small enough that all the germs are given on W by holomorphic function and e has zeros. Thus $X \cap W = V(\omega)$.

Proposition 4.6. *A hypersurface can locally be defined on $V(\omega)$ where ω is a polynomial in z_n Wierstrass polynomial at one of the points.*

Proof. Without the loss of generality, we my assume that ω has not multiple prime factor. $\square$

Theorem 4.2. *Let $f \in H_n[t]$ be monomial with no multiple prime factors and* $\deg(f) = k$*, suppose $f \in \mathcal{O}(U)[t]$. Then the discriminant $\mathcal{O}(U) \ni D_f \neq 0$.*

Set

$$X = V(f) \subseteq U \times \mathbb{C}, S = V(D_f) \subset U, U' = U \backslash S.$$

Furthermore, let

$$p : U \times \mathbb{C} \to U, \pi = p|_X, X' = X \backslash \pi^{-1}(S) = \pi^{-1}(U').$$

Then we have the following statements.

1. *$X' \xrightarrow{\pi} U'$ is an unlimited unbranched covering with k-sheets.*
2. *$X \xrightarrow{\pi} U$ is a covering.*
3. *X' is locally the graph of a holomorphic function.*
4. *X' is connected if and only if f is irreducible in $\mathcal{O}(U)^\circ[t]$ where X' is locally a graph of f.*

Proof. Clearly π is discrete and open. Take $(z_0, t_0) \in U \times \mathbb{C}$ such that $f(z_0, t_0) = 0$. Let W be a neighborhood of (z_0, t_0). Choose ε, δ according to Lemma 4.2 such that for $z \in D_\delta(z_0)$ there exists at least one zero of $f(z, f)$ in $D_\varepsilon(t_0)$. Take δ, ε small enough so that $D_\delta \times D_\epsilon \subseteq W$, and

$$\pi(W) \supseteq U.$$

Assume that X' is not connected. That is, $X' = X_1' \cup X_2'$ for disjoint open sets X_1', X_2'. Take $z_0 \in U'$, then we know that

$$\begin{aligned} f(z,t) &= \prod_{i=1,\cdots,k} (t - \varphi_i(z)), \text{ holomorphic close to } z_0, \\ &= \prod_{i=1,\cdots,k_1} (t - \varphi_i(z)) \prod_{i=k_1+1,\cdots,k} (t - \varphi_i(z)), \\ &= f_1(z,t) f_2(z,t), \text{ close to } z_0. \end{aligned}$$

Note that $\varphi_i(z)$ is holomorphically continued to all of U'. We obtain,

$$f(z,t) = (t^{k_1} + a_1^1(z) t^{k_1 - 1} \cdots)(t^{k_2} + a_2^1(z) t^{k_2 - 1} \cdots)$$

where a_i^j are elementary symmetric functions of $\{\varphi_i\}_{i=1,\cdots,k_1}$ or $\{\varphi_i\}_{i=k_1+1,\cdots,k}$, respectively with degree i. In particular, they are holomorphic on U'. By

Riemann-extension theorem, we obtain that $a_i^j(z)$ for $j = 1, 2$ extends to a holomorphic function on U. We have that

$$f(z,t) = f_1(z,t)f_2(z,t).$$

For the other direction. Suppose f is not irreducible that is $f(z,t) = f_1(z,t)f_2(z,t) \in \mathcal{O}(U)[t]$. THen

$$X' = V(f) = V(f_1) \cup V(f_2),$$

which is a disjoint union. $\square$

Remark 4.11. *The set S is considered as the set of points which are not regular, for example intersections of some curves.*

Remark 4.12. *All points in X' are regular. But there are also regular points in $X \backslash X' = \pi^{-1}(S)$.*

For the second part, $\pi : X \to U$ covering then $\pi|_{X'} : X' \to U'$ proper by the previous lemma and is locally homeomorphic. Thus it is an unlimited unbranched covering. Again by the lemma, at a point $(z_0, t_0) \in X'$, we have X' is given by the local holomorphic function $t = q_x(z)$ notation from the lemma.

Remark 4.13. *Let X_{reg} be the set of regular points. Then $X' \subseteq X_{\text{reg}} \subset X$. May of course happen that $X_{\text{reg}} \neq X'$.*

4.4 Functions on Analytic Sets

In this section, $X \subseteq \mathbb{C}^n$ is used to denote some analytic set, and $X_{\text{reg}} \subseteq X$ the set of regular points of X.

Definition 4.11. *A function $f : X \to \mathbb{C}$ is called holomorphic at a point $z_0 \in X$ if there is a neighborhood $U = U(z_0) \subseteq \mathbb{C}^n$ and a holomorphic function $F : U \to \mathbb{C}$ such that*

$$F|_{U \cap X} = f.$$

In particular, it is called holomorphic if it is holomorphic at all points of X.

A function $f : X \to \mathbb{C}$ is weakly holomorphic, if it is continuous and holomorphic at all regular points.

A function $f : X_{\text{reg}} \to \mathbb{C}$ is normal if f is holomorphic and locally bounded on X.

Remark 4.14. *We have the following implications,*

$$f \text{ is holomorphic} \Rightarrow f \text{ is weakly holomorphic} \Rightarrow f \text{ is normal.}$$

They are all equivalent if all points of X are regular.

Example 4.4. *Motivated by Example 4.2. Let $X \subseteq \mathbb{C}^2$ such that $X = V(z^3 - w^2)$. Consider the following holomorphic function,*

$$\mathbb{C} \ni t \overset{\varphi \ holomorphic}{\mapsto} (t^2, t^3) \in X \subseteq \mathbb{C}^2.$$

Observe that φ is a homeomorphism. For $(z,w) \in X$, set $t = \sqrt{z}$, then we get

$$\varphi(\pm t) = (z,w).$$

Consider $\varphi^{-1} : X \to \mathbb{C}$. φ^{-1} is continuous and holomorphic outside of the origin. Therefore, this is weakly holomorphic function. However, this is not holomorphic at the origin.

Suppose there is $h(z,w)$ in $U(0)$ such that

$$h|_X = \varphi^{-1}.$$

In other words,

$$(h|_X) \circ \varphi = \varphi^{-1} \circ \varphi.$$

We have,

$$h(t^2, t^3) = t.$$

Observe that $h(t^2,t^3)$ has 0 of order at least 2, but the right hand side shows that it has 0 of order 1.

Proposition 4.7. *Suppose X is an analytic set without singular points contained in a domain $U \subseteq \mathbb{C}^n$. X is a analytic submanifold of dimension $k \leq n$. That is to say, local properties of holomorphic functions on $\mathbb{C}^k$ are true for holomorphic functions on X. (e.g. Theorem 1.9(Identity Theorem), Theorem 1.5(Maximal Principle)).*

Proposition 4.8. *Let X be an analytic set such that X_{reg} is connected and dense in X. If f is weakly holomorphic on X and is identically 0 on some non-empty open subset U of X, then f is identically 0 on X.*

Proof. Note that X_{reg} is open by definition. As $U \cap X_{\text{reg}}$ is a non-empty open set and $f \equiv 0$ on $U \cap X_{\text{reg}}$, using Proposition 4.7, $f \equiv 0$ on X_{reg}. By the continuity of f, $f \equiv 0$ on X. □

Example 4.5. *Let $X = V(z_1 z_2) \subseteq \mathbb{C}^2$. The origin is the only singular point in X. Consider $f(z_1, z_2) = z_1$. However, this doesn't mean f is everywhere 0 on X. This shows that the condition of connectedness in the proposition above is essential*

Example 4.6. *Let $X = V(z_1 z_2) \subseteq \mathbb{C}^2$. Consider,*

$$f = \begin{cases} 1 & z_1 = 0, \\ 0 & z_2 = 0. \end{cases}$$

This function cannot be continued to the origin not even continuously. We see the case that Riemann extension theorem does not hold.

Theorem 4.3. *Let X be a hypersurface containing the origin and consider*

$$X = \{(z,t) \mid w(z,t) = 0\},$$

where

$$w(z,t) = t^k + a_1(z)t^{k-1} \cdots ,$$

which is defined on $D^m \times D$*. Let* f *be a a normal function on* X*. Then there are holomorphic functions*

$$a_1(z), \cdots , a_k(z),$$

on D *such that for all* $z \in X_{\mathrm{reg}}$*,*

$$f(z)^k + a_1(z)f(z)^{k-1} + \cdots + a_k(z) \equiv 0.$$

Proof. Let Δ be the discriminant of ω and $S = V(\Delta)$. Set $X' = \pi^{-1}(D \backslash S)$ where $\pi(z,t) = z$.

$$X' \xrightarrow{\pi} D' = D \backslash S,$$

is an unlimited unbranched cover. Let $z \in D'$, and $V(z) = V$ be a neighborhood. We have,

$$\pi^{-1}(V) = U_1 \cup \cdots \cup U_k,$$

which is a disjoint union. Consider,

$$\pi|_{U_i} : U_i \xrightarrow{\sim} V,$$

which is biholomorhphic. Note that

$$U_i = \{(z,t_i) \mid t_i \text{ is a zero of } \omega(z,t)\}.$$

Also we have,

$$U_i = \{(z, \varphi_i(z))\},$$

for some $\varphi_i \in \mathcal{O}(V)$ defined on V. We set,

$$f_i(z) = f(z, \varphi_i(z)).$$

We see,

$$\prod_{i=1,\cdots,k} (f(z,t) - f_i(z)) = 0,$$

on V. On the other hand, this is equal to

$$= f(z,t)^k + a_1(z)f(z,t)^{k-1} + \cdots$$

where $a_1, \cdots$ are defined in all of D. □

Remark 4.15. *The above theorem tells that there exists an integer algebraic equation for* f*. we have,*

Theorem 4.4 (Maximal principle)**.** *Let* X *be a hypersurface and* X_{reg} *connected. For a weakly holomorphic function* f*, if it is bounded then it is constant.*

Proof. Suppose $0 \in X$, and pick a coordinate such that $f(0)$ is real and greater than 0(ie. multiplying by $e^{i\theta}$). We know thre is sucu $a_1, \cdots$ that,

$$f^k + a_1 f^{k-1} \cdots = 0.$$

Previously we have seen that each a_i can be given by a symmetric functions of $f(z, t_i)$. In particular, we have,

$$\pm a_1 = \sum_{i=1}^{k} f(z, t_i).$$

Thus we have,

$$a_1(0) = kf(0).$$

Recall that $|f(z, t_i)| \leq f(0)$. a_1 has an absolute maximum? at 0, thus

$$a_1 \equiv kf(0).$$

In particular,

$$kf(0) = a_1(z) = \sum f(z, t_i) \leq \sum |f(z, t_i)| \leq kf(0).$$

Thus we have,

$$\sum |f(z, t_i)| = kf(0) \Rightarrow |f(z, t_i)| = f(0).$$

Furthermore,

$$\sum f(z, t_i) = \sum |f(z, t_i)| \Rightarrow f(z, t_i) = |f(z, t_i)| = f(0).$$

This tells us that close to the origin the functions $f(z, t_i)$ are constant. By the identity, we get,

$$f \equiv f(0).$$

□

4.5 More Algebras

Let R be a commutative ring with a unit.

Definition 4.12. *An ideal $\mathfrak{q} \subset R$ is primary if $fg \in \mathfrak{q}$ and $g \notin \mathfrak{q}$ then $f^l \in \mathfrak{q}$ for some power $l \in \mathbb{N}$.*

Definition 4.13. *Let $\mathfrak{a}, \mathfrak{b} \subseteq R$ be ideals we define,*

$$(\mathfrak{a} : \mathfrak{b}) = \{f \in R \mid f\mathfrak{b} \subseteq \mathfrak{a}\}.$$

Definition 4.14. *Let M be a R-module. We define,*

$$\text{Ann}(M) = \{f \in R \mid fM = 0\}.$$

Definition 4.15. *An ideal* $\mathfrak{a} \subseteq R$ *is irreducible if for any ideals* $\mathfrak{b}, \mathfrak{c} \subseteq R$, *we have,*

$$\mathfrak{a} = \mathfrak{b} \cap \mathfrak{c} \Rightarrow \mathfrak{a} = \mathfrak{b} \text{ or } \mathfrak{a} = \mathfrak{c}.$$

Remark 4.16. *Prime ideals are both primary and irreducible.*

Remark 4.17.

$$\sqrt{\sqrt{\mathfrak{a}}} = \sqrt{\mathfrak{a}}.$$

We also have,

$$\sqrt{\mathfrak{a} \cap \mathfrak{b}} = \sqrt{\mathfrak{a}} \cap \sqrt{\mathfrak{b}}$$

Remark 4.18. *It is an exercise to show that,*

$$\sqrt{\mathfrak{a}} = \bigcap_{\mathfrak{p} \in \operatorname{Spec}(R/\mathfrak{a})} \mathfrak{p}.$$

and

$$\mathfrak{q} \text{ is primary then there is } \mathfrak{p} \in \operatorname{Spec}(R) \text{ such that } \sqrt{\mathfrak{q}} = \mathfrak{p}.$$

Corollary 4.1.

$$\sqrt{(0)} = \bigcap_{\mathfrak{p} \in \operatorname{Spec} R} \mathfrak{p}.$$

Proof. Follows directly from Remark 4.18. ☐

Definition 4.16. *A ring is reduced if* $\sqrt{(0)} = (0)$.

Lemma 4.4 (Primary Decomposition)**.** *Let* R *be Noetherian. An ideal is primary if and only if it is irreducible. Furthermore, every ideal is finite intersection of primary ideals.*

Proof. Exercise. ☐

Definition 4.17. *Let* R *be Noetherian and* $\mathfrak{a} \subset R$ *be an ideal. The primary decomposition,*

$$\mathfrak{a} = \bigcap_{i=1,\cdots,k} \mathfrak{q}_i,$$

is irredundant if all the $\mathfrak{q}_i$ *are different.*

Theorem 4.5 (Lasker-Noether)**.** *Let* R *be Noetheria. Then any ideal* $\mathfrak{a} \subset R$ *admits an irredundant representation, i.e.*

$$\mathfrak{a} = \bigcap_{i=1,\cdots,k} \mathfrak{q}_i.$$

Furthermore, the number of primary ideals k *is uniquely determined and so is the radicals* $\mathfrak{p}_i$ *of* $\mathfrak{q}_i$ *which are uniquely determined.*

If $\mathfrak{p}$ is a prime ideal containing $\mathfrak{a}$ and it is minimal with respect to inclusion. Then the radials of $\mathfrak{q}_i$ is $\mathfrak{p}_i$ for some i and

If $\mathfrak{a} = \sqrt{\mathfrak{a}}$, then

$$\mathfrak{a} = \bigcap_{\substack{i=1,\cdots,k \\ \mathfrak{p}_i \in \mathrm{Spec}(R)}} \mathfrak{p}_i,$$

where $\mathfrak{p}_i$-s are minimal ideals containing $\mathfrak{a}l$

Definition 4.18. *Consider the primary decomposition of $\mathfrak{a} \subseteq R$ where R is Noetherian. Each $\mathfrak{p}_i$ of the decomposition is called the primary component of $\mathfrak{a}$. The minimal ones among such primary ideals are called either minimal/isolated. Otherwise they are called embedded.*

4.6 Analytic germs

Definition 4.19. *Let X be topological space and $A, B \subset X$ and $a \in X$. We say,*

$$A \sim_a B$$

if there exista a neighborhood of $a, U = U(a)$, $U \cap A = U \cap B$. We denote a class of this equivalence relation by A_a or B_a.

Definition 4.20. *An analytic germ at $a \in \mathbb{C}^n$ is a germ at a of an analytic set X. Thus we write X_a for such germ.*

Lemma 4.5. *Union, intersection, and inclusion of analytic germs are defined. Furthermore, finite unions/intersections of analytic germs are again analytic.*

Definition 4.21. *Let $a \in \mathbb{C}^n$ and considre germs at $\mathcal{O}_a$. We define,*

$$\mathfrak{a} = (f_1, \cdots, f_n) \subset \mathcal{O}_a,$$

an ideal generated by $f_1, \cdots, f_n$. such that f_i converges in $U(a)$ for all $i = 1, \cdots, k$. Then we define,

$$V(\mathfrak{a}) = \{x \in \mathbb{C}^n \mid \forall i = 1, \cdots, k, f_i(x) = 0\}_a,$$

which is a germ associated with a.

Lemma 4.6. *$V(\mathfrak{a})$ only depends on $\mathfrak{a}$.*

Definition 4.22. *Let X_a be an analytic germ at a represented by an analytic set $X \subseteq U(a)$ and $f \in \mathcal{O}_a$ given by holomorphic function on $U(a)$. We have,*

$$f|_{X_a} = 0 \Leftrightarrow f \equiv 0 \text{ on } X \cap U(a).$$

We define,

$$J(X_a) = \{f \in \mathcal{O}_a \mid f \text{ vanishes on } X_a\}.$$

So $J(X_a)$ is an ideal in $\mathcal{O}_a$.

Remark 4.19. *We want to show what relation we have with,*

$$\mathfrak{a} \mapsto V(\mathfrak{a}),$$
$$J(X) \leftarrow\!\shortmid X.$$

Proposition 4.9. *For any analytic sets X, Y, we have,*

$$VJ(X) = X.$$

Furthermore,

$$\begin{aligned} X \subseteq Y \Leftrightarrow J(X) \supseteq J(Y), \\ J(X \cap Y) = J(X) + J(Y), \\ J(X \cup Y) = J(X) \cap J(Y). \end{aligned}$$

And for any ideal $\mathfrak{a}$, we have,

$$JV(\mathfrak{a}) = \sqrt{\mathfrak{a}}.$$

Furthermore, if $\mathfrak{a}, \mathfrak{b} \subset R$, we have,

$$\begin{aligned} \mathfrak{a} \subseteq \mathfrak{b} \Leftrightarrow V(\mathfrak{a}) \supseteq V(\mathfrak{b}), \\ V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b}), \\ V(\mathfrak{a}, \mathfrak{b}) = V(\mathfrak{a}) \cap V(\mathfrak{b}). \end{aligned}$$

Proposition 4.10. *Consider a descending sequence of germs*

$$X_1 \supseteq X_2 \supseteq \cdots,$$

Then this stabilizes at finitely many steps.

Proof. Look at the ideals and use that $\mathbb{C}$ is a field thus $\mathbb{C}[x]$ is Noetherian. $\square$

Definition 4.23. *Let X be irreducible then there is no analytic sets Y, Z which are proper subsets of X such that $X = Y \cup Z$.*

Proposition 4.11. *X is irreducible if and only if $J(X)$ is prime.*

Proof. Suppose $J(X)$ is not prime then, there are $f, g \in \mathcal{O}_a$ with $fg \in J(X)$, such that $f, g \notin J(X)$. Then we have,

$$V(f) \cup V(g) = V(fg) \supseteq X.$$

Thus we have $X = (V(f) \cap X) \cup (V(g) \cap X)$ which neither of the factors are equal to X. Thus X is not irreducible.

The question is if $\mathfrak{p}$ is prime then is $V(\mathfrak{p})$ irreducible? We have,

$$JV(\mathfrak{p}) \supset \mathfrak{p}.$$

Recall Hilbert/Rückert Nullstellensatz,

$$JV(\mathfrak{p}) = \mathfrak{p}.$$

Accordingly, we have,

$$JV(\mathfrak{a}) = \sqrt{\mathfrak{a}}.$$

□

Definition 4.24. *We say,*

$$X = \bigcup_{i=1,\cdots,k} X_i,$$

is irredundant if no X_i can be skipped.

Theorem 4.6. *Every analytic germs is the irredundant union of finitely many prime gemrs*

$$X = \bigcup_{i=1,\cdots,k} X_i,$$

where each X_i is uniquely determiend by X.

Proof. Let $\mathfrak{M}$ be the set of germs which is not-empty and $\mathfrak{J}$ be the corresponding ideals. Since $\mathbb{C}[x]$ is Noetherian, we have, $\mathfrak{J}$ has a maximal element. Thus $\mathfrak{M}$ has a minimal element.

Consider $\mathfrak{M} = \{\ X$ where the statement fails.$\}$. Then by the argument above, there is a minimal element X^0 of $\mathfrak{M}$. Since X^0 is redundant we have $X^0 = X^1 \cup X^2$ where X^1, X^2 satisfies the statement. Then we can take X_i^0 so that X^0 also satisfies the statement.
The uniqueness follows from Lasker-Noether theorem. Indeed, we have,

$$X = X_1 \cup \cdots \cup X_r,$$

then consider

$$\mathfrak{a} = J(X) = J(X_1) \cap \cdots \cap J(X_r) = \mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_r.$$

□

Example 4.7. *Let X be an analytic set on $U \subseteq \mathbb{C}$. Consider a germ X_a at $a \in X$. We have,*

$$\mathcal{O}_a / J(X_a) = \mathcal{O}(X)_a.$$

We say $\mathcal{O}(X)_a$ is the associated ring of holomorphic germs on X.

Example 4.8. $X_1 = V(z_3) \subseteq \mathbb{C}^3$ *and* $X_2 = V(z_1 z_2) \subseteq \mathbb{C}^3$.

Example 4.9. *Consider $X = V(\underbrace{z^2 - uw^2}_{=f}) \subseteq \mathbb{C}^3$. Then f is irreducible but at $a = (0, 0, a_3 \neq 0)$ then $f = (z + w\sqrt{u})(z - w\sqrt{u})$. We see that X_0 is prime but X_a is not.*

5 Convexity

5.1 Holomorphic convexity

Definition 5.1. *Let $\mathcal{G} \subseteq \mathbb{C}^n$ be a domain. An extension pair is a tuple (U, V) of open sets such that U is non-empty and V is connected not equal to $\mathcal{G}$ satisfying,*

$$U \subseteq \mathcal{G} \cap V.$$

Definition 5.2. *Let $f\mathcal{O}(\mathcal{G})$. An extension of f is a triplet (U, V, F)*

i). (U, V) is an extension pairing,

ii). $F \in \mathcal{O}(V)$,

iii). $F|_U \equiv f|_U$.

Definition 5.3. *A domain $\mathcal{G} \subseteq \mathbb{C}^n$ is called a domain of holomorphy, if for each extension pair (U, V), there is a holomorphic functions $f \in \mathcal{O}(\mathcal{G})$, such that f cannot be extended to (U, V).*

Definition 5.4. *A domain $\mathcal{G} \subseteq \mathbb{C}^n$ is called a domain of existence if there is a holomorphic function $f \in \mathcal{O}(\mathcal{G})$ which is not extendable by any extension.*

Remark 5.1. *Clearly a domain of existence is a domain of holomorphy but the converse is not necessarily true.*

Definition 5.5. *Let f be a function defined on M, we set,*

$$|f|_M := \max\{|f(x)| \mid x \in M\}.$$

Definition 5.6. *Let $K \subseteq \mathcal{G}$ be a compact set. The holomorphic convex hull of K is*

$$\hat{K} = \{z \in \mathcal{G} \mid \forall f \in \mathcal{O}(\mathcal{G}), |f(z)| \leq |f|_K\}.$$

In particular, the domain $\mathcal{G}$ is holomorphically convex if $\hat{K}$ is compact for all compact sets $K \subseteq \mathcal{G}$.

Example 5.1. *For dimension 1, all domains are domains of holomorphy. Consider a domain $\mathcal{G}$, and $z_0 \in \partial\mathcal{G}$. Then the function,*

$$z \mapsto \frac{1}{z - z_0}$$

cannot be extended beyond the boundary.

Remark 5.2. *For $n \geq 1$, convex domains are domains of holomorphy and holomorphically convex.*

Proposition 5.1. *Products of domains of holomorphy/holomorphycally convex sets are again domains of holomorphy/holomorphically convex, respectively. In particular, disks and balls carry these properties, (disks are products of one dimensional domains and disks and balls are convex).*

Remark 5.3. *Let $\mathcal{G}$ be a domain and $K \subseteq \mathcal{G}$ be a compact set then $\mathcal{G}\backslash K$ is never a domain of holomorphy nor holomorphically convex.*

Definition 5.7. *Let $\mathcal{G}$ be a domain and $z \in \mathcal{G}$. We define,*

$$\delta(z) = \delta_{\mathcal{G}}(z) := \max\{r > 0 \mid D_r(z) \subseteq \mathcal{G}\}.$$

$\delta(z)$ is called the holomotphic distance of z.

For a compact subset $K \subseteq \mathcal{G}$, we define,

$$\delta(K) := \min\{\delta(z) \mid z \in K\}.$$

Definition 5.8. *For a compact set K of a domain $\mathcal{G}$ and $0 < \rho < \delta(K)$, we define*

$$K_\rho = \bigcup_{z \in K} D_\rho(z).$$

Remark 5.4. *δ is continuous and strictly positive or everywhere ∞. In particular, $\delta(K) > 0$ for any compact set $K \subseteq \mathcal{G}$.*

Theorem 5.1 (Cartan/Thullen)**.** *Let $\mathcal{G}$ be a domain and K be a compact subset of $\mathcal{G}$. Take $z_0 \in \hat{K}$. For a holomorphic function u on $\mathcal{G}$, let $p_{u,z_0}(z)$ be the power series of u around z_0. Then $p_{u,z_0}(z)$ converges on $D_{\delta(K)}(z_0)$.*

Proof. By assumption, we have,

$$p_{u,z_0}(z) = \sum_{\alpha \in \mathbb{N}^n} a_\alpha (z - z_0)^\alpha,$$

where a_α is given by,

$$a_\alpha = \frac{1}{\alpha!} D^\alpha u(z_0).$$

We have,

$$D^\alpha u(z) = \frac{\alpha!}{(2\pi i)^\alpha} \int_{|\zeta_1 - z_1| = \rho} \cdots \int_{|\zeta_n - z_n| = \rho} \frac{u(\zeta)}{(\underline{\zeta} - \underline{z})^{\alpha + (1, \cdots, 1)}} d\underline{\zeta}$$

Taking the modulo, we have,

$$|D^\alpha u(z)| \leq \alpha! \left(\frac{1}{2\pi}\right)^n |u|_{K_\rho} \frac{1}{\rho^{|\alpha| + n}} (2\pi\rho)^n = \alpha! \frac{|u|_{K_\rho}}{\rho^{|\alpha|}}.$$

This holds for all $z \in K$. Thus it must also hold for $z_0 \in \hat{K}$. That is,

$$|a_\alpha| \leq \frac{|u|_{K_\rho}}{\rho^{|\alpha|}}.$$

Note that $z \in D_\rho(z_0)$, then

$$\sum_{\alpha \in \mathbb{N}^n} a_\alpha (\overline{z} - \overline{z}_0)^\alpha,$$

with $|a_\alpha| \leq \frac{|u|_{K_\rho}}{\rho^{|\alpha|}} \rho^{|\alpha|} \leq |u|_{K_\rho}$. Thus $p_{u,z_0}(z)$ must converge in $D_\rho(z_0)$ for $\rho < r$. That is to say $p_{u,z_0}(z)$ converges in $D_{\delta(K)}(z_0)$. $\square$

Lemma 5.1. *Let $K \subseteq \mathcal{G}$ be compact and $z_0 \in \mathcal{G} \backslash \hat{K}$. For each $M, \varepsilon > 0$, there is a function $f \in \mathcal{O}(\mathcal{G})$ such that*

$$|f(z_0)| > M, |f|_K < \varepsilon.$$

Proof. Since $z_0 \notin \hat{K}$, there is $\tilde{f} \in \mathcal{O}(\mathcal{G})$ with

$$\tilde{f}(z_0) \geq |\tilde{f}|_K.$$

It is an exercise for the readers to show that, from the equation above, we can obtain $f \in \mathcal{O}(\mathcal{G})$ such that $f(z_0) > 1 > |f|_K$. Now take high powers of f, we obtain the claim. □

Proposition 5.2. *Let $\mathcal{G}$ be a domain which is holomorphically convex. Let $(x_j)_{j \in \mathbb{N}}$ be a sequence with accumulation points in $\mathcal{G}$. Then there is $f \in \mathcal{O}(\mathcal{G})$ unbounded on that sequence.*

Proof. Let $K_1 \subseteq K_2 \subseteq \cdots$ be a compact exhaustion of $\mathcal{G}$. That is for each compact set K of $\mathcal{G}$ is contained in one of K_i. This is justified since $\mathcal{G}$ is a domain inheritting topologies defined on $\mathbb{R}^n$. Clearly, we have,

$$\hat{K}_1 \subseteq \hat{K}_2 \subseteq \cdots.$$

By the assumption, this is again a compact exhaustion. From now on, let $K_j = \hat{K}_j$ without loss of generality. Pick a sequence $(j_i)_{i \in \mathbb{N}}$ such that

$$x_{j_i} \notin K_i, x_{j_i} \in K_{j_{i+1}}.$$

After renumbering replace $(x_i)_{i \in \mathbb{N}}$ with $(x_{j_i})_{i \in \mathbb{N}}$. Let $(f_i)_{i \in \mathbb{N}}$ be a sequence in $\mathcal{O}(\mathcal{G})$ such that

$$|f_1(x_1)| \geq \frac{1}{2} > |f|_{K_1},$$

for $i \geq 2$,

$$|f_i(x_i)| > i + \sum_{k=1}^{i-1} |f_k(x_i)|, |f_i|_{K_i} < \frac{1}{2^i}.$$

Now take,

$$f(x) = \sum_{j=1}^{\infty} f_j(x).$$

Observe that,

$$\begin{aligned} |f(x_m)| &= |\sum_{j \leq m} f_j(x_m) + \sum_{j > m} f_j(x_m)|, \\ &\geq |f_m(x_m)| - \sum_{j < m} |f_j(x_m)| - \sum_{j > m} |f_j(x_m)|, \\ &> m - 1. \end{aligned}$$

Thus f is unbounded on the sequence. □

Theorem 5.2. *The following are equivalent for a domain $\mathcal{G} \subseteq \mathbb{C}^n$.*

1). $\mathcal{G}$ is a domain of holomorphy.

2). $\mathcal{G}$ is holomorphically convex.

3). $\mathcal{G}$ is a domain of existence.

Proof. Let us first show that 1) $\Rightarrow$ 2). Let $\mathcal{G}$ be a domain of holomorphy. and K be a compact subset of $\mathcal{G}$. Set $z_0 \in \hat{K}$ and let

$$r_0 = \delta(z_0), r = \delta(K).$$

To obtain a contradiction, suppose that $r_0 < r$ then $(D_{r_0}(z_0), D_r(z_0))$ is an extension pair. Let $u \in \mathcal{O}(\mathcal{G})$, from Theorem 5.1, set $F := p_{u,z_0}(z)$ be the power series of u around z_0. Then F converges on $D_r(z_0)$ and $F \equiv u$ on $D_{r_0}(z_0)$. We obtained a extension $(D_{r_0}(z_0), D_r(z_0), F)$ of u which contradicts that $\mathcal{G}$ is a domain of holomorphy. We conclude $r_0 \geq r$. In particular, for any $z_0 \in \hat{K}$ we have,

$$\delta(z_0) \geq r.$$

This shows that $\hat{K}$ is bounded. Let $z_0 \in \overline{\hat{K}}$, then there are $(z_i)_{i \in \mathbb{N}}$ in $\hat{K}$ such that $z_i \to z_0$. If $z_0 \in \mathcal{G}$ then $z_0 \in \hat{K}$. If $z_0 \in \partial\mathcal{G}$ then $\delta(z_j) \to 0$. However, $\delta(z_j) \geq \delta(K) > 0$. Thus $z_0 \in \hat{K}$. We conclude K is closed and bounded, that is it is compact in $\mathbb{C}^n$.

For 2) $\Rightarrow$ 1) is shown in a Proposition 5.2. Indeed it constructs f such that it cannot be extended beyond $\partial\mathcal{G}$.

For 2) $\Rightarrow$ 3), consider a compact exhaustion $(K_i)_{i \in \mathbb{N}}$ of $\mathcal{G}$. From the argument of Proposition 5.2, we assume $K_i = \hat{K}_i$ without loss of generality. Let $\{x'_i\}_{i \in \mathbb{N}}$ be a sequence of rational points, which is everywhere dense in $\mathcal{G}$. Consider another sequence,

$$x'_1, x'_1, x'_2, x'_1, x'_2, x'_3, \cdots,$$

that each element occurs infinitely many times. Rename this to $(x_i)_{i \in \mathbb{N}}$ and set $D_i = D_{\delta(x_i)}(x_i)$. Observe that $\overline{D_i} \cap \partial\mathcal{G} \neq \emptyset$. Take $y_i \in D_i \backslash K_i$, and $\tilde{f} \in \mathcal{O}(\mathcal{G})$ with $|\tilde{f}(y_i)| > |\tilde{f}_j|_{K_i}$. As in the proof of the previous proposition we have $f_i \in \mathcal{O}(\mathcal{G})$, $f_i(y_i) = 1$ and $|f_i|_{K_i} < \frac{1}{2^i}$. We set,

$$f(z) = \prod_{i=1}^{\infty} (1 - f_i(z))^i.$$

Then for a compact set $K \subset K_{i_0}$ we have,

$$\sum_{i \geq i_0} i|f_i(z)| \leq \sum_{i \geq i_0} i2^{-i} < \infty.$$

This shows that f is holomorphic. We also note that at i_i, $f(y_i) = 0$ of order $\geq j$. Let U_3 be the union of all open subsets of $\mathcal{G} \cap U_2$ that meets U_1, then there exists $z \in \partial\mathcal{G} \cap \partial U_c a \cap U_2$ such that $D_r(z) \subset\subset U_2$.

Now assume that F holomorphic on U_2 and coincide with f on U_1. By identity theorem, $F \equiv f$ on U_3. In particular on $D_r(z) \cap U_3$. Thus F is holomorphic on $\overline{D_r(z)}$.

There is j such that $x_j \in D_{\frac{r}{2}}(z)$,

$$D = D_j := D_{\delta(x_j)}(x_j).$$

Let $x_{j_p} \to x$ for some subsequence let $y_{j_p} \in D_j(x)$. Since we have,

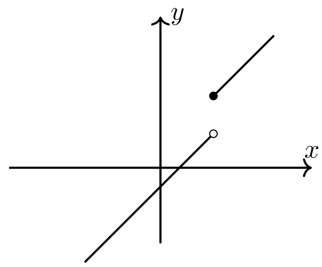
$$D_{\delta(x_j)}(x_j) \subset\subset D_r(z),$$

and $F \equiv f$ on $D_r(z)$. F has zeros of order j_p at $y_{j_p} \in \overline{D_r(z)}$. F is holomorphic function with unbounded zeros on compact set $\overline{D_r(z)}$. Such F does not exist thus we derived a contradiction. Therefore, f cannot be extended. $\square$

Definition 5.9. *A function $f : X \to \mathbb{R}$ is upper semicontinuous if for any $x_0 \in X$, we have,*

$$\limsup_{x \to x_0} f(x) \leq f(x_0).$$

Example 5.2. *The following function is upper semicontinuous.*



Remark 5.5. *Let $f \in \mathcal{O}(U)$ and $a \in U$. If we have,*

$$f = \sum_{i=0}^{\infty} f_i.$$

where f_i has pole at $z - a$ and $o(f) = k \Leftrightarrow f_k \neq 0$. Then the function,

$$o : U \to \mathbb{Z},$$

which is upper semicontinuous. Thus for a compact subset $K \subseteq U$, $o(f)$ is bounded on K.

5.2 Pseudoconvexity

Definition 5.10. *A function $h : U \to \mathbb{R}$ where $U \subset \mathbb{C}$ is harmonic if it is $\mathcal{C}^2$ and $\Delta h = 0$ where*

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = 4\frac{\partial^2}{\partial z \partial \overline{z}}.$$

Theorem 5.3 (Dirichlet problem)**.** *Let D be a disk and $\Gamma = \partial D$. Suppose $\varphi : \Gamma \to \mathbb{R}$ is continuous. Then there exist a continuous function $h : \overline{D} \to \mathbb{R}$ such that*

$$h|_{\Gamma} = \varphi,$$

and h is holomorphic on D.

Theorem 5.4 (Fejer's theorem)**.** *Let $\Gamma = \{z \in \mathbb{C} \,|\, |z| = 1\}$. Then the functions $\{\mathrm{Re}(p(z)) \,|\, p$ is a holomorphic polynomial$\}$ is dense in $\mathcal{C}^0_{\mathbb{R}}(\Gamma)$.*

Definition 5.11. *Let $U \subseteq \mathbb{C}$ and $\varphi : U \to \mathbb{R} \cup \{-\infty\}$. φ is said to be subharmonic if is it upper semi-continuous and for each $D \subset\subset U$, a disk, and for each continuous functional h on $\overline{D}$ which is holomorphic on D, we have, if $h \geq \varphi$ on Γ then $h \geq \varphi$ on all of $\overline{D}$.*

Remark 5.6. *In $\mathbb{R}$, we have a correspondence between,*

$$\textit{linear} \leftrightarrow \textit{harmonic}, \textit{convex} \leftrightarrow \textit{subharmonic}.$$

Remark 5.7. *If φ is subharmonic, we have, for all $z_0 \in U$, there exists $r_0 > 0$ such that for $r \leq r_0$,*

$$\varphi(z_0) \leq \frac{1}{2\pi} \int_0^{2\pi} \varphi(z_0 + re^{i\theta}) d\theta.. \tag{M}$$

Theorem 5.5. *The following statements are equivalent.*

1). φ is harmonic, then it is subharmonic.

2). If φ is subharmonic, we have, for all $z_0 \in U$, there exists $r_0 > 0$ such that for $r \leq r_0$, Equation (M) *holds.*

3). If φ is subharmonic then for all $z_0 \in U$, and any $r_0 > 0$ such that $D_{r_0}(z_0) \subset\subset U$, Equation (M) *holds.*

Remark 5.8. *We have the following facts*

1). φ is subhamornic then it satisfies locally mean principle.

2). If $\varphi \not\equiv -\infty$ which is subharmonic then φ is locally integrable.

3). Let φ, ψ be subharmonic then so is their sum $\varphi + \psi$.

4). φ is subharmonic if and only if at each point $z_0 \in U$, there is a neighborhood $V(z_0)$ of z_0 such that φ is subharmonic on $V(z_0)$.

5). Let $(\varphi_n)_{n \in \mathbb{N}}$ be a sequence of subharmonic functions which converges locally uniformly to φ, then φ is subharmonic.

6). Let $(\varphi_n)_{n \in \mathbb{N}}$ be a sequence of monotonically decreasing subharmonic functions converging to φ, then φ is again subharmonic.

7). Let f be holomorphic function then $\log|f|, |f|$ are subharmonic.

Theorem 5.6. *Let φ be subharmonic in a domain $\mathcal{G}$. Suppose $\varphi \not\equiv -\infty$. Then φ is locally integrable.*

Proof. Let $z_0 \in \mathcal{G}$ with $\varphi(z_0) > -\infty$ and $D = D_r(z_0) \subset\subset \mathcal{G}$. Assume that $h : \overline{D} \to \mathbb{R}$ is continuous and $h \geq \varphi$ on $\overline{D}$.

$$\begin{aligned}\int_{\overline{D}} h(z)d\lambda(z) &= \int_0^r \rho d\rho \int_0^{2\pi} h(z_0 + \rho e^{i\theta})d\theta, \\ &\geq \int_0^r \rho d\rho \int_0^{2\pi} \varphi(z + \rho e^{i\theta})d\theta\end{aligned}$$

Using Equation M, we have

$$\int_{\overline{D}} h(z)d\lambda(z) \geq \frac{r^2}{2}2\pi\varphi(z_0) = \pi r^2\varphi(z_0).$$

□

Remark 5.9. *Let $M = \{z \in U \mid \varphi$ is not integrable in $D(z)\}$. Then M is a closed set by definition.*

Remark 5.10. *Let $z_0 \in M$ where M is as above, $D_r(z_0)$ has points z with $\varphi(z) > -\infty$. Thus in the setting of the previous theorem, φ is integrable in some $D_\rho(z)$ containing z_0.*

Theorem 5.7. *Let φ be of $\mathcal{C}^2$, then φ is subharmonic if and only if $\Delta\varphi \geq 0$.*

Proof. Assume $\Delta\varphi(z_1) < 0$. That is there exist $r > 0$ and $z_1 \in U$ such that $\Delta\varphi < 0$ on $\overline{D_r(z_1)}$. Let h be continuous on $\overline{D_r(z_1)}$ which is harmonic on $D_r(z_1)$ and $h \equiv \varphi$ on $\partial D_r(z_1)$. Then

$$\varphi \leq h$$

on $\overline{D_r(z_1)}$ and on the boundary this is 0. Set $\psi := \varphi - h$. If φ were subharmonic then $\psi \leq 0$ on $\overline{D_r(z_1)}$. Suppose ψ takes minimum at z_0. We have on $D_r(z)$,

$$\Delta\psi = \Delta\varphi$$

We have,

$$\Delta\psi(z_0) < 0,$$

which is not possible thus φ is not subharmonic.

Assume $\Delta\varphi > 0$ and φ is not subharmonic. Then there exists a continuous function h on $\overline{D_r(z_0)}$ with $\varphi \leq h$ on $\partial D_r(z_0)$ but $\varphi > h$ somewhere. Set $\psi = \varphi - h$. Similar to the previous case we have $\psi \leq 0$ on $\partial D_r(z_0)$ therefore $\psi > 0$ somewhere on $D_r(z_0)$ thus has a maximum in $D_r(z_0)$ and there $\Delta\psi > 0$.

So φ is subharmonic if $\Delta\varphi > 0$. On the other hands, $\Delta\varphi \geq 0$ then

$$\varphi_\varepsilon := \varphi + \varepsilon|z|^2, \Delta\varphi_\varepsilon > 0.$$

φ_ε is subharmonic so is φ. □

Theorem 5.8. *Suppose f is holomorphic then $\log|f|$ and $|f|, |f|^2$ are all subharmonic.*

Proof. Note that
$$\log|f| \leq \mathrm{Re}(P(z)),$$
on $\Gamma = \partial D$ where P is a holomorphic polynomial. To show
$$\exp(\log|f|) \leq \exp(\mathrm{Re}(p(z))) = |e^{P(z)}|,$$
on Γ. Therefore, we have,
$$|f \exp^{-P(z)}| \leq 1$$
on Γ so on all of $\overline{D}$. Thus we conclude, on $\overline{D}$ we have,
$$\log|f| \leq \mathrm{Re}(P(z)).$$

For $|f|$ we have,
$$|f(z_0)|e^{i\alpha} = f(z_0) = \frac{1}{2\pi i}\int_0^{2\pi} \varphi(z_0 + re^{i\theta})d\theta.$$
That is
$$\begin{aligned}
|f(z_0)| &= \frac{1}{2\pi}\int_0^{2\pi} e^{-i\alpha} f(z_0 + re^{i\theta})d\theta, \\
&\leq \frac{1}{2\pi}\int_0^{2\pi} |f(z_0 + re^{i\theta})|d\theta, \\
&= \frac{1}{2\pi}\int_0^{2\pi} |f|(z_0 + re^{i\theta})d\theta.
\end{aligned}$$
For the last part, we have $|f|^2 = f\overline{f}$. Thus it is equivalent to show that the right hand side is subharmonic. Indeed,
$$\frac{\partial^2}{\partial z \partial \overline{z}}(f\overline{f}) = \left|\frac{\partial f}{\partial z}\right|^2 \geq 0.$$
□

5.3 Plurisubharmonic functions

Definition 5.12. *Let $U \subseteq \mathbb{C}^n$ and $\varphi : U \to \mathbb{R} \cup \{-\infty\}$. φ is plurisubharmonic if*

i). φ is upper semi-continuous,

ii). $\varphi|_L$ is subharmonic for any complex line $L : z = a + \omega t$ where $a \in U$ and $\omega \in \mathbb{C}^n \backslash \{0\}$.

Remark 5.11. *We have following facts about plurisubharmonic funcions.*

1). For any $c \in \mathbb{R}$, c is plurisubharmonic.

2). φ, ψ are plurisubharmonic so is the sum $\varphi+\psi$ and the positive scalar product of it.

3). φ, ψ are plurisubharmonic then $\max(\varphi, \psi)$ is also plurisubharmonic.

4). $(\varphi_i)_{i\in I}$ is a sequence of plurisubharmonic functions then its limit is also plurisubharmonic if it exists.

5). φ is locally plurisubharmonic then it is globally plurisubharmonic.

6). φ satisfies the local maximal principle.

Proposition 5.3. *Let $\varphi \neq -\infty$ be a plurisubharmonic function then φ is locally integrable.*

Proof. It is proven for $n = 1$. Use induction and the Fubini's theorem. $\square$

Example 5.3. *Suppose f is holomorphic then $|f|, \log|f|, |f|^2 = f\overline{f}$ are all plurisubharmonic.*

Example 5.4. *Suppose f is holomorphic and $V(f) = \{f = 0\}$. That is $\log|f| = -\infty$ on $V(f)$. Thus $V(f)$ has Lebesgue measure 0.*

Remark 5.12.

$$\{\varphi = -\infty \mid \varphi \text{ is plurisubharmonic}\}$$

is called the pluripolar set.

Definition 5.13 (Levi form)**.** *Let $\varphi \in \mathcal{C}^2(U, \mathbb{R})$. The Levi matrix of φ is*

$$L_\varphi(z) = \left(\frac{\partial^2 \varphi}{\partial z_j \partial \overline{z}_k}\right)_{j,k=1,\cdots,n}.$$

The Levi form of φ is

$$L_\varphi(z,t) = \sum_{j,k=1,\cdots,n} \frac{\partial^2 \varphi}{\partial z_j \partial \overline{z}_k}(z) t_j \bar{t}_k.$$

Remark 5.13. *$L_\varphi(z)$ is Hermitian and observe*

$$L_\varphi(z,t) = \begin{pmatrix} t_1 & \cdots & t_n \end{pmatrix} L_\varphi(z) \begin{pmatrix} \overline{t_1} \\ \vdots \\ \overline{t_n} \end{pmatrix}.$$

Proposition 5.4 (Change of coordinates)**.** *Let F be a biholomorphic map and $z = F(w)$. Then we have,*

$$L_{\varphi\circ F}(w) = (J_F^h)^t(w) L_\varphi(F(w)) \overline{J_F^h}(w).$$

Definition 5.14. *We define* $(\mathbb{C}^n, z)$ *to be the tangent space of* $\mathbb{C}^n$ *at* z.

Remark 5.14. *Let* F *be biholomorphic and* $z = F(w)$. *Then we have,*

$$(\mathbb{C}^n, w) \ni s \mapsto F_* s := J_F^h s.$$

That is

$$L_{\varphi \circ F}(w, s) = L_\varphi(F(w), F_* s).$$

Remark 5.15. *Rank and index of L are invariant under the changes of coordinate. All eigenvalues of L are real by a suitable holomorphic coordinate change, we obtain,*

$$L_\varphi(z) = \operatorname{diag}(\underbrace{1, \cdots, 1}_{r_+}, \underbrace{-1, \cdots, -1}_{r_-}, \underbrace{0, \cdots, 0}_{r_0}).$$

By a unitary transform we have,

$$L_\varphi(z) = \operatorname{diag}(\lambda_1, \cdots, \lambda_{r_+}, \mu_1, \cdots, \mu_{r_-}, 0, \cdots, 0).$$

Proposition 5.5. φ *is of* $\mathcal{C}^2$ *class if and only if* $L_\varphi(z) \geq 0$.

Proof. Let q be a complex line $q : z_0 + wt$ for $w \neq 0$. Note $\varphi|_q = \varphi(z_0 + wt)$ has to be such that

$$\frac{\partial^2}{\partial t \partial \bar{t}} \varphi(z_0 + wt)|_{t=0} = \sum_{j,k} \frac{\partial^2}{\partial z_j \partial \bar{z}_k} \varphi(z_0 + wt) w_j \overline{w}_k \geq 0.$$

□

Definition 5.15. φ *is called strictly plurisubharmonic if its Levi form is positive-definite (ie.* $L_\varphi(z) > 0$*).*

Example 5.5. $\varphi(z) = \|z\|^2$ *is strictly subharmonic. Indeed*

$$L_\varphi = I_n.$$

Theorem 5.9. *Let* $\mathcal{G}$ *be a domain and* $\mathcal{G}' \subset\subset \mathcal{G}$ *be relatively compact in* $\mathcal{G}$. *Let* $\varphi : \mathcal{G} \to \mathbb{R} \cup \{-\infty\}$ *be plurisubharmonic function. Then there is a sequence* $(\varphi_n)_{n \in \mathbb{N}}$ *of strictly smooth,plurisubharmonic functions on* $\mathcal{G}'$ *such that* $\varphi_n \to \varphi$ *on* $\mathcal{G}'$.

Theorem 5.10. *Let* $F : U \to V$ *be biholomorphic and* $\varphi : V \to \mathbb{R} \cup \{-\infty\}$ *is plurisubharmonic. Then* $\varphi \circ F$ *is also plurisubharmonic on* U.

Definition 5.16. *Let* $\mathcal{G} \subseteq \mathbb{C}^n$ *be a domain. It is said to be pseudoconvex if* $\mathcal{G}$ *has a strictly plurisubharmonic* $\mathcal{C}^2$ *exhaustive function* φ. *That is* $\varphi : \mathcal{G} \to \mathbb{R}$ *is a strictly plurisubharmonic and*

$$\mathcal{G}_c := \{z \in \mathcal{G} \mid \varphi(z) < c\} \subset\subset \mathcal{G}.$$

(Recall relatively compact, $A \subset\subset B$ *means* $\overline{A}$ *is compact and contained in* B*).*

Example 5.6. *Take $\mathcal{G} = \mathbb{C}^n$ and $\varphi(z) = \|z\|^2$, then $\mathcal{G}$ is pseudoconvex.*

Definition 5.17. *A function $\delta : \mathbb{C}^n \to \mathbb{R}_{\geq 0}$ is called a distance function if*

i). it is continuous,

ii). $\delta(z) = 0 \Leftrightarrow z = 0$,

iii). $\delta(cz) = |c|\delta(z)$ for any $c \in c \in \mathbb{R}$.

Definition 5.18. *Let δ be a distance function and $\mathcal{G}$ be a domain, we define,*

$$\delta_{\mathcal{G}}(z) = \inf\{\delta(z - w) \mid w \in \partial\mathcal{G}\}.$$

Theorem 5.11. *Let $\mathcal{G}$ be domain. Then the following are equivalent.*

1). It is pseudoconvex.

2). $\mathcal{G}$ has a continuous plurisubharmonic exhaustive function.

3). $\mathcal{G}$ has a strictly plurisubharmonic $\mathcal{C}^2$ exhaustive function.

4). There exists a distance function δ such that $u(z) = -\log \delta(z)$ is plurisubharmonic.

5). For any distance function δ, $u(z) = -\log \delta(z)$ is plurisubharmonic.

Definition 5.19. *Let $\mathcal{G} \subset\subset \mathbb{R}^n$ be a domain. It is said to have a smooth boundary if there exists $U = U(\partial\mathcal{G})$ and $r : U \to \mathbb{R}$ such that r is $\mathcal{C}^\infty$ and $dr(x) \neq 0$ for any $x \in U$. Furthermore, we have,*

$$U \cap \mathcal{G} = \{x \in U \mid r(x) < 0\}, \partial\mathcal{G} = \{r = 0\}.$$

Let $k \in \mathbb{N}$. $\mathcal{G}$ is said to have $\mathcal{C}^k$ boundary with a boundary function r such that if $\tilde{r}$ is another boundary function of $\mathcal{C}^k$-class, we have, $\tilde{r} = hr$ for some $h > 0$ where h is $\mathcal{C}^k$ outside $\partial\mathcal{G}$ and of $\mathcal{C}^{k-1}$ everywhere.

Remark 5.16. *For a domain $\mathcal{G} \subset\subset \mathbb{R}^n$, we have,*

$$\delta(x) = \begin{cases} \operatorname{dist}(x, \partial\mathcal{G}) & x \notin \overline{\mathcal{G}}, \\ -\operatorname{dist}(x, \partial\mathcal{G}) & x \in \overline{\mathcal{G}}. \end{cases}$$

If $\mathcal{G}$ has $\mathcal{C}^k$ boundary then $\delta(x)$ is also a boundary function.

Proposition 5.6. *Let $\mathcal{G} \subset\subset \mathbb{C}^n$ with $\mathcal{C}^k$ boundary with boundary function $r : U(\partial\mathcal{G}) \to \mathbb{R}$. If it is plurisubharmonic, then $\mathcal{G}$ is pseudoconvex.*

Proof. Let $\varphi(z) = -\log(-r)$. Then

$$\frac{\partial^2}{\partial z_j \partial \overline{z_k}}\varphi = \frac{\partial}{\partial z_j}\frac{-\frac{\partial r}{\partial \overline{z}_k}}{r} = -\frac{\frac{\partial^2}{\partial z_j \partial \overline{z_k}} r}{r} + \frac{\frac{\partial r}{\partial z_j}\frac{\partial r}{\partial \overline{z}_k}}{r^2}.$$

Therefore, we have,

$$\sum \frac{\partial^2}{\partial z_j \partial \overline{z_k}} \varphi t_j \bar{t}_k = -\frac{1}{r} \sum \frac{\partial^2}{\partial z_j \partial \overline{z_k}} r t_j \bar{t}_k + \frac{1}{r^2} \sum \frac{\partial r}{\partial z_j} t_j \frac{\partial r}{\partial \overline{z}_k} \bar{t}_k,$$
$$= \cdots + \frac{1}{r^2} \left(\left| \sum \frac{\partial r}{\partial z_j} t_j \right|^2 \right) \geq 0.$$

Extend φ to all of $\overline{\mathcal{G}}$ in $\mathcal{C}^k$. We have there exists a plurisubharmonic function close to the boundary. Now pass to

$$\varphi(z) + A\|z\|.$$

It is plurisubharmonic on all of $\mathcal{G}$ if A is large enough. □

Definition 5.20. *Let $\mathcal{G} \subset\subset \mathbb{C}^n$ and $\mathcal{G} = \{r < 0\}$. We say $\mathcal{G}$ satisfies the Levi condition if for any $z \in \partial\mathcal{G}$,*

$$\sum_{j,k=1}^{n} \frac{\partial^2}{\partial z_j \partial \overline{z}_k} r(z) t_j \bar{t}_k \geq 0, \tag{L}$$

for all $\underline{t} \in \mathbb{C}^n$ with

$$\sum \frac{\partial r}{\partial j}(z) t_j = 0.$$

Definition 5.21. *$\mathcal{G} = \{r < 0\}$ satisfies the strong Levi condtion if*

$$\sum_{j,k=1}^{n} \frac{\partial^2}{\partial z_j \partial \overline{z}_k} r(z) t_j \bar{t}_k > 0, \tag{SL}$$

for all $\underline{t} \in \mathbb{C}^n \backslash \{0\}$ with

$$\sum \frac{\partial r}{\partial j}(z) t_j = 0.$$

Remark 5.17. *If r satisfies either of* (L) *or* (SL) *so do all other boundary functions.*

Remark 5.18 (Geometric Interpretation)**.** *Consider $z \in \partial\mathcal{G}$ and $(\mathbb{C}^n, t)$ the real tangent space to $\partial\mathcal{G}$ at z. Then*

$$\sum \frac{\partial r}{\partial z_j}(z) t_j + \sum \frac{\partial r}{\partial \overline{z}_k}(z) \bar{t}_k = 0,$$

gives the maximal compact tangent space to $\partial\mathcal{G}$ at z singled out by

Proposition 5.7. *Let $\mathcal{G} \subset\subset \mathbb{C}^n$ with smooth boundary. It is pseudoconvex if and only if each boundary function satisfies* (L) *at all boundary points.*

Proof. See [4] Lemma 2.11. □

Definition 5.22. *Let $\mathcal{G} \subset\subset \mathbb{C}^n$ with smooth boundary. It is strictly pseudoconvex if the boundary function satisfies* (SL).

Proposition 5.8. *Let $\mathcal{G}$ be a domain. It is strictly pseudoconvex if and only if there exists a strictly plurisubharmonic boundary function.*

Proof. See [4]. □

Remark 5.19. *Suppose $\mathcal{G}$ is strictly pseudoconvex. For any boundary point of the domain, we can find a biholomorhphic function defined on a neighborhood of the point such that the image is convex.*

Remark 5.20. *All smooth domains in $\mathbb{C}$ are strictly pseudoconvex. For the detailed proof, see [5].*

Definition 5.23. *Let $\mathcal{G} \subset \mathbb{C}^n$ be pseudoconvex and φ be a strictly plurisubharmonic exhaustive function, set,*

$$\mathcal{G}_c = \{x \in \mathcal{G} \mid \varphi(x) < c\}.$$

c is called critical if $\varphi(x) = c$ then $d\varphi(x) \neq 0$.

Remark 5.21. *The set of critical values in $\mathbb{R}$ have measure 0. And there is always a sequence $(\mathcal{G}_r)_{r\in\mathbb{N}}$ where for each r, $\mathcal{G}_r \subset\subset \mathcal{G}$,$\mathcal{G}_r$ is strictly pseudoconvex, and $\mathcal{G}_r \subset\subset \mathcal{G}_{r+1}$. Moreover,*

$$\bigcup \mathcal{G}_r = \mathcal{G}.$$

5.4 The Levi Problems

Theorem 5.12. *Let $\mathcal{G}$ be holomorphically convex, then it is pseudoconvex.*

Proof. Without loss of generality, there is a compact exhaustion

$$K_j \subset K_{j+1}^\circ \subset K_{j+1},$$

such that $K_j = \hat{K}_j$, and each compact subset K of $\mathcal{G}$ is contained in some K_j. Fix j and consider $z \in K_{j+2} - K_{j+1}^\circ$. we find $f \in \mathcal{O}(\mathcal{G})$ such that $|f(z)|^2 > j$ and

$$|f|_{K_j}^2 < 1.$$

By continuity of f, this holds for some neighborhood of z. By the compactness, we find a finitely many functions $f_1, \cdots, f_m$ such that

$$\sum |f_i|^2 > j \quad \text{on } K_{j+2} - K_{j+1}^\circ,$$

and for each $|f_i|_{K_j} < 1$. Passing to f_i^N for some N large enough, obtain for each j,

$$f_1^j, \cdots, f_{m_j}^j,$$

with

$$\sum_{i=1}^{m_j} |f_i^j|^2 > j \quad \text{on } K_{j+2} - K_{j+1}^\circ,$$
$$\sum_{i=1}^{m_j} |f_i^j|^2 < \left(\frac{1}{2}\right)^j \quad \text{on } K_j.$$

Set

$$\varphi_j = \sum_{i=1}^{m_j} |f_i^j|^2,$$

and,

$$\varphi = \sum_{j=0}^{\infty} \varphi_j,$$

which is locally uniformly convergent and thus φ is obviously continuous.

We claim it is differentiable. Set,

$$\phi_j(z,w) = \sum f_i^j(z)\overline{f_i}^j(\overline{w}).$$

Then this is holomorphic in z and $\overline{w}$. Thus $\phi(z,w) = \sum_{j=0}^{\infty} \phi_j(z,w)$ which is locally uniformly convergent, is also holomorphic but

$$\varphi(z) = \phi(z, \overline{z}),$$

thus φ is differentiable. □

Our goal of the chapter is to study the following theorem

Theorem 5.13 (Levi Problem)**.** *Pseudoconvex domains are also holomorphically convex.*

Remark 5.22. *The Levi condition for $\mathcal{G} = \{r < 0\}$ in dimension 2 can be restated by*

$$\det \begin{pmatrix} 0 & \frac{\partial r}{\partial z_1} & \frac{\partial r}{\partial z_2} \\ \frac{\partial r}{\partial \overline{z}_1} & \frac{\partial^2}{\partial z_1 \partial \overline{z}_1} r & \frac{\partial^2}{\partial z_2 \partial \overline{z}_1} \\ \frac{\partial r}{\partial \overline{z}_2} & \frac{\partial^2}{\partial z_1 \partial \overline{z}_2} & \frac{\partial^2}{\partial z_2 \partial \overline{z}_2} \end{pmatrix} \leq 0.$$

6 Cauchy-Riemann Equations

We will consider (p,q)-forms which are given by

$$\begin{aligned} f &= \sum_{\substack{1 \leq i_1 < \cdots < i_p \leq n \\ 1 \leq j_1 < \cdots < j_q \leq n}} f_{i_1,\cdots,i_p,j_1,\cdots,j_q} dz_{i_1} \wedge \cdots \wedge dz_{i_p} \wedge d\overline{z}_{j_1} \wedge \cdots \wedge d\overline{z}_{j_q}, \\ &= \sum_{\substack{|I|=p \\ |J|=q}} f_{I,J} dz^I \wedge d\overline{z}^J. \end{aligned}$$

We set,

$$E^{p,q}(U) = \mathcal{C}^{\infty}_{p,q}(U).$$

Also we denote,

$$\overline{\partial} f = \sum \overline{\partial} f_{I,J} \wedge dz^I \wedge d\overline{z}^J.$$

That is for a function $f \in \mathcal{C}^{\infty}(U)$,

$$\overline{\partial} f = \sum \frac{\partial f}{\partial \overline{z}_j} d\overline{z}_j.$$

Note that $\overline{\partial} \circ \overline{\partial} = 0$.
Consider the ODE

$$\begin{cases} \overline{\partial} u = f, f \in E^{p,q}, \\ \overline{\partial} f = 0. \end{cases}$$

We call u is closed if $\overline{\partial} u = 0$ and exact if such f in the above exists.
If we get rid of p, we have,

$$f \leftrightarrow \binom{n}{p} \text{tuple of } (0,q)\text{-forms}.$$

6.1 Solving Cauchy-Riemann equation - First step

Recall Bochner-Martinelli integral formula.

$$B(\zeta, z) = \frac{1}{2\pi i} \frac{1}{\|\zeta - z\|^{2n}} \beta \wedge (\overline{\partial}_{\zeta} \beta)^{n-1}, \beta(\zeta, z) = \sum_{j=1,\cdots,m} (\overline{\zeta}_j - \overline{z}_j) d\zeta_j.$$

Set $\mathcal{G}, \partial\mathcal{G} \subset\subset \mathbb{C}^n, f \in \mathcal{C}^1(\overline{\mathcal{G}}), z \in \mathcal{G}$.

$$f(z) = \int_{\partial \mathcal{G}} f(\zeta) B(\zeta, z) - \int_{\mathcal{G}} \overline{\partial} f(\zeta) \wedge B(\zeta, z).$$

Theorem 6.1. *Let $f \in \mathcal{C}^1_{0,1,C}(\mathbb{C}^n), \overline{\partial} f = 0$ Set*

$$u(z) = -\int_{\mathbb{C}^n} f(\zeta) \wedge B(\zeta, z).$$

Then u is $\mathcal{C}^1$ ($\mathcal{C}^k$ if $f \in \mathcal{C}^k$) and $\overline{\partial} u = f$. If $n > 1$ then u has compact support.

Proof. The first question we pose is that whether u is differentiable or not. We have,

$$\begin{aligned} f(\zeta) &= \sum f_j(\zeta) d\overline{\zeta}_j, \\ u(z) &= \sum u_j(z), \\ u_j(z) &= -\int_{\mathbb{C}^n} f_j d\overline{\zeta}_j \wedge B(\zeta, z), \\ &= \text{constant} \int \frac{f_j(\zeta)(\overline{\zeta}_j - \overline{z}_j)}{\|\zeta - z\|^{2n}} dV(\zeta). \end{aligned}$$

Let L_z be a linear partial differnetial opeartor (first order) with constant coefficients.

$$\begin{aligned} L_z u_j(z) &= \text{constant}\, L_z \int \frac{f(\zeta)(\overline{\zeta}_j - \overline{z}_j)}{\|\zeta - z\|^{2n}} dV(\zeta), \\ &= \text{constant}\, L_z \int_{\mathbb{C}^n} \frac{f(z+w)w_j}{\|w\|^{2n}} dV(w), \\ &= \text{constant} \int_{\mathbb{C}^n} \frac{L_z f(z + w\overline{w}_i)}{\|w\|^{2n}} dV(w), \\ &= \text{constant} \int \frac{L_\zeta f(\zeta)(\overline{\zeta}_j - \overline{z}_j)}{\|\zeta - z\|^{2n}} dV(\zeta). \end{aligned}$$

Now we want to show that $\overline{\partial} u = f$, that is

$$\frac{\partial u}{\partial \overline{z}_j} = f_j.$$

For $j = 1$, we see,

$$\begin{aligned} \frac{\partial u}{\partial \overline{z}_1} &= -\int \frac{\partial f}{\partial \overline{\zeta} 1} \wedge B(\zeta, z), \\ &= -\int \left(\frac{\partial f_1}{\partial \overline{\zeta}_1} d\overline{\zeta}_1 + \cdots + \frac{\partial f_n}{\partial \overline{\zeta}_n} d\overline{\zeta}_n \right) \wedge B(\zeta, z). \end{aligned}$$

By the assumption $\overline{\partial} f = 0$ thus we have,

$$\frac{\partial f_1}{\partial \overline{\zeta}_j} = \frac{\partial f_j}{\partial \overline{\zeta}_1}.$$

Thus we have,

$$\begin{aligned} \frac{\partial u}{\partial \overline{z}_1} &= -\int \underbrace{\left(\frac{\partial f_1}{\partial \overline{\zeta}_1} d\overline{\zeta}_1 + \cdots + \frac{\partial f_1}{\partial \overline{\zeta}_n} d\overline{\zeta}_n \right)}_{\overline{\partial} f_1} \wedge B(\zeta, z), \\ &= -\int \overline{\partial} f_1 \wedge B(\zeta, z) = f_1(z). \end{aligned}$$

Add-on: $n > 1$, u is holomorphic outside the support of f then $u(z) \to 0$ for $z \to \infty$. Thus $u \equiv 0$ on an open set outside of the support of f. That is it is everywhere 0 on unbounded component of complement of $\operatorname{supp} f$. $\square$

Remark 6.1. *The Kugelsatz follows from Theorem 6.1.*

Theorem 6.2. *Let $\mathcal{G}, \partial\mathcal{G} \subset\subset \mathbb{C}$ and $f \in \mathcal{C}^1(\overline{\mathcal{G}})$. Let*

$$u(z) = \frac{1}{2\pi i} \int_{\mathcal{G}} \frac{f(\zeta)}{\zeta - z} d\zeta \wedge d\overline{\zeta}, z \in \mathcal{G}.$$

Then $\frac{\partial u}{\partial \overline{z}} = f(z)$.

Proof. Let $z_0 \in \mathcal{G}$ and $D_r(z_0) \subset\subset \mathcal{G}$. Let φ be a smooth function such that

$$\varphi(z) = \begin{cases} 0, & z \notin D_r(z_0), \\ 1, & z \in D_{\frac{r}{2}}(z_0), \\ \text{in } [0,1], & \text{elsewhere.} \end{cases}$$

We see,

$$u(z) = \underbrace{\int \frac{\varphi(\zeta)f(\zeta)}{\zeta - z} d\zeta \wedge d\overline{\zeta}}_{u_1(z)} + \underbrace{\int \frac{(1-\varphi(\zeta))f(\zeta)}{\zeta - z} d\zeta \wedge d\overline{\zeta}}_{u_2(z)}$$

For $z \in D_{\frac{r}{2}}(z_0)$, we have,

$$\frac{\partial u}{\partial \overline{z}} = \frac{\partial u_1}{\partial \overline{z}} = \varphi(z)f(z) = f(z).$$

$\square$

Remark 6.2. *In the above theorem, f depends on additional parameters, then u depends on them in the same way. More precisely, if f and u are related such that there is a linear operator T with*

$$u = Tf,$$

and we have dependencies on f such that

$$f(z; t_1, \cdots, t_k), t = (t_1, \cdots, t_k) \in V \subseteq \mathbb{R}^k,$$

then

$$D_{t_i} u(z; t_1, \cdots, t_k) = T(D_{t_j} f).$$

Such T is the inverse to $\overline{\partial}$ if it exists and called the solution operator.

Suppose we have constructed a solution operator T. For Tf, we want $f \in \mathcal{C}^1(\mathcal{G}) \cap L^1(\mathcal{G})$. You can have estimates in terms of L^1, L^p, L^∞. But if $f \in \mathcal{C}^\infty(\mathcal{G})$, there is $u \in \mathcal{C}^\infty(\mathcal{G})$, $\frac{\partial u}{\partial \overline{z}} = f$, we have weak range approximation. For more detail, see [1].

Definition 6.1. *Let $f \in \mathcal{C}^\infty_{0,q}(D), \overline{\partial} f = 0$. We denote $f \in E(k)$ or say f is of complexity k if $f_J = 0$ for all J that contains l greater than k.*

Remark 6.3. *Clearly $E(0) = 0$, and $E(n) = \mathcal{C}^\infty_{0,q}(D)$.*

Theorem 6.3 (Dolbeault lemma)**.** *Let $f \in \mathcal{C}^\infty_{0,q}(D), \overline{\partial} f = 0$ where $D \subseteq \mathbb{C}^n$ is a polydisk. Then there is $u \in \mathcal{C}^\infty_{0,q-1}(D)$ with $\overline{\partial} u = 0$.*

Note that D is arbitrary, and even $\mathbb{C}^n$ works. u is not given by a linear operator.

Suppose $D'' \subset\subset D$" Show $f \in \mathcal{C}^\infty_{0,q}(D'), \overline{\partial} f = 0$, then there is $u \in \mathcal{C}^\infty_{p,q-1}(D'')$ with $\overline{\partial} f$ on D''.

Proof. Suppose $f = \sum_{|J|=q} f_J d\overline{z}^J \in \mathcal{C}^\infty_{0,q}(D')$. We will prove the statement by the induction on k for $E(k)$.

For $k = 0$, there is nothing to show.

Suppose the statement holds for $k-1$, then

$$f = \sum_{|J|=q} f_J d\overline{z}^J \in E(k), \overline{\partial} f = 0.$$

Then take $g, h \in E(k-1)$ such that $f = d\overline{z}_k \wedge g + h$. And

$$0 = \overline{\partial f} = -d\overline{z}_k \wedge \overline{\partial} g + \overline{\partial} h.$$

Thus we obtain,

$$d\overline{z}_k \wedge \overline{\partial} g = \overline{\partial} h,$$

and we have,

$$g = \sum_{|K|=q-1} g_K d\overline{z}^K.$$

Suppose this is not the case, then on the left, $l > k$, $d_k \wedge d\overline{z}_l \wedge \cdots$ but not on the right. We claim that g_K are holomorphic in $z_{k+1}, \cdots, z_n$. By Theorem 6.2, we found G_k of $\frac{\partial G_k}{\partial \overline{z}_k} = g_k$ on D''. As holomorphic in $z_{k+1}, \cdots, z_n$ smooth in all variables.

$$\begin{aligned}
G &= \sum_{|K|=q-1} G_K d\overline{z}^K. \\
\overline{\partial} G &= \sum_{K, l<k} \frac{\partial G_k}{\partial \overline{l}} d\overline{z}_l \wedge d\overline{z}^K + \sum_K \frac{\partial G_k}{\partial \overline{k}} \wedge d\overline{z}^K, \\
&= h_1 + d\overline{z}_k \wedge g.
\end{aligned}$$

Sum them up we have,

$$f = \overline{\partial} G + \underbrace{(h - h_1)}_{\in E(k-1)} \quad \text{on } D''.$$

We observe further that,

$$\overline{\partial}(h - h_1) = 0, \tag{1}$$
$$\tag{2}$$

and there is u_1 defined on D'' such that $\overline{\partial} u_1 = h - h_1$. Thus we have,

$$f = \overline{\partial}(G + u_1) \quad \text{on } D''.$$

Let $D_0 \subset\subset D_1 \subset\subset D_2 \subset\subset \cdots \subset\subset D$ be a sequence of polydisks that exhausts D. We consider two cases

Case 1. $q > 1$.

Find $u_0 \in \mathcal{C}^\infty_{0,q-1}(D)$ such that $\overline{\partial} u_0 = f$ on D_0. Assume $u_0, u_1, \cdots, u_d$ are such that

$$\overline{\partial} u_i = f \text{ on } D_i, u_i = u_{i-1} \text{ on } D_{i-2}.$$

Find $u'_{d+1} \in \mathcal{C}^\infty_{0,q-1}(D)$ such that

$$\overline{\partial} u'_{d+1} = f \text{ on } D_d.$$

Find $v \in \mathcal{C}^\infty_{0,q-2}(D)$ such that

$$\overline{\partial} v = u'_{d+1} - u_d \text{ on } D_{d-1}.$$

Set $u_{d+1} = u'_{d+1} - \overline{\partial} v$ Then we have,

$$\overline{\partial} u_{d+1} = f \text{ on } D_{d+1}.$$

Furthermore,

$$u_{d+1} - u_d = u'_{d+1} - \overline{\partial} v - u_d = 0 \text{ on } D_{d-1}.$$

Inductively, we get $u_0, u_1, \cdots$. Set $u = \lim_{d\to\infty} u_d$. On D_k we have

$$u \equiv u_d, d \text{ sufficiently large}, \overline{\partial} u = \overline{\partial} u_d = f.$$

Assume that we found $u_0, u_1, u_2, \cdots, u_d$ with

$$\overline{\partial} u_i = f \text{ on } D_i, |u_i - u_{i-1}|_{D_{i-2}} \leq \frac{1}{2^i}.$$

Construct u_{d+1} by taking $u'_{d+1} \in \mathcal{C}^\infty(D)$ such that

$$\overline{\partial} u'_{d+1} = f \text{ on } D_{d+1}.$$

We also have

$$\overline{\partial}(\underbrace{u'_{d+1} - u_d}_{\text{a holomorphic function on } D_d}) = 0 \text{ on } D_d.$$

Let p be a Taylor polynomial of $u'_{d+1} - u_d$ about 0 degree chosen sufficiently large so that

$$|u'_{d+1} - u_d - p|_{D_d - 1} < \frac{1}{2^{l+1}}.$$

Set $u_{d+1} = u'_{d+1} - p$. Continus this procedure get a sequence $u_0, u_1, \cdots$. Take $u = \lim_{d\to\infty} u_d$. We observe this is locally uniform. Therefore, u is continuous.

Consider $u - u_m$ then

$$\lim_{\substack{d\to\infty \\ d>m}} (\overbrace{u_d - u_m}^{\text{holomorphic function for } d \text{ large enough}}).$$

Thus u is of $\mathcal{C}^\infty$. In particular,

$$\overline{\partial} u = \underbrace{\overline{\partial}(u - u_m)}_{=0} + \overline{\partial} u_m = f.$$

$\square$

Remark 6.4. *In the above proof, we can assume the solution u is $\mathcal{C}^\infty_{0,q-1}(D')$ but $\overline{\partial} u = f$ is true only on D''.*

Definition 6.2 (Dolbeault Cohomology)**.** *Let D be a polydisk. We define, the Dolbeault cohomology as*

$$H^{0,q}_{\mathrm{Dol}}(D) = \frac{\{f \in \mathcal{C}^\infty_{0,q} \mid \overline{\partial} f = 0\}}{\{\overline{\partial} u \mid u \in \mathcal{C}^\infty_{0,q}\}}.$$

Remark 6.5. *Theorem 6.3 can be restated by the following way.*

$$H^{0,q}_{\mathrm{Dol}}(D) = 0,$$

for any $q \geq 1$. This is proved by first showing the case when $q = 1$, then b the induction for higher cases.

For $n = 1$ and $\mathcal{G}$ be an arbitrary domain. Suppose we want to solve the equation,

$$\begin{cases} \overline{\partial} u = f, \\ \frac{\partial u(z)}{\partial \overline{z}} = f(z) \end{cases}.$$

Take $\mathcal{G}_0 \subset\subset \mathcal{G}_1 \subset\subset \cdots \subset\subset \mathcal{G}$ such that it exhausts $\mathcal{G}$. Suppose $\partial \mathcal{G}$ is reasonable(ie. the Stoke's theorem can be applied). Solve $u_j \in \mathcal{C}^\infty(\mathcal{G})$ such that

$$\frac{\partial u_j}{\partial \overline{z}_j}(z) = f(z) \text{ on } \mathcal{G}_j.$$

To do so, we need an approximation theorem. Namely, the Range Approximation Theorem. Then it follows that

$$\exists u \in \mathcal{C}^\infty(\mathcal{G}), \frac{\partial u}{\partial \overline{z}} = f.$$

u was not constructed by a linear operator,

$$\mathcal{C}^\infty_{0,q} \ni f \overset{T}{\mapsto} Tf \in \mathcal{C}^\infty_{0,q-1}, \overline{\partial} Tf = f.$$

Using Functional analysis, we know that such T does not exists. Let $f \in \mathcal{C}^\infty_{0,q}, \overline{\partial} f = 0$, bounded on D.

6.2 Applying the Cauchy-Riemann equation

Definition 6.3 (Meromorphic functions)**.**

Definition 6.4. *Let U be an open set. We denote $\mathcal{K}(U)$ to be the set of meromorphic functions on U.*

Remark 6.6. *We have $\mathcal{O}_x \subset \mathcal{K}_x$ and $\mathcal{K}(U) = \bigsqcup_{x \in U} \mathcal{K}_x$.*

Definition 6.5. *Let $\mathcal{G} \subseteq \mathbb{C}^n$ be a domain. Cousin-I-distribution is $(U_i, f_i)_{i \in I}$, such that* *i).* $(U_i)_{i\in I}$ *is an open cover of* $\mathcal{G}$*,*

ii). $f_i \in \mathcal{K}(U_i)$ *is a meromorphic function for each* $i \in I$ *such that* $f_j - f_i = f_{ij}$ *is a holomorphic function on* $U_{ij} = U_i \cap U_j$*.*

A solution to such a distribution is a meromorphic function f *on* $\mathcal{G}$ *such that* $f - f_i$ *is holomorphic on* U_i *for each* $i \in I$*.*

We now want to find a meromorphic function f on $\mathcal{G}$ such that $f - f_i$ is holomorphic on U_i.

Definition 6.6. *A divisor of* $\mathcal{G}$ *is a pair* $\Delta = (U_i, f_i)_{i\in I}$ *such that*

i). $(U_i)_{i\in I}$ *is an open covering of* $\mathcal{G}$*,*

ii). $f_i \not\equiv 0$ *is meromorphic in* U_i *for each* $i \in I$*,*

iii). $f_j/f_i = f_{ij}$ *is holomorhpic without zeros on* U_{ij} *for each pair* $i, j \in I$*.*

Definition 6.7. *Let* Δ_1, Δ_2 *be two divisors. They are said to coincide if* $\Delta_1 \cup \Delta_2$ *is also a divisor.*

Definition 6.8. *A divisor of the form* $(\mathcal{G}, f)$ *is called principle.*

Definition 6.9. *A solution of such a* $\Delta = (U_i, f_i)_{i\in I}$ *is a meromorphic function* f *on* $\mathcal{G}$ *with* f/f_i *is holomorphic without zeros on* U_i*. That is a meromorphic function* f *on* $\mathcal{G}$ *such that* Δ *and* $(\mathcal{G}, f)$ *concincide.*

Second Cousin Problem Are all divisors of a domain $\mathcal{G}$ principal?

Third Cousin Problem Suppose $\mathcal{G}$ be a domain containing the origin. Let $\mathcal{O}(\mathcal{G})$ be an algebra of holomorphic functions and set

$$M_0 = \{f \in \mathcal{O}(\mathcal{G}) \mid f(0) = 0\}.$$

Is it true that $M_0 = (z_1, \cdots, z_n)$? That is for any $f \in \mathcal{O}(\mathcal{G})$ with $f(0) = 0$, do we find a solution to the following Diophantine equation,

$$\sum_{j=1}^{n} f_j(z) z_j = f(z).$$

Fourth Cousin Problem Let $M \subseteq \mathcal{G}$ be anaanalytic set and f be a holomorphic function on M. Is there a holomorphic function F on $\mathcal{G}$ such that $F|_M = f$?

Fifth Cousin Problem Let M be an analytic set of codimension 1. Is there. a holomorphic function f on $\mathcal{G}$ such that $V(f) = M$.

$(U_i, f_i)_{i\in I}$ be a principal part distribution such that $U_i \subset\subset \mathcal{G}$. Let $\{q_i\}_{i\in I}$ be a partition of unity subordinate to $\{U_i\}_{i\in I}$. Set

$$g_i = \sum \varphi_i f_{ji} \text{ on } U_i, \text{ where } f_{ji} = f_i - f_j \text{ on } U_{ij}.$$

Then we have,

$$\begin{aligned}\overline{\partial}g_i - \overline{\partial}g_k &= \overline{\partial}\sum \varphi_j f_{ji} - \sum \varphi_j f_{jk},\\ &= \overline{\partial}\sum \varphi_j (f_{ji} - f_{jk}),\\ &= \overline{\partial}\sum_j f_{ki}\varphi_j,\\ &= f_{ki} \underbrace{\sum \overline{\partial}\varphi_j}_{=0 \text{as } \overline{\partial} \text{ is a closed } (0,1) \text{ form}} = 0.\end{aligned}$$

Set $r = \overline{\partial}g_i$ on U_i. Globally defined on all of $\mathcal{G}$.

Assume Cauchy-Riemann equation is solvable, that is there is $u \in \mathcal{C}^\infty(\mathcal{G})$ such that

$$\overline{\partial}u = r.$$

Set $f = f_i - g_i + u$ on U_i. Then $\overline{\partial}f = 0$. Note that f is globally defined. Indeed,

$$(f_i - g_i + u) - (f_k - g_k + u) = f_i - f_k - (g_i - g_k) = 0.$$

Particularly, $f - f_i$ is holomorphic thus f solves the First Cousin problem. That is on a Disk, the First cousin problem is solvable.

Theorem 6.4. *If* $(\mathrm{CR})_{0,q}$ *is solvable then* $M_0 = (z_1, \cdots, z_n)$.

Definition 6.10. *Let* $\mathcal{G}$ *be a domain in* $\mathbb{C}^n$ *and* $a \in \partial\mathcal{G}$. *A boundary poitn* $c \in \partial\mathcal{G}$ *is called attainable if there exists* $\varepsilon > 0$ *such that for some* $b \in B_\varepsilon(a) \cap \mathcal{G}$, *and* δ *the Euclidean distance between* b *and* $\partial\mathcal{G}$, *we have,* $c \in \partial B_\delta(b) \cap \partial\mathcal{G} \cap B_{2\varepsilon}(a)$.

Remark 6.7. *For each attainable boundary point* c, *we find a holomorphic function on* $\mathcal{G}$ *which is not extendable beyond* c.

Example 6.1. *Consider* $\mathcal{G} \cap \{z_n = 0\} = \mathcal{G}_0$ *not empty and* $0 \in \partial\mathcal{G}$. *Consider* $j : \mathcal{G}_0 \hookrightarrow \mathcal{G}$ *and* $p : \mathbb{C}^n \to \mathbb{C}^{n-1}$. *Consider*

$$M = \{z \in \mathcal{G} \mid p(z) \notin \mathcal{G}_0\}.$$

This is relatively closed in $\mathcal{G}$. *Also* $\mathcal{G}_0$ *is relatively closed in* $\mathcal{G}$. *We have* $M \cap \mathcal{G}_0 =$. *So there isa* $\mathcal{C}^\infty$ *function on* $\mathcal{G}$,$\varphi : \mathcal{G} \to [0,1]$ *with* $\varphi \equiv 1$ *on* $V(\mathcal{G}_0)$ *and* $\varphi \equiv 0$ *on* M.

Theorem 6.5. *If* $\mathcal{G}$ *is such that* $(\mathrm{CR})_{0,q}$ *are always solvable then* $\mathcal{G}$ *is a domain of holomorphy.*

Proof. Let $\mathcal{G} \subseteq \mathbb{C}^n$. The proof is due to induction. For $n = 1$, we have that all domains are holomorphically convex. Suppose we have the case for $n - 1$.

Let $a \in \partial\mathcal{G}$ and $\mathcal{G}_0 = \{z_n = 0\} \cap \mathcal{G}$. Without loss of generality, we assume that $0 \in \partial\mathcal{G}_0$ in $\mathbb{C}^{n-1}$. Let

$$p : \mathbb{C}^n \to \mathbb{C}^{n-1}, j : \mathcal{G}_0 \to \mathcal{G},$$

be canonical projection and injection, respectively. We set,

$$M := \{z \in \mathcal{G} \mid p(z) \notin \mathcal{G}_0\}.$$

Then $M, \mathcal{G}_0$ are closed in $\mathcal{G}$ and $M \cap \mathcal{G}_0 = \emptyset$. Take $\varphi \in \mathcal{C}^\infty(\mathcal{G})$ such that $0 \le \varphi \le 1$ and

$$\varphi(\mathcal{G}_0) = \{1\}, \varphi(M) = \{0\}.$$

We first show that $f \in \mathcal{C}^\infty_{0,q}(\mathcal{G}_0), \overline{\partial} f = 0$ then there is $F \in \mathcal{C}^\infty_{0,q}(\mathcal{G})$ such that $\overline{\partial} F = 0$ and $F|_{\mathcal{G}_0} = f$ that is $F \circ j = f$.

Indeed $\varphi(f \circ p) \circ j = f$. Therefore take

$$F = \varphi(f \circ p) - z_n u,$$

where $u \in \mathcal{C}^\infty_{0,q}(\mathcal{G})$ so that

$$\overline{\partial} F = 0.$$

That is

$$0 = \overline{\partial} F = \overline{\partial}\varphi \circ (f \circ p) - z_n \overline{\partial} v. \Rightarrow \qquad \overline{\partial} u = \frac{1}{z_n} \overline{\partial}\varphi \circ (f \circ p) \in \mathcal{C}^\infty_{0,q+1}(\mathcal{G}).$$

This is $\overline{\partial}$-closed. Thus there is $v \in \mathcal{C}^\infty_{0,q}$ such that $\overline{\partial} F = 0$ and $F \circ j = f$.

We now show that $\mathcal{G}_0$ satisfies the assumption in $\mathbb{C}^{n-1}$.

Indeed If $f \in \mathcal{C}^\infty_{0,q}(\mathcal{G}_0)$ and $\overline{\partial} f = 0$ extend f to F as before. There is $u \in \mathcal{C}^\infty_{0,q-1}(\mathcal{G})$ with $\overline{\partial} u = F$. Take $v = u \circ j$, we then have,

$$\overline{\partial} v = f.$$

For the induction hypothesis, there is $f \in \mathcal{O}(\mathcal{G}_0)$ not extendable beyond 0. As we have shown, f extends to a holomorphic function F on $\mathcal{G}$ and F cannot be extended beyond 0. $\square$

Our next goal is to study $(\mathrm{CR})_{0,q}$ solvability on pseudoconvex domains, namely

$$H^{p,q}_{\mathrm{Dol}}(\mathcal{G}) = 0$$

for $q \ge 1$ and all p.

7 Sheaves of cohomology

The main reference if [6] but the exposition with more details will be found in [2].

7.1 Sheaves

Definition 7.1. *Let S, X be topological spaces. A pair $(\mathcal{S}, \pi : \mathcal{S} \to X)$ is called a sheaf on X if $\pi : \mathcal{S} \to X$ is a local homeomorphism. We will denote it simply by $\mathcal{S}$.*

Definition 7.2. *Let $U \subseteq X$ be an open subset. For a sheaf $\mathcal{S}$ on X. The section of $\mathcal{S}$ over U is a continuous map $s : U \to \mathcal{S}$ such that $\pi \circ s = \mathrm{id}$. The stalk of $\mathcal{S}$ at $x \in X$ is the preimage $S_x = \pi^{-1}(x)$.*

Proposition 7.1 (Fundamental Properties of Sheaves)**.** *Let S be a sheaf on X.*

1). For a section $s : U \to S$, $s(U)$ is open. In particular, $\pi|_{s(U)} : s(U) \to U$ is a homeomorphism.

2). $\{s(U)\}_{U \subset X \text{ open}}$ forms a basis of the topology of S.

3). S_x carries the discrete topology.

4). $\sigma \in S_x$ then there is a neighborhood $U = U(x)$ of x and a section $s : U \to S$ such that $s(x) = \sigma$.

5). Let s, t be sections such that $s(x_0) = t(x_0)$. Then there is a neighborhood $U = U(x_0)$ such that $s \equiv t$ on U.

Definition 7.3. *A subsheaf $(\mathcal{S}', \pi')$ of a sheaf $\mathcal{S} \xrightarrow{\pi} X$ is a pair such that $\mathcal{S}' \subseteq \mathcal{S}$ is an open subset and $\pi' = \pi|_{\mathcal{S}}$.*

In particular, for a open subset$M \subseteq X$, $\pi^{-1}(M)$ is a subsheaf which is called a restriction of $\mathcal{S}$ to M.

Notation 7.1. *We denote the set of all sections over U as $\Gamma(U, S)$.*

Remark 7.1. *Clearly if $S' \subseteq S$ is a subsheaf we have,*

$$\Gamma(U, S') \subseteq \Gamma(U, S).$$

Definition 7.4. *Let $(\mathcal{S}, \pi_{\mathcal{S}}), (\mathcal{T}, \pi_{\mathcal{T}})$ be sheaves on X. A morphism $\varphi : \mathcal{S} \to \mathcal{T}$ of sheaves is a continuous map such that the following diagram commute,*

$$\begin{tikzcd} \mathcal{S} \arrow[rr, "\varphi"] \arrow[dr, "\pi_{\mathcal{S}}"'] & & \mathcal{T} \arrow[dl, "\pi_{\mathcal{T}}"] \\ & X & \end{tikzcd}$$

This induces a map between stalks and for each $x \in X$, we denote,

$$\varphi_x := \varphi|_{\mathcal{S}_x} : \mathcal{S}_x \to \mathcal{T}_x.$$

Remark 7.2. *Given a morphism of sheaves, $\varphi : \mathcal{S} \to \mathcal{T}$, this induces a map*

$$\Gamma(U, \mathcal{S}) \ni s \mapsto \varphi \circ s \in \Gamma(U, \mathcal{T}).$$

We will denote $\varphi|_U(s) = \varphi \circ s$.

Remark 7.3. *For any morphism of sheaves $\varphi : \mathcal{S} \to \mathcal{T}$, $\varphi(\mathcal{S})$ is a subsheaf of $\mathcal{T}$.*

Definition 7.5. *A sheaf of abelian group is a sheaf $(\mathcal{S}, \pi)$ on X such that,*

1. *For each $x \in X$, $(\mathcal{S}_x, +)$ is an abelian group.*
2. *For $(\Gamma(U, \mathcal{S}), +)$ is a group with group operation operation defined point-wise. That is*

$$\Gamma(U, \mathcal{S}) \times \Gamma(U, \mathcal{S})(s, t) \mapsto s + t = [U \ni x \mapsto s(x) + t(x)] \in \Gamma(U, \mathcal{S}).$$

Proposition 7.2. *Let $\mathcal{S}$ be a sheaf of abelian groups. Then the map,*

$$\mathcal{S} \underset{\pi}{\times} \mathcal{S} \xrightarrow{\pm} \mathcal{S},$$

is continuous where

$$\mathcal{S} \underset{\pi}{\times} \mathcal{S} = \{(f, g) \in \mathcal{S} \times \mathcal{S} \mid \pi(f) = \pi(g)\} \subseteq \mathcal{S} \times \mathcal{S}.$$

Definition 7.6. *$\mathcal{S}' \subseteq \mathcal{S}$ is a subsheaf of abelian groups if*

i). it is a subsheaf of sets,

ii). $\mathcal{S}'_x$ is a subgroup of $\mathcal{S}_x$ for each $x \in X$.

Definition 7.7. *Let $\mathcal{S}, \mathcal{T}$ be sheaves of abelian groups. A sheaf homomorphism $\varphi : \mathcal{S} \to \mathcal{T}$ is such that*

1. *φ is a homomorphism of sheaves of sets,*
2. *$\varphi_x : \mathcal{S}_x \to \mathcal{T}_x$ is a group homomorphism for each $x \in X$.*

Remark 7.4. *We have the following trivial statements.*

1). $\Gamma(U, \mathcal{S})$ is an abelian group for any open subset $U \subseteq X$.

2). $\varphi(\mathcal{S}) \subseteq \mathcal{T}$ is a subsheaf of $\mathcal{T}$.

Definition 7.8. *Let $\varphi : \mathcal{S} \to \mathcal{T}$ be a homomorphism of abelian sheaves. Then we define,*

$$\operatorname{Ker} \varphi = \bigsqcup_{x \in X} \{\sigma \in \mathcal{S}_x \mid \varphi_x \sigma = 0 \in \mathcal{T}_x\}.$$

Proposition 7.3. $\operatorname{Ker} \varphi$ *is a subsheaf of $\mathcal{S}$.*

Proof. Let $s : U \to \mathcal{S}$ be a section such that $s(x) \in \operatorname{Ker} \varphi$ for some $x \in X$. Then □

Remark 7.5. *For a sheaf of abelian group $\mathcal{S}$, the zero subsheaf is a subsheaf $\underline{0}$ such that*

$$\underline{0} = \bigsqcup_{x \in X} \{0_{\mathcal{S}_x}\}.$$

This is initial and final.

Lemma 7.1. *For an open set $U \subseteq X$, $\varphi|_U : \Gamma(U, \mathcal{S}) \to \Gamma(U, \mathcal{T})$ is a group homomorphism.*

Definition 7.9. *A sheaf of commutative ring is a sheaf of abelian group $\mathcal{R}$ such that*

i). For all $x \in X$, $\mathcal{R}_x$ is a commutative ring.

ii). Multiplication is continuous that is for any open set $U \subseteq X$,

$$\Gamma(U, \mathcal{R}) \times \Gamma(U, \mathcal{R}) \ni (s, t) \mapsto st = [U \ni x \mapsto s(x)t(x)]$$

is a continuous map.

iii). There is a unit section $x \mapsto 1_x \in \mathcal{R}_x$.

Remark 7.6. $\Gamma(U, \mathcal{R})$ *is a commutative ring for all $U \subset X$ open.*

Definition 7.10. *Let R be a commutative ring and X be a topological space. We define a constant sheaf $\underline{R}$ of R over X to be*

$$X \times R \xrightarrow{\pi} X,$$

where R is equipped with the discrete topology, $X \times R$ is equipped with product topology, and π is a canonical projection.

Remark 7.7. *If $U \subseteq X$ is connected then $\Gamma(U, \underline{R}) = R$.*

Example 7.1. *The following are example of sheaves of rings.*

1). $\mathcal{O}$ a sheaf of germs of holomorphic functions. This is a Hausdorff space which can be proven using the identity theorem.

2). For $k \in \mathbb{N}$, $\mathcal{C}^k(\mathbb{C}^n)$ is a sheaf of ring. Furthermore, we have a descending chain,

$$\mathcal{C}^0(\mathbb{C}^n) \supseteq \cdots \supseteq \mathcal{C}^\infty(\mathbb{C}^n) \supseteq \mathcal{O}.$$

3). The sheaf of meromorphic function $\mathcal{K}$.

4). $\mathcal{O}^*$ is the subsheaf of $\mathcal{O}$ consisting of nowhere 0 holomorphic functions and $\mathcal{O}^*_x$ forms an abelian group under multiplication.

5). $\mathcal{K}^*$ is defined similary.

Definition 7.11. *Let $\mathcal{R}$ be a sheaf of commutative rings. We define a sheaf of $\mathcal{R}$-modules $\mathcal{A}$ to be such that*

i). it is a sheaf of abelian groups,

ii). $\mathcal{A}_x$ is an $\mathcal{R}_x$-module for each $x \in X$,

iii). for each open $U \subseteq X$, $\Gamma(U, \mathcal{R}) \times \Gamma(U, \mathcal{A}) \ni (r, a) \mapsto ra = [U \ni x \mapsto r(x)a(x)]$ is an element of $\Gamma(U, \mathcal{A})$ ie, defines a section over U.

Definition 7.12. *A presheaf F of sets is a contravariant functor on the category of open sets of X with values in the category of sets. That is for two open sets $V \subseteq U \subseteq X$, recall that the inclusion relation is represented by an inclusion map $\iota : V \to U$. Since F is contravariant, this induces a morphism of set,*

$$\rho_{UV} := F(V \hookrightarrow U),$$

which we call a restriction map.

Definition 7.13. *A homomorphism of presheaves is a functorial morphism (ie. a natural transformation between two contravariant functors.) That is a collection of map $(\varphi_U)_U : \mathcal{S} \to \mathcal{T}$ such that the following diagram commutes for all $V \subseteq U$ open,*

$$\begin{tikzcd} \mathcal{S}(U) \arrow[r, "\varphi_U"] \arrow[d, "\rho^{\mathcal{S}}_{UV}"'] & \mathcal{T}(U) \arrow[d, "\rho^{\mathcal{T}}_{UV}"] \\ \mathcal{S}(V) \arrow[r, "\varphi_V"'] & \mathcal{T}(V) \end{tikzcd}$$

Definition 7.14. *Let $\mathcal{S} \xrightarrow{\pi} X$ be a sheaf. The canonical presheaf associated $\mathcal{S}$ is a a functor $X \supseteq U \mapsto \Gamma(U, \mathcal{S})$.*

Definition 7.15. *A presheaf $\mathcal{S}$ is a sheaf if it is isomorphic to canonical presheaf of a sheaf.*

Example 7.2. *Set $\mathcal{K}_x = \mathrm{Frac}(\mathcal{O}_x)$ for each $x \in \mathbb{C}^n$ that is $\mathcal{K}_x$ is the germ of meromorphic functions. We define a presheaf $\mathcal{K}^{\mathrm{pre}} := \bigsqcup_{x\in\mathbb{C}^n} \mathcal{K}_x \to \mathbb{C}^n$. The topology on $\mathcal{K}^{\mathrm{pre}}$ is define by the following way. Take an open subset $U \subseteq \mathbb{C}^n$ and sections $f, g \in \Gamma(U, \mathcal{O})$ such that g is nowhere vanishing on U. Then the topology is generated by base element of the form $\left\{\frac{f_x}{g_x} \mid x \in U\right\}$. This does not define a sheaf.*

Definition 7.16. *Let $\mathcal{S}$ be a presheaf on X and $x \in X$. Let $\mathcal{U}(x) = \{U \subseteq X \mid U$ is open and $x \in U\}$. Then we define,*

$$\tilde{\mathcal{S}}_x := \bigsqcup_{U\in\mathcal{U}(x)} \mathcal{S}(U).$$

Let $f \in \mathcal{S}(U)$ and $g \in \mathcal{S}(V)$ for some open sets $U, V \subseteq X$. We define the equivalence relation $f \sim g$ if there is an open set $W \subseteq U \cap V$ such that

$$\rho_{UW}(f) = \rho_{VW}(g).$$

The stalk of $\mathcal{S}$ at x is then defined as,

$$\mathcal{S}_x := \tilde{\mathcal{S}}_x / \sim .$$

Furthermore, the sheafification of $\mathcal{S}$ is defined as,

$$(\widehat{\mathcal{S}}, \pi) := \bigsqcup_{x\in X} \mathcal{S}_x \xrightarrow{\pi} X,$$

where $\mathcal{S}_x \ni (f,x) \overset{\pi}{\mapsto} x \in X$. *We equip* $\widehat{\mathcal{S}}$ *with the topology by defining the base open sets as*

$$\forall U \subseteq X \text{ open}, s \in \mathcal{S}(U), \{\rho_{Ux}(f) = f(x) \mid x \in U\}.$$

Proposition 7.4. *For any presheaf* $\mathcal{S}$, $(\widehat{\mathcal{S}}, \pi)$ *is a sheaf.*

Example 7.3. *For any sheaf* $\mathcal{S}$ *we have* $\mathcal{S} \cong \widehat{\mathcal{S}}$.

Definition 7.17. *Let* $\mathcal{S}$ *be a sheaf on* X. *The presheaf of sections of* $\mathcal{S}$ *is* $\Gamma_{\mathcal{S}}$ *such that*

$$X \supseteq U \mapsto \Gamma_{\mathcal{S}}(U) = \Gamma(U, \mathcal{S}).$$

Remark 7.8.

$$\widehat{\Gamma_{\mathcal{S}}} \cong \mathcal{S}.$$

Proposition 7.5. $\widehat{(\cdot)}$ *is a functor sending presheaves to sheaves. That is given a morphism of presheaves,*

$$\mathcal{S} \overset{\varphi}{\to} \mathcal{T},$$

the sheafification then induces a morphism of sheaves,

$$\widehat{\mathcal{S}} \overset{\widehat{\varphi}}{\to} \widehat{\mathcal{T}}.$$

Suppose $\mathcal{S}$ *is a subpresheaf of* $\mathcal{T}$ *then*

$$\widehat{\mathcal{S}} \subseteq \widehat{\mathcal{T}}.$$

For a presheaf $\mathcal{S}$ *and its sheafification* $\widehat{\mathcal{S}}$, *there is a canonical morphism*

$$\mathcal{S} \to \Gamma_{\widehat{\mathcal{S}}}$$

By the first assertion, we have,

$$\widehat{\mathcal{S}} \to \widehat{\Gamma_{\widehat{\mathcal{S}}}}.$$

Remark 7.9. *The proposition above shows that we can justify denoting a sheaf* $\mathcal{S}$ *and its canonical presheaf* $\Gamma_{\mathcal{S}}$ *with the same letter.*

Definition 7.18. *Let* $\mathcal{G} \subseteq \mathbb{C}^n$ *be a domain. The structure sheaf* $\mathcal{O}_{\mathcal{G}}$ *is a sheaf of holomorphic functions on* $\mathcal{G}$.

Definition 7.19. *A ringed space is a pair* $(X, \mathcal{O})$ *such that*

i). X, *is a topological space,*

ii). $\mathcal{O} \subseteq \mathcal{C}^0(X)$ *a subsheaf of rings of continuous functio*

Given two ringed spaces $(X, \mathcal{O}), (Y, \mathcal{R})$, *a morphism of ringed spaces* $(X, \mathcal{O}) \overset{f}{\to} (Y, \mathcal{R})$ *is a map such that* *i). $f : X \to Y$ is continuous,*

ii). for any $U \subset Y$ open, and $s \in \mathcal{R}(U)$, then $s \circ f \in \mathcal{O}(f^{-1}(U))$.

Definition 7.20. *A n-dimensional compact manifold is a ringed space $(X, \mathcal{O}_X)$ such that*

i). X is Hausdorff,

ii). $\forall x \in X$, there is a neighborhood $U = U(x)$ and an isomorphism $\varphi : (U\mathcal{O}_U) \to (V, \mathcal{O}_V)$ where $(V, \mathcal{O}_V)$ is an open set in $\mathbb{C}^n$ with $\mathcal{O}_V$ a sheaf of germs of holomorphic functions on V.

Definition 7.21. *Let $\mathcal{S} \subseteq \mathcal{T}$ be sheaves of abelian groups. We define the factor sheaf $\mathcal{T}/\mathcal{S}$ to be such that*

$$\forall x \in X, (\mathcal{T}/\mathcal{S})_x := \mathcal{T}_x/\mathcal{S}_x,$$

and thus

$$\mathcal{T}/\mathcal{S} := \bigsqcup_{x \in X} (\mathcal{T}/\mathcal{S})_x.$$

The topology on $\mathcal{T}/\mathcal{S}$ then defined as follows. For $U \subseteq X$ and $s \in \mathcal{T}(U)$, the base open sets are of the form,

$$\{s(x) | x \in U\}.$$

Remark 7.10. *$\mathcal{T}/\mathcal{S}$ with above construction is a sheaf. Verifying it is left as an exercise to the readers.*

Alternatively, we could define $\mathcal{T}/\mathcal{S}$ to be the sheafification of the following presheaf,

$$X \supseteq U \mapsto (\mathcal{T}/\mathcal{S})(U) = \mathcal{T}(U)/\mathcal{S}(U).$$

Remark 7.11. *There is a surjective homomorhpism,*

$$\mathcal{T} \twoheadrightarrow \mathcal{T}/\mathcal{S},$$

which is induced by

$$\Gamma_{\mathcal{T}}(U) \twoheadrightarrow \Gamma_{\mathcal{T}}(U)/\Gamma_{\mathcal{S}}(U).$$

We then have a canonical morphism,

$$\Gamma_{\mathcal{T}}(U)/\Gamma_{\mathcal{S}}(U) \to (\mathcal{T}/\mathcal{S})(U).$$

We would expect it to be isomorphism, however, it turnes out that the map is neither injective nor surjective.

This leads to the following question. Given a surjective map $\alpha : \mathcal{T} \to \mathcal{S}$, is it true that the associated map $\Gamma(\alpha) : \Gamma_{\mathcal{T}} \to \Gamma_{\mathcal{S}}$ is again surjective?

This question then leads us to the theory of cohomology. That is we want to study when these canonically associated morphisms turn out to be isomorphisms.

Definition 7.22. *Let M be an analytical set of a domain $\mathcal{G} \subseteq \mathbb{C}^n$. We define the associated sheaf of ideal to be,*

$$\mathcal{G} \supseteq U \mapsto \mathcal{I}(U) = \{f \in \mathcal{O}(U) \mid f|_{M \cap U} \equiv 0\}.$$

Clearly, $\mathcal{I} \subseteq \mathcal{O}_{\mathcal{G}}$ as sheaves of abelian groups. The associated quotient sheaf is then defined as,

$$\tilde{\mathcal{O}_M} := \mathcal{O}_{\mathcal{G}}/\mathcal{I}|_M.$$

Remark 7.12. *The pair $(M, \tilde{\mathcal{O}}_M)$ is a ringed space.*

Definition 7.23. *A complex space is a ringed space $(X, \mathcal{O}_X)$ such that for all $x \in X$ there is a neighborhood $U = U(x)$ of x and an isomorphism $\varphi : (U, \mathcal{O}_X|_U) \to (M, \mathcal{O}_M)$ where $(M, \mathcal{O}_M)$ is an analytic with its structure sheaf in some open set $V \subseteq \mathbb{C}^n$ for some $n \in \mathbb{N}$. A morphism of ringed spaces between complex spaces are called a holomorphic map.*

Example 7.4. *Let $(M, \mathcal{O}_M)$ be a complex space, then $(M, \tilde{\mathcal{O}}_M)$ is realized as collections of weakly holomorphic functions, We note that $\tilde{\mathcal{O}}_M \subseteq \mathcal{O}_M$. We can even prove that $(M, \tilde{\mathcal{O}}_M)$ is a complex space but this is not a trivial statement.*

Remark 7.13. *Let $\hat{\mathcal{O}}_M$ be the sheaf of germs of normal functions. This is not a complex space but we can study how it is associated to a complex space. This lecture will not talk about this association however, the readers can look into more details in [2].*

7.2 Cohomology

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be $\mathcal{O}_X$-modules.

Definition 7.24. *Let $\mathscr{U} = \{U_i\}_{i \in I}$ be an open covering of X, $\mathcal{A}$ be a sheaf on X. A q-cochain in $\mathcal{A}$ with respect to $\mathscr{U}$ is a map $a : I^q \to \mathcal{A}$ such that*

$$a(i_0, \cdots, i_q) \in \mathcal{A}\left(U_{i_0 \cdots i_q} = \bigcup_{l=0}^{q} U_{i_l}\right),$$

which is alternating in the indices, that is

$$a(i_0, \cdots, i_j, \cdots, i_k, \cdots, i_q) = -a(i_0, \cdots, i_k, \cdots, i_j, \cdots, i_q).$$

We can view a as an element of $\prod_{(i_0, \cdots, i_q) \in I^q} \mathcal{A}(U_{i_0 \cdots i_q})$. We denote the set of all q-cochains as

$$\mathcal{C}^q(\mathscr{U}, \mathcal{A}),$$

which is an abelian group and even an $\mathcal{O}_X$-module.

Notation 7.2. *Let $V \subset U$ be open and $a \in \mathcal{A}(U)$ then we will denote,*

$$a|_V^U := \rho_{UV}(a).$$

Definition 7.25. *A coboundary is an $\mathcal{O}_X$-module homomorphism $\delta : \mathcal{C}^q(\mathscr{U}, \mathcal{A}) \to \mathcal{C}^{q+1}(\mathscr{U}, \mathcal{A})$ such that*

$$\delta(a)(i_0, \cdots, i_q, i_{q+1}) = \sum_{j=0}^{q+1} (-1)^j a(i_0, \cdots, i_{j-1}, i_{j+1}, \cdots, i_{q+1})|^{U_{i_0 \cdots i_{j-1} i_{j+1} \cdots i_{q+1}}}_{U_{i_0 \cdots i_{q+1}}}.$$

Proposition 7.6. *Let δ be a coboundary then,*

$$\delta \circ \delta = 0.$$

Proof. This can be shown with direct calculations. □

Definition 7.26. *A sequence*

$$0 \to \mathcal{C}^0(\mathscr{U}, \mathcal{A}) \xrightarrow{\delta} \mathcal{C}^1(\mathscr{U}, \mathcal{A}) \xrightarrow{\delta} \mathcal{C}^0(\mathscr{U}, \mathcal{A}) \xrightarrow{\delta} \cdots$$

is called the cochain complex of $\mathcal{A}$ with respect to $\mathscr{U}$. We will denote this by

$$(\mathscr{C}^\bullet(\mathscr{U}, \mathcal{A})).$$

Definition 7.27. *Given a cochain complex of $(\mathscr{C}^\bullet(\mathscr{U}, \mathcal{A}))$. The q-cocycle is*

$$Z^q(\mathscr{U}, \mathcal{A}) := \operatorname{Ker} \delta = \operatorname{Ker}(\mathcal{C}^q(\mathscr{U}, \mathcal{A}) \to \mathcal{C}^{q+1}(\mathscr{U}, \mathcal{A})).$$

The q-coboundary is

$$B^q(\mathscr{U}, \mathcal{A}) := \operatorname{Im} \delta = \operatorname{Im}(\mathcal{C}^{q-1}(\mathscr{U}, \mathcal{A}) \to \mathcal{C}^q(\mathscr{U}, \mathcal{A})).$$

Remark 7.14. *By Proposition 7.6, we have $B^q(\mathscr{U}, \mathcal{A}) \subseteq Z^q(\mathscr{U}, cA)$.*

Definition 7.28. *q-th cohomology of $\mathcal{A}$ with respect to $\mathscr{U}$ is defined as*

$$H^q(\mathscr{U}, \mathcal{A}) = Z^q(\mathscr{U}, \mathcal{A})/B^q(\mathscr{U}, \mathcal{A}).$$

Proposition 7.7. *$\mathcal{C}^q(\mathscr{U}, \cdot), \mathscr{C}^\bullet(\mathscr{U}, \cdot)$, and $H^\bullet(\mathscr{U}, \cdot)$ are functorial that is for given a $\mathcal{O}_X$ homomorphism $\mathcal{A} \xrightarrow{f} \mathcal{B}$, this induces*

$$\mathcal{C}^q(\mathscr{U}, \mathcal{A}) \overset{\mathcal{C}^q(\mathscr{U}, f)}{\to} \mathcal{C}^q(\mathscr{U}, \mathcal{B}),$$
$$\mathcal{C}^\bullet(\mathscr{U}, \mathcal{A}) \overset{\mathscr{C}^\bullet(\mathscr{U}, f)}{\to} \mathcal{C}^\bullet(\mathscr{U}, \mathcal{B}),$$
$$H^q(\mathscr{U}, \mathcal{A}) \overset{H^q(\mathscr{U}, f)}{\to} H^q(\mathscr{U}, \mathcal{B}).$$

This notion can be further generalized from $\mathcal{O}_X$ modules to $\mathcal{O}_X$ presheaves. In particular, $H^\bullet(\mathscr{U}, \cdot)$ is a covariant functor from a category of $\mathcal{O}_X$ presheaves to $\mathcal{O}_X$ modules.

Furthermore, we have the following commutative diagram,

$$
\begin{array}{ccccc}
\mathcal{A} & \longrightarrow & \mathcal{C}^\bullet(\mathscr{U},\mathcal{A}) & \longrightarrow & H^\bullet(\mathscr{U},\mathcal{A}) \\
\Big\downarrow{\scriptstyle \alpha} & & \Big\downarrow{\scriptstyle \mathscr{C}^\bullet(\alpha)} & & \Big\downarrow{\scriptstyle H^\bullet(\alpha)} \\
\mathcal{B} & \longrightarrow & \mathcal{C}^\bullet(\mathscr{U},\mathcal{B}) & \longrightarrow & H^\bullet(\mathscr{U},\mathcal{B})
\end{array}
$$

Definition 7.29. *Let X be a topological space and $\mathscr{U} = (U_i)_{i\in I}, \mathscr{V} = (V_j)_{j\in J}$ be two covering. $\mathscr{V}$ is finer than $\mathscr{U}$ if for each $i \in I$ there is $j \in J$ such that $V_j \subseteq U_i$.*

In that case $\mathscr{V}$ is called a refinement of $\mathscr{U}$. We furthere define a refinement map which is a map $\varphi : J \to I$ such that $V_j \subseteq U_{\varphi(i)}$.

Definition 7.30. *Let $\mathscr{U}, \mathscr{V}$ be open covering of a topological space X and $a \in \mathcal{C}^q(\mathscr{U},\mathcal{A})$. Supposer $\mathscr{V}$ is a refinement of $\mathscr{U}$ with a refinement map $\phi : J \to I$. Then we define,*

$$\varphi^q a(j_0, \cdots, j_q) = a(\varphi(j_0), \cdots, \varphi(j_q))|^{U_{\varphi(j_0)\cdots\varphi(i_q)}}_{V_{j_0\cdots j_q}}.$$

This defines maps $\varphi^\bullet : \mathcal{C}^\bullet(\mathscr{U},\mathcal{A}) \to \mathcal{C}^\bullet(\mathscr{V},\mathcal{A})$ and

$$h^\bullet_\varphi : H^\bullet(\mathscr{U},\mathcal{A}) \to H^\bullet(\mathscr{V},\mathcal{A}).$$

Lemma 7.2. *$h^\bullet_\varphi$ does not depend on φ.*

Proof. If φ, ψ are two refinement maps then $\varphi^\bullet, \psi^\bullet$ are homotopic. Details can be found in [6]. □

Suppose $\mathcal{A}$ is a sheaf and $\Gamma_{\mathcal{A}}$ be its canonical presheaf.

Proposition 7.8. *The map,*

$$h^1 : H^1(\mathscr{U},\mathcal{A}) \to H^1(\mathscr{V},\mathcal{A}),$$

is injective.

Proof. Details can be found in [3]. □

The above proposition can be rephrased by the following corollary.

Corollary 7.1. *Suppose we are given $c \in Z^1(\mathscr{U},\mathcal{A})$, a cocycle. Under a refinement map φ, we have $\varphi^1 c \in Z^1(\mathscr{V},\mathcal{A})$.*

If $\varphi^1 c = \delta b$ for some $b \in \mathcal{C}^0(\mathscr{V},\mathcal{A})$, then there is $\tilde{b} \in \mathcal{C}^0(\mathscr{U},\mathcal{A})$ such that $\delta\tilde{b} = c$.

Notation 7.3. *Motivated by Lemma 7.2, we introduce the following notation.*

$$h^\bullet(\mathscr{U},\mathscr{V}) : H^\bullet(\mathscr{U},\mathcal{A}) \to H^\bullet(\mathscr{V},\mathcal{A}).$$

Obviously we have,

$$h^\bullet(\mathscr{U},\mathscr{U}) = \mathrm{id}\,.$$

Suppose we have refinements $\mathscr{W} \leq \mathscr{V} \leq \mathscr{U}$, we have,

$$h^\bullet(\mathscr{U},\mathscr{W}) = h^\bullet(\mathscr{V},\mathscr{W}) \circ h^\bullet(\mathscr{U},\mathscr{V}).$$

Definition 7.31. *Let $\mathscr{S}$ be a set of open covering of X such that for any open covering $\mathscr{U}$ of X, there is a refinement $\mathscr{V}$ of $\mathscr{U}$ contained in $\mathscr{S}$. We then define the inductive limit,*

$$H^\bullet(X,\mathcal{A}) = \varinjlim_{\substack{\mathscr{U} \\ \text{open covering}}} H^\bullet(\mathscr{U},\mathcal{A}),$$

such that for each $\mathscr{U} \in \mathscr{S}$, there is a well-define morphism,

$$h^\bullet(\mathscr{U}) : H^\bullet(\mathscr{U},\mathcal{A}) \to H^\bullet(X,\mathcal{A}),$$

such that for any refinement map $\varphi : \mathscr{V} \to \mathscr{U}$, we have a commutative diagram,

$$\begin{tikzcd} H^\bullet(\mathscr{U},\mathcal{A}) \arrow[rr, "\varphi^\bullet"] \arrow[dr, "h^\bullet(\mathscr{U})"'] & & H^\bullet(\mathscr{V},\mathcal{A}) \arrow[dl, "h^\bullet(\mathscr{V})"] \\ & H^\bullet(X,\mathcal{A}) & \end{tikzcd}$$

Each element $\gamma \in H^q(X,\mathcal{A})$ is represented by an element $c \in H^q(\mathscr{U},\mathcal{A})$ such that $\delta(c) = 0$, for some $\mathscr{U} \in \mathscr{S}$ such that if $c' \in H^q(\mathscr{U}',\mathcal{A})$ where $\mathscr{U}' \in \mathscr{S}$ and $\delta(c') = 0$ also represents γ then there is a refinement $\mathscr{U}'' \in \mathscr{S}$ such that

$$\exists b \in \mathcal{C}^{q-1}(\mathscr{U}'',\mathcal{A}), \delta b = h(\mathscr{U},\mathscr{U}'')c - h(\mathscr{U}',\mathscr{U}'')c'.$$

We note that the map $H^\bullet(X,\cdot)$ is functorial. That is given $\alpha : \mathcal{A} \to \mathcal{B}, \beta : \mathcal{B} \to \mathcal{C}$, we have,

$$h^\bullet(\alpha) : H^\bullet(X,\mathcal{A}) \to H^\bullet(X,\mathcal{B}).$$

Furthermore $h^\bullet(\beta \circ \alpha) = h^\bullet(\beta) \circ h^\bullet(\alpha)$.

If $\mathcal{A}$ is a sheaf then we have,

$$\mathcal{A}(X) = \Gamma(X,\mathcal{A}) = H^0(X,\mathcal{A}).$$

In the case of $\mathcal{O}_X$-modules, $H^\bullet$ and all the other ones turn out to be $\mathcal{O}_X$-modules.

Remark 7.15. *For $a \in \mathcal{C}^0(\mathscr{U},\mathcal{A})$, we have $a(i_0) \in \mathcal{A}(U_{i_0})$ and $\delta a = 0$ that is $a(i_0) - a(i_1) = 0$ on $U_{i_0 i_1}$.*

With this we may define a global section $\tilde{a} \in \mathcal{A}(X)$ glued from open cover $\{U_i\}_{i\in I}$ and $(a_i \in U_i)_{i\in I}$ that is

$$\tilde{a}|^X_{U_i} = a_i.$$

Definition 7.32. *Let $\mathcal{A}, \mathcal{B}$ be presheaves. We say*

$$\alpha : \mathcal{A} \to \mathcal{B}$$

is $\begin{cases} \textit{injective,} \\ \textit{surjective} \end{cases}$ *if for any open set U of X the map $\alpha(U) : \mathcal{A}(U) \to \mathcal{B}(U)$ is* $\begin{cases} \textit{injective,} \\ \textit{surjective} \end{cases}$ *We define the following presheaves.*

1). $\ker\alpha = [U \mapsto \ker\alpha(U)]$,

2). $\operatorname{im}\alpha = [U \mapsto \operatorname{im}\alpha(U)]$.

Definition 7.33. *Let $\mathcal{A}, \mathcal{B}$ be sheaves. A morphism $\alpha : \mathcal{A} \to \mathcal{B}$ is* $\begin{cases} \textit{injective}, \\ \textit{surjective} \end{cases}$ *if for any $x \in X$, the map $\alpha_x : \mathcal{A}_x \to \mathcal{B}_x$ is* $\begin{cases} \textit{injective}, \\ \textit{surjective.} \end{cases}$ *We define* $\ker\alpha, \operatorname{im}\alpha$ *to be the sheaf with stalks*

$$(\ker\alpha)_x = \ker\alpha_x, (\operatorname{im}\alpha)_x = \operatorname{im}\alpha_x.$$

Remark 7.16. *Given $\mathcal{A} \xrightarrow{\alpha} \mathcal{B}$ a morphism of presheaves and its corresponding sheaf morphism $\hat{\mathcal{A}} \xrightarrow{\hat{\alpha}} \hat{\mathcal{B}}$. All the above properties of α is transfered to $\hat{\alpha}$. The question remaining is is it true the other way around.*

Proposition 7.9. *If $\hat{\alpha}$ is injective then so is α.*

Remark 7.17. *$\hat{\alpha}$ being surjective does not imply α is surjective.*

Definition 7.34. *A sequence*

$$0 \longrightarrow \mathcal{A} \longrightarrow \mathcal{B} \longrightarrow \mathcal{C} \longrightarrow 0$$

is exact if for any point x of X, the sequence

$$0 \longrightarrow \mathcal{A}_x \longrightarrow \mathcal{B}_x \longrightarrow \mathcal{C}_x \longrightarrow 0$$

is exact.

For an exact sequence of presheaves,

$$0 \longrightarrow \mathcal{A} \longrightarrow \mathcal{B} \longrightarrow \mathcal{C} \longrightarrow 0$$

.

We have

$$0 \longrightarrow \mathcal{C}^\bullet(\mathscr{U}, \mathcal{A}) \xrightarrow{\mathcal{C}^\bullet(\alpha)} \mathcal{C}^\bullet(\mathscr{U}, \mathcal{B}) \xrightarrow{\mathcal{C}^\bullet(\beta)} \mathcal{C}^\bullet(\mathscr{U}, \mathcal{A}) \longrightarrow 0$$

By algebra, we have a long exact sequence,

$$\begin{tikzcd}
0 \arrow[r] & H^0(\mathscr{U}, \mathcal{A}) \arrow[r] & H^0(\mathscr{U}, \mathcal{B}) \arrow[r] & H^0(\mathscr{U}, \mathcal{C}) \arrow[dll, "\delta_0^*"] \\
& H^1(\mathscr{U}, \mathcal{A}) \arrow[r] & H^1(\mathscr{U}, \mathcal{B}) \arrow[r] & H^1(\mathscr{U}, \mathcal{C}) \arrow[dll, "\delta_1^*"] \\
& H^2(\mathscr{U}, \mathcal{A}) & &
\end{tikzcd}$$

Proposition 7.10. *Proposition 7.9 and Remark 7.17 can be restated as follows.*

Given a exact sequence of sheaves

$$0 \longrightarrow \mathcal{C}^\bullet(\mathscr{U}, \mathcal{A}) \xrightarrow{\mathcal{C}^\bullet(\alpha)} \mathcal{C}^\bullet(\mathscr{U}, \mathcal{B}) \xrightarrow{\mathcal{C}^\bullet(\beta)} \mathcal{C}^\bullet(\mathscr{U}, \mathcal{A}) \longrightarrow 0$$

We have a following exact sequence,

$$0 \longrightarrow \Gamma_{\mathcal{A}} \longrightarrow \Gamma_{\mathcal{B}} \longrightarrow \Gamma_{\mathcal{C}}$$

Let us denote $\widetilde{\Gamma_{\mathcal{C}}} = \Gamma_{\mathcal{A}}/\Gamma_{\mathcal{B}}$. Then we have an exact sequence.

$$0 \longrightarrow \Gamma_{\mathcal{A}} \longrightarrow \Gamma_{\mathcal{B}} \longrightarrow \widetilde{\Gamma_{\mathcal{C}}} \longrightarrow 0.$$

The question is do we have the following equalities,

$$H^q(X, \mathcal{C}) = H^q(X, \Gamma_{\mathcal{C}}) = H^q(X, \widetilde{\Gamma_{\mathcal{C}}}).$$

Definition 7.35. *A Hausdorff space is said to be paracompact if each open covering admits a locally finite refinement.*

Theorem 7.1. *Let X be a paracompact space and $\mathcal{A}$ be a presheaf. Suppose $\hat{\mathcal{A}} = 0$ then we have $H^q(X, \mathcal{A}) = 0$ for all q.*

Proposition 7.11. *Let $\mathcal{S}$ be a presheaf and X be a paracompact space. Then we have,*

$$H^q(X, \mathcal{S}) = H^q(X, \hat{\mathcal{S}}).$$

Proof. Let $\Gamma_{\hat{\mathcal{S}}}$ be its canonical presheaf and $\tau : \mathcal{S} \to \Gamma_{\hat{\mathcal{S}}}$ be a caonnical homomorphism.

$$0 \longrightarrow \ker \tau \longrightarrow \mathcal{S} \longrightarrow \operatorname{im} \tau \longrightarrow 0$$

$$0 \longrightarrow \operatorname{im} \tau \longrightarrow \Gamma_{\hat{\mathcal{S}}} \longrightarrow \Gamma_{\hat{\mathcal{S}}}/\operatorname{im} \tau \longrightarrow 0$$

By definition, we have $\ker \tau$ is a 0-sheaf, similarly for $\Gamma_{\hat{\mathcal{S}}}/\operatorname{im} \tau$. By Theorem 7.1, we have,

$$H^q(X, \ker \tau) = 0, H^q(X, \Gamma_{\hat{\mathcal{S}}}/\operatorname{im} \tau) = 0.$$

That is we have the following isomorphisms,

$$H^q(X, \mathcal{S}) \xrightarrow{\sim} H^q(X, \operatorname{im} \tau), H^q(X, \operatorname{im} \tau) \xrightarrow{\sim} H^q(X, \widehat{\Gamma_{\hat{\mathcal{S}}}}).$$

□

Definition 7.36. *Let $\mathscr{U}$ be an open covering of a topological space X and $\mathcal{A}, \mathcal{B}$ be sheaves of abelian groups. $\mathscr{U}$ is called $\mathcal{A}$-acyclic if it is globally finite and $H^q(U_{i_0, \cdots, i_p}, \mathcal{A}) = 0$ for all $p \geq 0$ and $q \geq 1$.*

Theorem 7.2 (Leray)**.** *If $\mathscr{U}$ is acyclic for $\mathcal{A}$, then $H^q(X, \mathcal{A}) \cong H^q(\mathscr{U}, \mathcal{A})$.*

Proof. See [6]. ∎

Definition 7.37. *A sheaf of abelian group $\mathcal{A}$ is acyclic if $H^q(X, \mathbb{A}) = 0$ for all $q \geq 1$.*

Definition 7.38. *An acyclic resolution of a sheaf of abelian groups $\mathcal{A}$ is a long exact sequence,*

$$0 \longrightarrow \mathcal{A} \xrightarrow{h} \mathcal{A}^0 \xrightarrow{h^0} \mathcal{A}^1 \xrightarrow{h^1} \mathcal{A}^2 \longrightarrow$$

such that each $\mathcal{A}^p$ is acyclic for $p \geq 0$.

Now consider,

$$0 \longrightarrow \mathcal{A}(X) \xrightarrow{h_X} \mathcal{A}^0(X) \xrightarrow{h^0_X} \mathcal{A}^1(X) \xrightarrow{h^1_X} \mathcal{A}^2(X) \longrightarrow$$

This is no longer an exact sequence. Thus we consider a cohomology of $\mathbb{A}^0(X)$.

Theorem 7.3. *Consider an acyclic resolution,*

$$0 \longrightarrow \mathcal{A} \xrightarrow{h} \mathcal{A}^0 \xrightarrow{h^0} \mathcal{A}^1 \xrightarrow{h^1} \mathcal{A}^2 \longrightarrow$$

We have

$$H^q(X, \mathcal{A}) = H^q(\mathcal{A}^0(X)).$$

Proof. For $q = 0$, we have ,

$$H^0(X, \mathcal{A}) = \mathcal{A}(X) = \ker h^0_X = \ker[\mathcal{A}^0(X) \to \mathcal{A}^1(X)] = H^0(\mathcal{A}^0(X)).$$

Denote $\overline{\mathcal{A}} = \operatorname{im} h^0 \subseteq \mathcal{A}^1$. Now,

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{A} \arrow[r] & \mathcal{A}^0 \arrow[r, "h^0"] \arrow[d, "h^0"] & \mathcal{A}^1 \arrow[r] & {} \\
& & \overline{\mathcal{A}} \arrow[ur, hook, "i"'] \arrow[d] & & \\
& 0 \arrow[ur] & 0 & &
\end{tikzcd}$$

We then have,

$$(I) \qquad 0 \longrightarrow \mathcal{A} \longrightarrow \mathcal{A}^0 \longrightarrow \overline{\mathcal{A}} \longrightarrow 0$$

$$(II) \qquad 0 \longrightarrow \overline{\mathcal{A}} \longrightarrow \mathcal{A}^1 \longrightarrow \mathcal{A}^2 \longrightarrow \cdots$$

THus we have,

$$\mathcal{A}^0(X) \longrightarrow \overline{\mathcal{A}}(X) \longrightarrow H^1(X, \mathcal{A}) \longrightarrow \underbrace{H^1(X, \mathcal{A}^0)}_{=0}$$

This means that

$$H^1(X,\mathcal{A}) = \overline{\mathcal{A}}(X)/h^0(\mathcal{A}^0(X)) = \frac{\ker\{h^1 : \mathcal{A}^1(X) \to \mathcal{A}^2(X)\}}{h^0(\mathcal{A}(X))} = H^1(\mathcal{A}^0(X)).$$

For $q :, 1 \leq q \leq \tilde{q}$, we set, $\mathcal{B} = \mathcal{A}^{q+1}$. We then have another acyclic resolution of $\overline{\mathcal{A}}$. $0 \longrightarrow \overline{\mathcal{A}} \longrightarrow \mathcal{B}^0 \longrightarrow \mathcal{B}^1 \longrightarrow \cdots$ Thus we obtain $H^i(\mathcal{B}^\bullet(X)) = H^{q+1}(\mathcal{A}^\bullet(X))$. Therefore,

$$H^{q+1}(X,\mathcal{A}) \cong H^q(X,\overline{\mathcal{A}}).$$

By induction on the exact cohomology sequence, we have,

$$H^q(X,\overline{\mathcal{A}}) = H^q(\mathcal{B}^\bullet(X)) \cong H^{q+1}(X,\mathcal{A}^\bullet(X)).$$

□

7.3 deRahm and Dolbeault Theorems

Theorem 7.4. *Let X be a smooth manifold (ie. $\mathcal{G} \subseteq \mathbb{C}^n$). Let $\mathcal{A}$ be a sheaf of germs of $\mathcal{C}^\infty(X)$. Then $\mathcal{A}$ is acyclic that is $H^q(X,\mathcal{A}) = 0$ for $q \geq 1$.*

Proof. Let $\mathscr{U}$ be an open covering of X with infinite index set I.

Claim 1.

$$q \geq 1 \Rightarrow H^q(\mathscr{U},\mathcal{A}) = 0.$$

Proof: Let $(\varphi_i)_{i\in I}$ be a smooth partition of unity subordinates to $\mathscr{U}$. Let $c \in \mathcal{C}^q(\mathscr{U},\mathcal{A})$ be such that $\delta(c) = 0$. We now define

$$b(i_0,\cdots,i_{q-1}) = \sum_{i\in I} \varphi_i c(i,i_0,\cdots,i_{q-1}).$$

Then we have,

$$\begin{aligned}
\delta b &= \sum_{i\in I}\sum (-1)^j c(i,i_0,\cdots,i_{j-1},i_{j+1},\cdots,i_q). \\
(\delta b)(i_0,\cdots,i_q) &= \sum_{i\in I}\varphi_i \sum_j (-1)^j c(i,\cdots,i_{j-1},i_{j+1},\cdots,i_q), \\
&= \sum_{i\in I}\varphi_i c(i_0,\cdots,i_q), \\
&= \underbrace{\left(\sum_{i\in I}\varphi_i\right)}_{=1} c(i_0,\cdots,i_q), \\
&= c(i_0,\cdots,i_q).
\end{aligned}$$

■ The statement follows from the claim. □

Remark 7.18. *Let $\mathcal{E}^p$ be a smooth p-forms on X. Then they are acyclic.*

Theorem 7.5. *Let $\mathcal{G} \subseteq \mathbb{R}^N$ be a convex domain and $f \in \mathcal{C}_q^\infty(\mathcal{G})$, with $df = 0$. Then $f = du$ for some $u \in \mathcal{C}_{q-1}^\infty(\mathcal{G})$.*

We have an exact sequence,

$$0 \longrightarrow \mathbb{C} \longrightarrow \mathcal{E}^0 \xrightarrow{d} \mathcal{E}^1 \xrightarrow{d} \mathcal{E}^2 \xrightarrow{d} \cdots \xrightarrow{d} \mathcal{E}^N \xrightarrow{d} 0$$

This is an acyclic resolution of $\mathbb{C}$. Then we have, for X connected

$$H^q(X, \mathbb{C}) \cong H^q(\mathcal{E}^\bullet(X)) = \frac{\{d\text{-closed } q\text{-forms}\}}{\{d\text{-exact } q\text{-forms}\}} = H^q_{\mathrm{dR}}(X).$$

Theorem 7.6 (de Rham)**.**

$$H^q(X, \mathbb{C}) \cong H^q_{\mathrm{dR}}(X).$$

Remark 7.19.

$$H^q(X, \mathbb{C}) \cong H^q_{\mathrm{sing}}(X),$$

where the left hand side is a sheaf cohomology and the right hand side is from topology.

Definition 7.39. *Let X be a compact manifold (eg. $\mathcal{G} \subseteq \mathbb{C}^n$). We define $\Omega^p(X)$ to be a sheaf of germs of holomorphic p-forms.*

Remark 7.20. $\Omega^0(X) = \mathcal{O}(X)$ *a structure sheaf.*

Remark 7.21. *For each p, we have an acyclic resolution,*

$$0 \longrightarrow \Omega^p \longrightarrow \mathcal{E}^{p,0} \xrightarrow{\overline{\partial}} \mathcal{E}^{p,1} \xrightarrow{\overline{\partial}} \mathcal{E}^{p,2} \xrightarrow{\overline{\partial}} \cdots \xrightarrow{\overline{\partial}} \mathcal{E}^{p,n} \longrightarrow 0$$

Theorem 7.7 (Dolbeault)**.**

$$H^q(X, \Omega^p) \cong H^{p,q}(\mathcal{E}^{p,\bullet}(X)) = H^{p,q}_{\mathrm{Dol}}(X).$$

Remark 7.22. *For $p = 0$, we have $\Omega^0(X) = \mathcal{O}(X)$. Thus we hae,*

$$H^q(X, \mathcal{O}) \cong H^{0,q}_{\mathrm{Dol}}(X).$$

That is for all $f \in \mathcal{E}^{0,q}(X)$, if $\overline{\partial} f = 0$ there is $u \in \mathcal{E}^{0,q-1}(X)$ with $\overline{\partial} u = f$.

Theorem 7.8. *Let $\mathcal{E}_{pq}$ be smooth forms on $\mathcal{G} \subseteq \mathbb{C}^n$. Then $H^p(\mathcal{G}, \mathcal{E}_{pq}) = 0$ for $q \geq 1$.*

Proof. Let $\mathscr{U}$ be a locally finite open covering with index set I. Let $(\varphi_i)_{i \in I}$ be a partition of unity subordinates to $\mathscr{U}$. Let $f \in Z^q(\mathscr{U}, \mathcal{E})$, then

$$a(i_0, \cdots, i_{q-1}) := \sum_{i \in I} \varphi_i f(i, i_0, \cdots, i_{q-1}).$$

Then

$$
\begin{aligned}
(\delta a)(i_0, \cdots, i_q) &= \sum_{j=0}^{q} (-1)^j a(i_0, \cdots, i_{j-1}, i_{j+1}, \cdots, i_q), \\
&= \sum_{i\in I} \sum_{j=0}^{q} (-1)^j \varphi_i f(i, i_0, \cdots, i_{j-1}, i_{j+1}, \cdots, i_q), \\
&= \sum_{i\in I} \varphi_i \sum_{j=0}^{q} (-1)^j f(i, i_0, \cdots, i_{j-1}, i_{j+1}, \cdots, i_q), \\
&= -\left(\sum_{i\in I} \varphi_i\right) f(i_0, \cdots, i_q), \\
&= -f.
\end{aligned}
$$

Therefore f is also exact. For $q = 1$, we have, $f(i_0, i_1)$,

$$
a(i_0) = \sum_{\varphi_i} f(i, i_0), a(i_1) = \sum \varphi_i f(i, i_1).
$$

Therefore,

$$
(\delta a)(i_0, i_1) = a(i_1) - a(i_0) = \sum \varphi_i (f(i, i_1) - f(i, i_0)).
$$

The readers are encouraged to do the case when $q = 1$ manually to see make sense of this proof. □

7.4 Global Problems

We will start with the first cousin problem. Using the terminologies we have introduced, we can rewrite the problem as follows.

Let $\mathcal{H} = \mathcal{K}/\mathcal{O}$. Then we have the exact sequene,

$$
0 \longrightarrow \mathcal{O} \hookrightarrow \mathcal{K} \twoheadrightarrow \mathcal{H} \longrightarrow 0.
$$

Is it true that we have,

$$
0 \longrightarrow \mathcal{O}(X) \hookrightarrow \mathcal{K}(X) \twoheadrightarrow \mathcal{H}(X) \longrightarrow 0
$$

still exaft?

Theorem 7.9. *If $H^1(X, \mathcal{O}) = 0$ then the first Cousin problem is solvable.*

Proof. We have a cohomology sequence,

$$
0 \longrightarrow \mathcal{O}(X) \longrightarrow \mathcal{K}(X) \longrightarrow \mathcal{H}(X) \longrightarrow H^1(X, \mathcal{O}) \longrightarrow \cdots
$$

Thus the assumption directly shows the statement. □

Remark 7.23. *For any polydisk X, we have $H^1(X, \mathcal{O}) = 0$.*

Definition 7.40. *We define the sheaf of germs of divisors to be $\mathcal{D} = \mathcal{K}^*/\mathcal{O}^*$.*

Recall the second cousin problem, stating that is all divisors principle. We have an exact sequence,

$$0 \longrightarrow \mathcal{O}^* \longrightarrow \mathcal{K}^* \longrightarrow \mathcal{D} \longrightarrow 0$$

$$0 \longrightarrow \mathcal{O}^*(X) \longrightarrow \mathcal{K}^*(X) \longrightarrow \mathcal{D}(X) \longrightarrow H^1(X, \mathcal{O}^*) \longrightarrow \cdots$$

Consider the map $\mathcal{O} \ni \mapsto \exp(2\pi i f) \in \mathcal{O}^*$. The kernel of this map is $\mathbb{Z}$ thus induces a following exact sequence,

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathcal{O} \xrightarrow{f \mapsto \exp(2\pi i f)} \mathcal{O}^* \longrightarrow 0$$

From this we get a cohomology sequence,

$$\cdots \longrightarrow H^1(X, \mathcal{O}) \longrightarrow H^1(X, \mathcal{O}^*) \longrightarrow H^2(X, \mathbb{Z}) \longrightarrow H^2(X, \mathcal{O}) \longrightarrow \cdots$$

If we have $H^1(X, \mathcal{O}), H^2(X, \mathcal{O}) = 0$ then $H^1(X, \mathcal{O}^*) \cong H^2(X, \mathbb{Z})$. Then for $H^2(X, \mathbb{Z}) = 0$, that is all divisors are principal.

Remark 7.24. *The above argument can be further generalized. if D is a domain then we have a following map.*

$$D \mapsto \underbrace{\tilde{D}}_{\in H^1(X, \mathcal{O}^*)} \mapsto \underbrace{c(D)}_{\in H^2(X, \mathbb{Z})}.$$

Such $c(D)$ is called the Chern class of D.

Theorem 7.10. *If $H^1(X, \mathcal{O}), H^2(X, \mathcal{O}) = 0$, then precisely the divisors with $c(D) = 0$ are principal.*

Remark 7.25. *For example, above conditions are attained when X is a polydisk.*

Now we will examine extorsion problems. Let M be an analytic set on X and $\mathcal{I}_M$ be the corresponding ideal sheaf of M. Then we have,

$$0 \longrightarrow \mathcal{I}_M \longrightarrow \mathcal{O} \longrightarrow \mathcal{O}_M = \mathcal{O}/\mathcal{I}_M \longrightarrow 0$$

The question is, is the sequence, $\mathcal{O}(X) \longrightarrow \mathcal{O}_M(X) \longrightarrow 0$ exact?

Theorem 7.11. *If $H^1(X, \mathcal{I}_M) = 0$, then holomorphic functions on M are restrictions of holomorphic functions on X.*

Proof. From the sequence,

$$0 \longrightarrow \mathcal{I}_M \longrightarrow \mathcal{O} \longrightarrow \mathcal{O}_M = \mathcal{O}/\mathcal{I}_M \longrightarrow 0$$

, we get a cohomology sequene,

$$\cdots \longrightarrow \mathcal{O}(X) \longrightarrow \mathcal{O}_M(X) \longrightarrow H^1(X, \mathcal{I}_M) \longrightarrow \cdots$$

$\square$

Let $x \in \mathbb{C}^n$ and define and $\mathcal{G} \subseteq \mathbb{C}^n$ be a domain containing the origin. We then define

$$J_{0,x} = \begin{cases} \mathcal{O}_x, & x \neq 0, \end{cases}$$

We then obtain the following exact sequence,

$$0 \longrightarrow \mathcal{K} \longrightarrow \mathcal{O}^n \xrightarrow{(f_1,\cdots,f_n)\mapsto\sum_{j=1}^n f_j z_j} \mathcal{J}_0 \longrightarrow 0$$

The question is for $f \in \mathcal{O}(\mathcal{G})$ with $f(0) = 0$, does there exists $f_j \in \mathcal{O}(\mathcal{G})$ such that

$$\sum f_j z_j = f.$$

This turns out to be true if $H^1(\mathcal{G}, \mathcal{K}) = 0$. Furthermore if $H^q(\mathcal{G}, \mathcal{O}) = 0$ for $q \geq 1$, we also get the answer to be positive.

8 Vanishing of Dolbeault Spaces

Now we turn our interest to solving Cauchy-Riemann equations.

Theorem 8.1. *If $\mathcal{G}$ is pseudoconvex then $(\mathrm{CR})_q$ is solvable for $q \geq 1$. That is for all $f \in \mathcal{C}^\infty_{0,q}(\mathcal{G})$, $\overline{\partial} f = 0$ implies that there is $u \in \mathcal{C}^\infty_{0,q-1}(\mathcal{G})$ such that $\overline{\partial} u = f$.*

This section studies the general solutions of these equations.

8.1 Spaces of forms

Through out this subsection, we will denote $\mathcal{G} \subseteq \mathbb{C}^n$ as a domain and E be some function spaces. We will denote,

Notation 8.1. *E_{pq} will denote a (p,q)-forms with coefficient in E. That is*

$$f = \sum_{\substack{|I|=p \\ |J|q}} f_{I,J} dz^I d\overline{z}^J,$$

where $I \in \{1, \cdots, n\}^p, J \in \{1, \cdots, n\}^q$.

Remark 8.1. *For $I \in \{1, \cdots, n\}^p$, we will denote I' to be the tuple where $I'_1 < \cdots < I'_p$. Then we denote*

$$f_{IJ} = \varepsilon^{IJ}_{I'J'} f_{I'J'},$$

where $\varepsilon^{IJ}_{I'J'}$ is the product of signs of permutations which send $I \to I', J \to J'$. Under this notation, we get

$$f = \frac{1}{p!q!} \sum f_{IJ} dz^I d\overline{z}^J.$$

Definition 8.1. *For one dimensional case $\mathbb{R}$, in order to define a distribution. Let us first consider the set of test functions, that is*

$$\mathcal{D} := \mathcal{C}^\infty_C(\mathbb{R}).$$

A sequence $(\varphi_n)_{n\in\mathbb{N}}$ in $\mathcal{D}$ converges to 0 in $\mathcal{D}$ if $(\varphi_n)_{n\in\mathbb{N}}$ converges uniformly to 0 and $\operatorname{supp}\varphi_n$ is contained in a finite measure interval for any $n \in \mathbb{N}$.

Definition 8.2. *A distribution is a linear continuous map $f : \mathcal{D} \to \mathbb{R}$ where we consider. Thus the space of distributions of $\mathcal{D}$ is precisely the continuous dual $\mathcal{D}'$ of $\mathcal{D}$. The topology on $\mathcal{D}'$ is defined as follows. $(f_n)_{n\in\mathbb{N}}$ converges to 0 in $\mathcal{D}'$ if $f_n(t) \to 0$ for each $t \in \mathcal{D}$.*

Example 8.1. *Any locally integrable function defines a continuous dual. Let f be a locally integrable funciton. Then we define,*

$$\tilde{f}[t] = \int_{-\infty}^{\infty} f(x)t(x)dx.$$

Definition 8.3. *The Dirac δ-distribution at the origin is defined as.*

$$\delta[f] = f(0).$$

Remark 8.2. *Such δ is not induced by any function.*

Definition 8.4. *Let $t \in \mathcal{D}, f \in \mathcal{D}$ then we define,*

$$f'[t] = -f[t'].$$

We refer to [8] for more details about distributions.

Theorem 8.2. *Let $(f_n)_{n\in\mathbb{N}}$ be a sequence in $\mathcal{D}'$ then*

$$f_n \overset{\mathcal{D}'}{\to} f \Rightarrow f'_n \overset{\mathcal{D}'}{\to} f'.$$

Definition 8.5. *Let us take $E = \mathcal{D}$ the set of test functions (more generally, we can take $\mathcal{D}'$ the space of distributions). We define a bilinear form for $f, q \in \mathcal{D}'_{p,q}$ by*

$$\langle f, g\rangle(z) = \sum f_{IJ}(z)\overline{g}_{IJ}(z).$$

Let φ be a continuous function which is everywhere positive on $\mathcal{G}$. We then define a scalar product,

$$(f, g)_\varphi := \int_{\mathcal{G}} \langle f, g\rangle(z) e^{-\varphi(z)} d\lambda(z).$$

We will denote,

$$\|f\|_\varphi^2 = (f, f)_\varphi.$$

We call $(\cdot, \cdot)_\varphi$ a scalar product with respect to the weight function φ.

Definition 8.6. *We denote $L^2_{p,q,\mathrm{loc}}(\mathcal{G})$ to be the set of locally square integrable functions/forms on $\mathcal{G}$. We also define,*

$$L^2_{p,q}(\mathcal{G}, \varphi) = \{f \in \mathcal{D}'_{p,q} \mid \|f\|_\varphi < \infty\}.$$

Remark 8.3. *$(L^2_{p,q}(\mathcal{G}, \varphi), (\cdot, \cdot)_\varphi)$ is a Hilbert space and we have,*

$$L^2_{p,q}(\mathcal{G}, \varphi) \subseteq L^2_{p,q,\mathrm{loc}}(\mathcal{G}).$$

Proposition 8.1. *$\mathcal{D}_{p,q}$ is dense in $L^2_{p,q}(\mathcal{G}, \varphi)$.*

Lemma 8.1. *Let $f \in L^2_{p,q,\mathrm{loc}}(\mathcal{G})$, then there is a weight function φ, such that $f \in L^2_{p,q}(\mathcal{G}, \varphi)$.*

Proof. Choose φ_0 such that $1 \in L^2(\mathcal{G}, \varphi_0)$. That is

$$\int_{\mathcal{G}} e^{-\varphi(z)} \lambda(z) < \infty.$$

Let us take φ such that

$$\varphi \geq \varphi_0 + \log |f|^2.$$

Then

$$\|f\|_\varphi^2 = \int_{\mathcal{G}} |f|^2 e^{-\varphi} d\lambda \leq \int_{\mathcal{G}} |f|^2 e^{-\varphi_0 - \log |f|^2} d\lambda \leq \|1\|_{\varphi_0} < \infty.$$

□

Let $f \in L^2_{p,q}(\mathcal{G}, \varphi)$ then $\overline{\partial} f \in \mathcal{D}_{p,q+1}(\mathcal{G}, \varphi)$. Let φ be a weight such that

$$L^2_{p,q+1}(\mathcal{G}, \varphi) \subseteq \mathcal{D}'_{p,q+1}.$$

If $\overline{\partial} f \in L^2_{p,q+1}(\mathcal{G}, \varphi)$, then we define,

$$T : L^2_{p,q}(\mathcal{G}, \varphi) \to L^2_{p,q+1}(\mathcal{G}, \varphi), f \mapsto \overline{\partial} f.$$

We have,

$$\operatorname{Dom} T = \{f \mid \overline{\partial} f = Tf \in L^2_{p,q}(\mathcal{G}, \varphi)\} \subseteq \mathcal{D}_{p,q}.$$

T is densely defined.

Lemma 8.2. *Such T is a closed operator that is the graph $\Gamma_T \subseteq L^2_{p,q}(\mathcal{G}, \varphi) \times L^2_{p,q+1}(\mathcal{G}, \varphi)$ is closed.*

Proof. Let $(f_n)_{n\in\mathbb{N}}$ be a sequence in $\operatorname{Dom} T$ such that $(f_n, Tf_n) \xrightarrow{L^2} (f, g)$. We want to show that $f \in \operatorname{Dom} T$ and $g = Tf$. Observe that L^2-convergence implies $\mathcal{D}'$ convergence. Therefore,

$$Tf_n = \overline{\partial} f_n \to \overline{\partial} f.$$

Therefore, we conclude $f \in \operatorname{Dom} T, g = \overline{\partial} f = Tf$. □

Proposition 8.2. *T is closed densely defined linear operator between Hilbert spaces.*

Definition 8.7. *We define the null space $\mathcal{N}_T$ of T to be*

$$\mathcal{N}_T := \{x \in \operatorname{Dom} T \mid Tx = 0\}.$$

And the range $\mathcal{R}_T$ of T to be

$$\mathcal{R}_T := \{Tx \mid x \in \operatorname{Dom} T\}.$$

Remark 8.4. *$\mathcal{N}_T$ is closed but $\mathcal{R}_T$ is not necessarily closed.*

Let $\varphi_1, \varphi_2, \varphi_3$ be weight functions. Consider

$$L^2_{p,q}(\mathcal{G}, \varphi_1) \xrightarrow{T} L^2_{p,q+1}(\mathcal{G}, \varphi_2) \xrightarrow{S} L^2_{p,q+2}(\mathcal{G}, \varphi_3)$$

where T, S are both defined by $\overline{\partial}$. We have $\mathcal{N}_S$ is closed, and $\overline{\partial} \circ \overline{\partial} = 0$, therefore

$$\mathcal{R}_T \subseteq \mathcal{N}_S.$$

The natural question to ask is when do we have an equality to the above inclusion. This question leads us to find general solutions of $(\mathrm{CR})_{p,q+1}$.

8.2 Unbounded Operators in Hilbert Spaces

We first recall the definition of linear operators between Hilbert spaces.

Definition 8.8. *Let H_1, H_2 be Hilbert spaces. A linear operator is a linear map $T : H_1 \to H_2$ which is defined on some subspace $\operatorname{Dom} T$ of H_1. (Thus T need not be defined in the whole space.) Similarly as before, $\mathcal{R}_T$ is defined as the range of T which is a subspace of H_2.*

Definition 8.9. *A linear operator $T : H_1 \to H_2$ is densely defined if $\operatorname{Dom} T$ is dense in H_1. T is closed if the graph,*

$$\Gamma_T := \{(f, Tf) \mid f \in \operatorname{Dom} T\}$$

is closed in $H_1 \times H_2$.

Proposition 8.3. *If $T : H_1 \to H_2$ is closed then $\mathcal{N}_T$ is closed in H_1.*

Proposition 8.4. *Suppose $T : (H_1, \langle\cdot,\cdot\rangle_1) \to (H_2, \langle\cdot,\cdot\rangle_2)$ be a linear operator which is densely defined. Let $x \in \operatorname{Dom} T, y \in H_2$. Let us define,*

$$l_y(x) \to \langle Tx, y\rangle_2.$$

If l_y is bounded then l_y extends to H_1. Using Riesz representation theorem, we obtain a linear map $T^* : H_2 \to H_1$ such that

$$l_y(x) = \langle Tx, y\rangle_2 = \langle x, T^*y\rangle_1.$$

We have,

$$\operatorname{Dom} T^* = \{y \in H_2 \mid \exists c \,\text{s. t.}\, |\langle Tx, y\rangle_2| \leq c\|x\|_1\}.$$

Definition 8.10. *Let $T : H_1 \to H_2$ be a linear operator. The adjoint operator $T^* : H_2 \to H1$ of T is a linear operator such that*

$$\langle Tx, y\rangle_2 = \langle x, T^*y\rangle.$$

Proposition 8.5. *Let $T : H_1 \to H_2$ be a densely defined closed linear operator. Then its adjoint $T^* : H_2 \to H_1$ is also densely defined and closed. Moreover $T^{**} = T$. Furthermore, in this case, we have,*

$$H_1 = \mathcal{N}_T \oplus \overline{\mathcal{R}_{T^*}}, H_2 = \mathcal{N}_{T^*} \oplus \overline{\mathcal{R}_T}.$$

Proof. Left as an exercise to the readers. □

The question we would like to study is what are the conditions for

$$\mathcal{R}_T = \overline{\mathcal{R}_T}, \mathcal{R}_{T^*} = \overline{\mathcal{R}_{T^*}},$$

that is when the range of either T or T^* closed?

Theorem 8.3. *Let $T : (H_1, \langle\cdot,\cdot\rangle_1) \to (H_2, \langle\cdot,\cdot\rangle_2)$ be a densely defined closed operator such that $\mathcal{R}_T \subseteq F$ for some closed subspace F. We have the following are equivalent.*

1). $\mathcal{R}_T = F$,

2). $\exists c > 0$ such that for all $y \in \operatorname{Dom} T^* \cap F$,

$$\|y\|_2 \leq C\|T^*y\|_1.$$

If $\mathcal{R}_T$ is closed then we have,

3). T^* has its range closed.

4). $\forall x \in \mathcal{R}_{T^*}$ there is $y \in \operatorname{Dom} T^*$ such that

$$\|y\|_2 \leq c\|x\|_1.$$

Proof. The complete proof can be found in [3] 4.1.1 and 4.1.2. □

8.3 The Basic Estimates

Definition 8.11. *Let $\mathcal{G} \subseteq \mathbb{C}^n$ be a domain and $\varphi_1, \varphi_2, \varphi_3 \in \mathcal{C}_c^\infty(\mathcal{G})$ be everywhere positive test functions. Consider the sequence*

$$L^2_{p,q}(\mathcal{G}, \varphi_1) \xrightarrow{T} L^2_{p,q+1}(\mathcal{G}, \varphi_2) \xrightarrow{S} L^2_{p,q+2}(\mathcal{G}, \varphi_3)$$

We say the basic estimate holds for the sequence if there is a constant $C > 0$ such that for all $f \in \operatorname{Dom} T^* \cap \operatorname{Dom} S$, we have

$$\|f\|^2_{\varphi_2} \leq C^2(\|T^* f\|^2_{\varphi_1} + \|Sf\|^2_{\varphi_3}).$$

Proposition 8.6. *Suppose the basic estimate holds, then for $f \in \mathcal{N}_S$, we have,*

$$\|f\|^2_{\varphi_2} \leq C^2 \|T^* f\|^2_{\varphi_1}.$$

Theorem 8.4. *If the basic estimate holds then $\mathcal{R}_T = \mathcal{N}_S$. That is for any $f \in \mathcal{N}_S$ there is $u \in L^2_{p,q}(\varphi_1)$ such that $Tu = f$.*

Proof. Follows from Proposition 8.6 together with Theorem 8.3. □

Remark 8.5. *Theorem 8.4 tells us that this solves $(\mathrm{CR})_{p,q+1}$ in the sense of distributions.*

Remark 8.6. *In the case of constants, Definition 8.11 says, for $a, b, c > 0$ we have,*

$$c \leq a + b \Rightarrow c^2 \leq 2(a^2 + b^2),$$
$$c^2 \leq a^2 + b^2 \Rightarrow c \leq a + b.$$

We now move on to find conditions for the basic estimate. Let φ, ψ be everywhere positive test functions. Set,

$$\varphi_1 = \varphi - 2\psi,$$
$$\varphi_2 = \varphi - \psi,$$
$$\varphi_3 = \varphi$$

Let us impose some conditions on ψ. Consider a chain of smooth functions

$$\eta_1 \leq \eta_2 \leq \cdots,$$

such that $\eta_i \in [0, 1]$ for each $i \in \mathbb{N}$ and for any compact set K, there is $i \in \mathbb{N}$ such that

$$K \subseteq \{\eta \equiv 1\}.$$

Definition 8.12. *ψ satisfies the following equation,*

$$|\partial\eta_n|^2 \leq e^{\psi}, \tag{H}$$

then we say ψ satisfies (H)*.*

Lemma 8.3. (H) *can be fulfilled.*

Proof. Let K be a compact subset of $\mathcal{G}$. For ψ to satisfy finitely many (construction)? Suppose $K \subset K'$ is also compact then ψ can be extended without changing on K to satisfy (H). $\square$

Definition 8.13. *φ, ψ be given. If the equation,*

$$L_\varphi(z;t) \geq 2(|\partial\psi(z)|^2 + e^{\psi(z)})\|t\|^2, \tag{E}$$

where L_φ is the Levi form is satisfied, we say φ satisfies condition (E) *with respect to ψ.*

Remark 8.7. *If φ satisfies* (E)*, then it is necessarily strictly plurisubharmonic. Furthermore, it admits plurisubharmonic smooth exhaustive functions. Therefore $\mathcal{G}$ is necessarily pseudoconvex. This follows from Theorem 5.11*

Theorem 8.5 (Hörmander)**.** *Let φ, ψ be such that together they satisfy* (H) *and* (E)*. Let us put,*

$$\begin{aligned} \varphi_1 &= \varphi - 2\psi, \\ \varphi_2 &= \varphi - \psi, \\ \varphi_3 &= \varphi. \end{aligned} \tag{D}$$

Then the basic estimates hold with constant $C = 1$. That is we have,

$$\|f\|^2_{\varphi_2} \leq \|T^* f\|^2_{\varphi_1} + \|Sf\|^2_{\varphi_3}.$$

Theorem 8.6 (Friedrich/Hörmander)**.** *Suppose ψ satisfies* (H) *and $\varphi_1, \varphi_2, \varphi_3$ are given as in Equation* (D)*. Then $\mathcal{D}_{p,q+1}$ is dense in $\operatorname{Dom} T^* \cap \operatorname{Dom} S$.*

Moreover, $f \in \operatorname{Dom} T^* \cap \operatorname{Dom} S$ then there is a sequence $(f_j)_{j\in\mathbb{N}}$ in $\mathcal{D}_{p,q+1}$ such that

$$\begin{aligned} f_j &\xrightarrow{L^2} f, \\ \overline{\partial} f_j &\xrightarrow{L^2} Sf, \\ T^* j &\xrightarrow{L^2} T^* f. \end{aligned}$$

Proof. The complete proof can be found in [3] 4.1.3. $\square$

Theorem 8.7. *Let ψ, φ satisfy* (H) *and* (E)*. Suppose $\varphi_1, \varphi_2, \varphi_3$ are given as in Equation* (D)*. Then, for each $f \in \mathcal{D}_{p,q+1}$, we have*

$$\|f\|^2_{\varphi_2} \leq \|T^* f\|_{\varphi_1} + \|Sf\|_{\varphi_3}.$$

That is, it satisfies the basic estimate.

8.4 Proof of Basic Estimate formel

Since $\mathbb{C}^n$ admits a global coordinate therefore we can assume $p = 0$ without the loss of genearlity.

$$(\mathcal{D}_{0,q}, \langle \cdot, \cdot \rangle_{\varphi_1}) \underset{T^*}{\overset{T=\overline{\partial}}{\rightleftarrows}} (\mathcal{D}_{0,q+1}, \langle \cdot, \cdot \rangle_{\varphi_2})$$

Compute T^* explicitly that is, for $fin\mathcal{D}_{0,q+1}$ such that

$$f = \sum_{|J|=q+1} f_J d\overline{z}^J.$$

Then consider,

$$u = \sum_{|K|=q} u_K d\overline{z}^K \in \mathcal{D}_{0,q}.$$

We want

$$(T^* f, u)_{\varphi_1} = (f, \overline{\partial} u)_{\varphi_2}.$$

Write,

$$T^* f = \sum_{|K|=q} (T^* f)_K d\overline{z}^K.$$

Note that we have,

$$\overline{\partial} u = \sum_K \sum_{j=1}^n \frac{\partial u_K}{\partial \overline{z}_j} d\overline{z}_j \wedge d\overline{z}^K.$$

Using the equation above, we have,

$$\begin{aligned}
(T^* f, u)_{\varphi_1} &= \int \sum_K (T^* f)_K \overline{u}_K e^{-\varphi_1} d\lambda. \\
(f, \overline{\partial} u) &= \int \sum_K \sum_{j=1}^n \frac{\partial u_K}{\partial \overline{z}_j} f_{(j,K)} d\overline{z}_j \wedge d\overline{z}^K e^{-\varphi_2} d\lambda, \\
&= - \int \sum_K \sum_j \frac{\partial (f_{(j,K)} e^{-\varphi_1})}{\partial z_j} \overline{u}_K e^{\varphi_1} e^{-\varphi_2} d\lambda.
\end{aligned}$$

Compute $T^* f = e^{\varphi_1} \sum_{K,j} \frac{\partial (f_{(j,K)} e^{-\varphi_1})}{\partial z_j} d\overline{z}^K$. We set,

$$e^{\varphi_1} (\theta (e^{-\varphi_1} f)) := e^{\varphi_1} \sum_{K,j} \frac{\partial (f_{(j,K)} e^{-\varphi_1})}{\partial z_j} d\overline{z}^K.$$

Compute $|\overline{\partial} f|^2$. Note we have,

$$\overline{\partial} f = \sum_{|J|=q+1} \frac{\partial f_j}{\partial \overline{z}_j} d\overline{z}_j \wedge d\overline{z}^J.$$

Therefore,

$$\begin{aligned}
|\overline{\partial} f|^2 &= \langle \overline{\partial} f, \overline{\partial} f \rangle, \\
&= \sum_{|J|=q+1} \sum_{|L|=q+1} \sum_{j} \sum_{l} \frac{\partial f_J}{\partial \overline{z}_j} \overline{\frac{\partial f_L}{\partial \overline{z}_l}} \cdot \varepsilon^{jJ}_{lL}.
\end{aligned}$$

where,

$$\varepsilon^{jJ}_{lL} = \begin{cases} 0, & j \in J, \\ 0, & l \in L, \\ 0, & (j,J) \neq (l,L) \text{ as sets}, \\ \operatorname{sgn} \binom{jJ}{lL}. \end{cases}$$

Thus decompose into two signs, where $j = l$ or $\varepsilon^{jJ}_{lL} = 1$. For the first case,

$$\sum_{j,J} \left| \frac{\partial f_j}{\partial \overline{z}_j} \right|^2.$$

For $j \neq l$, we have $j \in L$ and $l \in J$, set $L \backslash \{j\} = J \backslash \{l\} := K$. Then,

$$\sum_{J,L} \sum_{j \neq l} \frac{\partial f_J}{\partial \overline{z}_j} \overline{\frac{\partial f_L}{\partial \overline{z}_l}} \varepsilon^{j}_{lL} J.$$

Observe that we have,

$$\varepsilon^{jJ}_{lL} = \varepsilon^{jJ}_{jlK} \varepsilon^{jlK}_{ljK} \varepsilon^{ljK}_{lL} = -\varepsilon^{jJ}_{jlK} \varepsilon^{ljK}_{lL} = -\varepsilon^{J}_{lK} \varepsilon^{jK}_{L} = -1.$$

Thus,

$$-\sum_{|K|q} \sum_{j \neq l} \frac{\partial f_{lK}}{\partial \overline{z}_j} \overline{\frac{\partial f_{jK}}{\partial \overline{z}_l}}.$$

Combining two cases together we obtain,

$$|\overline{\partial} f|^2 = \sum_{J,j} \left| \frac{\partial f_J}{\partial \overline{z}_j} \right| - \sum_{|K|=q} \sum_{j,k} \frac{\partial f_{jK}}{\partial \overline{z}_k} \overline{\frac{\partial f_{kK}}{\partial \overline{z}_j}}.$$

Choose $\varphi, \psi > 0$ be smooth. Choose $\varphi_1, \varphi_2, \varphi_3$ be such that

$$\begin{aligned}
\varphi_1 &= \varphi - 2\psi, \\
\varphi_2 &= \varphi - \psi, \\
\varphi_3 &= \varphi.
\end{aligned} \tag{D}$$

And set ψ so that it satisfies (H), that is there is a sequence $(\eta_n)_{n \in \mathbb{N}}$ such that $\eta_n(\mathcal{G}) \subseteq [0,1]$, and for any compact set there is $n \in \mathbb{N}$ such that

$$K \subseteq \{\eta_n = 1\}.$$

And ψ satisfies

$$|\partial\eta_n|^2 \leq e^{\psi}. \tag{H}$$

Let v be a function on $\mathcal{G}$ and φ be a weight function,

$$\begin{aligned}\delta_j v &= e^{\varphi}\frac{\partial}{\partial z_j}(e^{-\varphi}v),\\ &= \frac{\partial v}{\partial z_j} - \frac{\partial \varphi}{\partial z_j}v.\end{aligned}$$

Insert δ in T^*f to obtain,

$$e^{\varphi}T^*f = -\sum_{|K|=q}\left(\sum_j \delta_j f_{jK} + \frac{\partial\psi}{\partial z_j}f_{jK}\right)d\overline{z}^K.$$

Now, we compute the following scalar product. All integrals are now over $\mathbb{C}^n$ and have compact support in $\mathcal{G}$.

$$\begin{aligned}\int\sum_K\sum_{j,k}\delta_j f_{jK}\overline{\delta_k f_{kK}}e^{-\varphi}d\lambda &= \int\left|e^{\psi}T^*f + \sum\frac{\partial\psi}{\partial z_j}f_{jK}\right|^2 e^{-\varphi}d\lambda,\\ &\leq 2\int e^{2\psi}|T^*f|^2e^{-\varphi}d\lambda + 2\int|\partial\psi|^2|f|^2e^{-\varphi}d\lambda\end{aligned}$$

Now add $\int|\partial f|^2e^{-\varphi}d\lambda$ to both sides and thi yields,

$$\begin{aligned}\int\sum_K\left(\sum_{j,k}\delta_j f_{jK}\overline{\delta_k f_{kK}} - \frac{\partial f_{jK}}{\partial\overline{z}_k}\overline{\frac{\partial f_{kK}}{\partial\overline{z}_j}}\right)e^{-\varphi}d\lambda + \int\sum_J\sum_j\left|\frac{\partial f_J}{\partial\overline{z}_j}\right|e^{-\varphi}d\lambda, \quad (*)\\ \leq \|T^*f\|^2_{\varphi_1} + \|Sf\|^2_{\varphi_2} + 2\int|\partial\psi|^2|f|^2e^{-\varphi}d\lambda.\end{aligned}$$

Lemma 8.4. *Let $v_1, v_2 \in \mathcal{D}_0$ we have,*

$$\int v_1\overline{\frac{\partial}{\partial\overline{z}_k}v_2}e^{-\varphi}d\lambda = -\int\delta_k v_1\overline{v_2}e^{-\varphi}d\lambda.$$

Moreover, we have,

$$\delta_j\frac{\partial}{\partial\overline{z}_k} - \frac{\partial}{\partial\overline{z}_k}\delta_j = \frac{\partial^2\varphi}{\partial z_j\partial\overline{z}_k}.$$

Using the lemma above, Equation $(*)$, becomes

$$-\int\sum_K\sum_{j,k}\left(\frac{\partial}{\partial\overline{z}_k}\delta_j f_{jK} - \delta_j\frac{\partial f_{jK}}{\partial\overline{z}_k}\right)\overline{f}_{kK}e^{-\varphi}d\lambda = \int\sum_K\sum_{j,k}\frac{\partial^2\varphi}{\partial z_j\partial\overline{z}_k}f_{jK}\overline{f_{kK}}e^{-\varphi}d\lambda.$$

Now this leads us to

$$\int \sum_K \sum_{j,k} \frac{\partial^2 \varphi}{\partial z_j \partial \overline{z}_k} f_{jK} \overline{f}_{kK} e^{-\varphi} d\lambda + \int \sum_J \sum_j \left| \frac{\partial f_j}{\partial \overline{z}_j} \right|^2 e^{-\varphi} d\lambda \qquad (**)$$

$$\leq 2\|T^* f\|_{\varphi_1}^2 + \|Sf\|_{\varphi_3}^2 + 2\int |f|^2 |\partial \psi|^2 e^{-\varphi} d\lambda$$

In particular for $q = 0$, we have,

$$\sum_{|J|=q+1} \left| \frac{\partial f_j}{\partial \overline{z}_j} \right|^2 \leq \|T^* f\|^2 + \|Sf\|^2.$$

Let φ be strictly plurisubharmonic, then

$$L_\varphi(z;t) \geq c(z)\|t\|^2,$$

where c is a positive function on $\mathcal{G}$.

$$\int (c(z) - |\partial \psi(z)|^2)|f|^2 e^{-\varphi} d\lambda \leq 2\|T^* f\|_{\varphi_1}^2 + \|Sf\|_{\varphi_3}^2. \qquad (***)$$

Finally introduce the condition (E) on c, that is

$$c(z) > 2(|\partial \psi(z)|^2 + e^{\psi(z)}) \Leftrightarrow c(z) - 2|\partial \psi|^2 > 2e^{\psi}.$$

Then Equation $(***)$ is now,

$$2\|f\|_{\varphi_2}^2 \leq 2\|T^* f\|_{\varphi_1}^2 + \|Sf\|_{\varphi_3}^2.$$

Thus we conclude,

$$\|f\|_{\varphi_2}^2 \leq \|T^* f\|_{\varphi_1}^2 + \|Sf\|_{\varphi_3}^2.$$

Proposition 8.7. *Suppose φ, ψ together satisfy conditions* (H),(D),(E), *and the triplet $(\varphi_1, \varphi_2, \varphi_3)$ is defined as in* (D), *then the basic estimate holds in $\mathcal{D}_{0,q}$. Namely,*

$$\|f\|_{\varphi_2}^2 \leq \|T^* f\|_{\varphi_1}^2 + \|Sf\|_{\varphi_3}^2.$$

Theorem 8.8. (H),(D),(E) *yield the basic estimate in* $\operatorname{Dom} T^* \cap \operatorname{Dom} S$.

Remark 8.8. *The theorem tells us that $\mathcal{G}$ must be pseudoconvex.*

8.5 Solving (CR)

Theorem 8.9. *Let $f \in L^2_{0,q+1,\mathrm{loc}}(\mathcal{G})$ where $\mathcal{G}$ is pseudoconvex. Suppose*

$$\overline{\partial} f = 0.$$

Then there is $u \in L^2_{0,q,\mathrm{loc}}(\mathcal{G})$ with

$$\overline{\partial} u = f.$$

Proof. Will be shown in the coming lecture on this friday. Roughly, we will show there are $(\varphi_1, \varphi_2, \varphi_3)$ such that $f \in L^{0,q,\text{loc}}(\mathcal{G}, \varphi_2)$ and the basic estimate holds. By the previous subsection, we know $f \in \mathcal{R}_T$ that is $f = \overline{\partial} u$ for some u. Let p be strictly plurisubharmonic exhaustion function such that

$$L_p(z;t) \geq m(z)\|t\|,$$

is continuous positive function. Let $\chi : \mathbb{R} \to \mathbb{R}$ be such that

$$\begin{aligned} &s \geq 0 \Rightarrow \chi(s) \equiv 0, \\ &s > 0 \Rightarrow \chi(s) > 0, \chi'(s) > 0, \chi''(s) > 0 \end{aligned}$$

We then set $\varphi = \chi \circ p$. Then,

$$L_\varphi(z;t) \geq \chi'(p(z))m(z)\|t\|^2.$$

Choose ψ at to satisfy (H). To satisfy (E), we need,

$$L_\varphi(z;t) \geq \chi'(p(z))m(z)\|t\|^2 \geq (|\partial\psi|^2 + e^{\psi})\|t\|^2.$$

Note that

$$\chi(p(z))m(z) \geq |\partial\psi|^2 + e^{\psi(z)}, \chi'(p(t)) \geq \frac{|\partial\psi|^2 + e^{\psi(z)}}{m(z)}.$$

We then consider the following condition.

$$\chi'(s) \geq \sup_{p(z)\leq s} \frac{|\partial\psi(s)|^2 + e^{\psi(z)}}{m(z)}. \tag{E'}$$

We then have (E') implies (E). χ' can be found that $\chi'' > 0, \chi > 0$. We cann still enlarge φ and free p (H),(**??**),(E). Thus

f given $(\overline{\partial} f = 0)$, there are φ, ψ so that $f \in L^2_{0,q+1}(\mathcal{G}, \varphi_2)$ where $\varphi_2 = \varphi - \psi$. Thus the basic estimate holds and $f \in \mathcal{R}_T$ by the previous subsection(be more specific). Thus there is $u \in L^2_{0,q}(\mathcal{G}, \varphi_1)$ with $\overline{\partial} u = f$. □

Definition 8.14. *Let $s = 0, 1, \cdots$. Then*

$$W^s := \{f \in L^2(\mathbb{C}^n) \mid \textit{all derivatives of order at most } s \textit{ are in } L^2(\mathbb{C}^n)\}.$$

And

$$W^s_{\text{loc}}(\mathcal{G}) := \left\{ f \mid {\textit{All derivatives of } f \textit{ of order less than } s \textit{ are in } L^2(K) \atop \textit{where } K \textit{ runs through all compacts sets of } \mathcal{G}} \right\}.$$

Furthermore,

$$W^s_{0,q,\text{loc}}(\mathcal{G})$$

be $(0,q)$-forms with coefficients in $W^s_{\text{loc}}(\mathcal{G})$. Note that $W^0_{0,q,\text{loc}}(\mathcal{G}) = L^2_{0,q,\text{loc}}(\mathcal{G})$.

Theorem 8.10.
$$W^{s+2n}_{\text{loc}}(\mathcal{G}) \subseteq \mathcal{C}^s(\mathcal{G}).$$
where $\mathcal{G} \subseteq \mathbb{C}^n$.

Proof. See any functional analysis book. $\square$

Definition 8.15. *Consider*
$$f = \sum_{|J|=q+1} f_J d\overline{z}^J,$$

then we define,
$$\vartheta f = \sum_{|K|=q} \left(\sum_j \frac{\partial f_j}{\partial z_j} \right) d\overline{z}^K.$$
Note that $\vartheta : \mathcal{D}'_{0,q+1} \to \mathcal{D}'_{0,q}$.

Remark 8.9. *Check that* $\vartheta\vartheta = 0$. *(Exercise)*

Definition 8.16. *A function* $\chi : \mathbb{C}^n \to \mathbb{R}$ *is called radial is for any* $z, z' \in \mathbb{C}^n$,
$$\|z\| = \|z'\| \Rightarrow \chi(z) = \chi(z').$$

Lemma 8.5. *Let* $f \in L^2_{0,q+1}(\mathcal{G})$ *with compact support. Suppose* $\overline{\partial} f$ *and* ϑf *belong to* L^2. *Then* $f \in W^1_{0,q+1,\text{loc}}(\mathcal{G})$.

Proof. Let $f \in \mathcal{D}_{0,q+1}$. Recall $(**)$,
$$\int \sum_{|J|=q+1} \sum_j \left| \frac{\partial f_j}{\partial \overline{z}_j} \right| d\lambda \leq (\|\overline{\partial} f\|^2 + \|\vartheta f\|^2).$$

So the $\overline{z}$-derivatives are uniformly estimated.

In general regular regularise. That is, we consider a smooth radial function χ
$$\chi(z) \geq 0, \int_{\mathbb{C}^n} \chi(z) d\lambda(z) = 1.$$
Set,
$$\chi_\varepsilon(z) = \left(\frac{1}{\varepsilon^{2n}} \right) \chi\left(\frac{z}{\varepsilon} \right).$$
Then we set,
$$f_\varepsilon(z) := f * \chi_\varepsilon(z) = \int f(\zeta)\chi_\varepsilon(\zeta - z) d\lambda(\zeta) = \int f(z - \varepsilon\zeta)\chi(\zeta) d\lambda(\zeta).$$

Then f_ε is smooth and approximate f in L^2-norm uniformly and furthermore, the convergence commutes with $\overline{\partial}$. $\square$

Lemma 8.6. *Let $f \in L^2_{0,0}, \overline{\partial} f \in L^2_{0,1}$ with compact support. Then we also have $\partial f \in L^2_{1,0}$.*

Proof. Assume $f \in \mathcal{D}_{0,0}$.

$$\int \left| \frac{\partial f}{\partial \overline{z}_j} \right| d\lambda(z) = -\int \frac{\partial^2 f}{\partial \overline{z}_j \partial z_j} \overline{f}(z) d\lambda(z) = \int \left| \frac{\partial f}{\partial z_j} \right|^2 d\lambda(z).$$

FOr the general case, regularize as above. $\square$

Theorem 8.11. *Let $\mathcal{G}$ be pseudoconvex. Let $f \in W^s_{0,q+1,\mathrm{loc}}(\mathcal{G})$ with $\overline{\partial} f = 0$. Then there is $u \in W^{s+1}_{0,q,\mathrm{loc}}(\mathcal{G})$ with $\overline{\partial} u = f$. For $q = 0$, all solutions are $W^{s+1}_{0,q,\mathrm{loc}}(\mathcal{G})$.*

Proof. For $q = 0$, we have some solution $\overline{\partial} u = f$. And $u \in L^2_{0,q,\mathrm{loc}}(\mathcal{G}) = W^0_{0,q}(\mathcal{G})$. Choose a smooth function χ with compact support. Then we examine, χu.

Assume $u \in W^\sigma$ whre $0 \leq \sigma \leq s$. We already now that u belongs to $\sigma = 0$. Then consider,

$$\frac{\partial}{\partial \overline{z}_j} \chi u = \frac{\partial \chi}{\partial \overline{z}_j} u + \chi \frac{\partial u}{\partial \overline{z}_j} \in W^\sigma.$$

Let D be a differential operator with constant coefficient of order σ. Then,

$$\frac{\partial}{\partial \overline{z}_j} D \chi u = D \underbrace{\frac{\partial}{\partial \overline{z}_j} \chi u}_{\in W^\sigma} \in W^0$$

By Lemma 8.6, $\frac{\partial}{\partial z_j} D \chi u \in W^0$. Thus $\chi u \in W^{\sigma+1}$ Therefore, $u \in W^{\sigma+1}_{\log}(\mathcal{G})$.

For $q > 0$. We use $H_1 \overset{T,T^*}{\leftrightarrow} H_2$. We know,

$$H_1 = \mathcal{N}_T \oplus \mathcal{R}_{T^*}, H_2 = \mathcal{N}_{T^*} \oplus \mathcal{R}_T.$$

Let $f \in \mathcal{R}_T$ then there is $u \in H_1$ such that $Tu = f$. Set

$$u = u_0 + u_1,$$

where $Tu_1 = f$ and $u_0 \in \mathcal{N}_T$. Thus $u \in \mathcal{R}_{T^*}$ therefore, there is v such that $u_1 = T^* v$. We then have two equations,

$$Tu_1 = f, u_1 = T^* v.$$

In our case, we want to find u with

$$\overline{\partial} u = f, u = T^* v.$$

$$\begin{aligned}
\vartheta(e^{-\varphi_1} u) &= \vartheta(e^{-\varphi_1} T^* v), \\
&= \vartheta(e^{-\varphi_1} e^{\varphi_1} \vartheta(e^{-\varphi_2} v)), \\
&= \vartheta \circ \vartheta(e^{-\varphi_2} v) = 0. \\
\vartheta(e^{-\varphi_1} u) &= e^{-\varphi_1} \vartheta u + Au.
\end{aligned}$$

where A is a differential operator of order 0. Let $f \in W^s$ assume $u \in W^\sigma$ for some $0 \le \sigma \le s$. We will show that $u \in W^{\sigma+1}$. Let χ be as above case when $q = 0$.

$$\begin{aligned}\overline{\partial}(\chi u) &= \overline{\partial}\chi a u + \chi\overline{\partial}u \in W^\sigma,\\ \vartheta(\chi u) &= \chi\vartheta u + bu = \chi a u + bu \in W^\sigma.\end{aligned}$$

Continue as for $q = 0$, and for a differential operator D of order σ, we have,

$$\begin{aligned}\overline{\partial}D(\chi u) &= D\overline{\partial}\chi u \in W^0,\\ \vartheta D(\chi u) &= D\vartheta\chi u \in W^0.\end{aligned}$$

By Lemma 8.5, we have $D\chi u \in W^1$ thus $\chi u \in W^{\sigma+1}$. □

Theorem 8.12. *$\mathcal{G}$ is pseudoconvex and $f \in \mathcal{C}^\infty_{0,q+1}(\mathcal{G})$ with $\overline{\partial}f = 0$.*

1). for $q = 0$, each solution $\overline{\partial}u = f$ is smooth,

2). for $q > 0$, there is some $u \in \mathcal{C}^\infty_{0,q}$ with $\overline{\partial}u = f$.

Proof. $f \in W^s_{\text{loc}}(\mathcal{G})$ for all $s > 2n$. Find $u \in W^{s+1}_{\text{loc}}(\mathcal{G})$. FoR $u \in W^{s+1} \subseteq \mathcal{C}^{s+1-2n}$. u is uniformly differentiable and $u \in \mathcal{C}^\infty_{0,q}(\mathcal{G})$. □

Remark 8.10. *$\overline{\partial}$ is an elliptic operator on functions. $\overline{\partial}$ is not elliptic for $q > 0$. In that case a smooth solutions will be singled out by an adjoint operator (T^* essentially θ), $\overline{\partial} \oplus \theta$ is elliptic.*

8.6 Applications

Definition 8.17. *Let $f \in \mathcal{C}^\infty_{p,0}(\mathcal{G})$ with $\overline{\partial}f = 0$. The p-form is holomorphic it can be written as*

$$f = \sum_{\substack{\alpha\in\mathbb{N}^n\\ |\alpha|=p}} f_\alpha dz^\alpha,$$

where f_α are all holomorphic. We write,

$$\Omega^p = \{\textit{germs of holomorphic p-forms}\}.$$

Remark 8.11.

$$\Omega^0 = \mathcal{O}, \Omega^p \cong \mathcal{O}^{\binom{n}{p}}.$$

Theorem 8.13. *For $\mathcal{G}$ a domain in $\mathbb{C}^n$, the following are equivalent.*

1). $\mathcal{G}$ is pseudoconvex.

2). $\mathcal{G}$ is holomorphically convex.

3). $\mathcal{G}$ is a domain of holomorphy.

4). $\mathcal{G}$ is a domain of existence.

5). $H^{p,q}_{\mathrm{Dol}}(\mathcal{G}) = 0$ for all $q \geq 1$.

6). $H^q(\mathcal{G}, \Omega^p) = 0$ for all $q \geq 1$, and solves the Levi problems.

Theorem 8.14. *The first Cousin problem is solvable in a domain of holomorphy. Namely,*

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{M} \longrightarrow \mathcal{K} = \mathcal{M}/\mathcal{O} \longrightarrow 0$$

$$0 \longrightarrow \mathcal{O}(\mathcal{G}) \longrightarrow \mathcal{M}(\mathcal{G}) \longrightarrow \mathcal{K}(\mathcal{G}) \longrightarrow H^1(\mathcal{G}, \mathcal{O}) = 0? \longrightarrow \cdots$$

Consider the second Cousin problem. That is let $\mathcal{G}$ be a domain and $\mathscr{U} = \{U_i \mid i \in I\}$ be an open covering. Consider,

$$f_i \in \mathcal{M}^*(U_i), f_i/f_j = h_{ji} \in \mathcal{O}^*(U_{ij}).$$

Cousin II distribution ion is a solution to the second Cousin problem which is a element $f \in \mathcal{M}^*(\mathcal{G})$ such that

$$f/f_i = h_i \in \mathcal{O}^*(U_i).$$

Consider,

$$0 \longrightarrow \mathcal{O}^* \longrightarrow \mathcal{M}^* \longrightarrow \mathcal{D} = \mathcal{M}^*/\mathcal{O}^* \longrightarrow 0$$

Note that $\mathcal{D}(\mathcal{G})$ is a group of divisors. Consider for $f \in \mathcal{M}^*(\mathcal{G})$, the representative (f) in $\mathcal{D}$ is called a principal divisor. (i.e. the Cousin II distribution solvable?).

$$\cdots \longrightarrow \mathcal{M}^*(\mathcal{G}) \longrightarrow \mathcal{D}(\mathcal{G}) \longrightarrow H^1(\mathcal{G}, \mathcal{O}^*) \longrightarrow \underbrace{H^1(\mathcal{G}, \mathcal{M}^*)}_{=0} \longrightarrow \cdots$$

Since $\mathcal{G}$ is pseudoconvex, we have $\mathcal{D}(\mathcal{G}) \to H^1(\mathcal{G}, \mathcal{O}^*) = 0$.

We also have what so-called exponential sequence,

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathcal{O} \xrightarrow{f \mapsto \exp(2\pi i f)} \mathcal{O}^* \longrightarrow 0$$

This gives a rise to a cohomology sequence,

$$\cdots \longrightarrow H^1(\mathcal{G}, \mathcal{O}) \longrightarrow H^1(\mathcal{G}, \mathcal{O}^*) \longrightarrow H^2(\mathcal{G}, \mathbb{Z}) \longrightarrow H^2(\mathcal{G}, \mathcal{O}) \longrightarrow \cdots$$

Imposing the assumption that $\mathcal{G}$ is pseudoconvex, we get

$$H^1(\mathcal{G}, \mathcal{O}^*) \cong H^2(\mathcal{G}, \mathbb{Z}).$$

From these observation, we have the following theorem.

Theorem 8.15. *The Second Cousin problem on a pseudoconvex domain $\mathcal{G}$ is solvable if and only if $H^2(\mathcal{G},\mathbb{Z})=0$.*

Theorem 8.16 (Oka-Grauert Principle)**.** *On a pseudoconvex domain, problems that can be expressed in terms of cohomology are solvable if and only if they admit continuous solutions. Or equivalently, the obstructions to solving these problems come from the topology of the domains.*

Example 8.2. *We give a counterexample to the Second Cousin problem where $\mathcal{G}$ pseudoconvex and simply connected. Set,*

$$\mathcal{G}=\{|z_1^2+z_2^2+z_3^2-1|<1\}\subset\subset\mathbb{C}^3.$$

Then $\mathcal{G}$ is simply connected but $H^2(\mathcal{G},\mathbb{Z})=\mathbb{Z}$. The idea to show this equality is the following. Observe first that

$$\mathcal{G}\cong D_1(0)\times Q,$$

where $Q=\{z_1^2+z_2^2+z_3^2=1\}$. Note that Q can be retracted to S^2. Thus the second cohomology of Q is the same as S^2 which is $\mathbb{Z}$. But the unit disk is contractible therefore, we obtain the claim.

Theorem 8.17 (Poincaré Problem)**.** *Let $\mathcal{G}$ be a pseudoconvex domain and $H^2(\mathcal{G},\mathbb{Z})=0$ If $f\in\mathcal{M}(\mathcal{G})$ then there exist $g,h\in\mathcal{O}(\mathcal{G})$ such that $f=g/h$.*

Let $D\in\mathcal{D}(\mathcal{G})$. Then it is locally given by the Cousin II distribution. Explicitlly, for $\mathscr{U}=\{U_i\}_{i\in I}$ an open covering such that $f_i\in\mathcal{M}(U_i)$ is meromorphic and it is given by $f_i=\frac{g_i}{h_i}$. We can assume that $g_{i,x},h_{i,x}$ are relatively prime for certain point x. On $U_{ij}=U_i\cap U_j$, there is u_{ij} such that

$$f_i=u_{ij}f_j.$$

where u_{ij} is a holomorphic function without zeros. This tells us that there exist u_{ij}^g,u_{ij}^h such that on U_{ij}, we have,

$$g_i=u_{ij}^g g_j, h_i=u_{ij}^h h_j,$$

and they are nowhere zero holomorphic functions. Note that collections,

$$\{(U_i,g_i)\mid \text{Cousin II}\}\overset{\text{defines}}{\to}D^+,\{(U_i,h_i)\mid \text{Cousin II}\}\overset{\text{defines}}{\to}D^-.$$

Definition 8.18. *The divisor D is positive if it is in the image of*

$$\tilde{\mathcal{O}}\to\mathcal{M}^*,$$

where $\tilde{\mathcal{O}}_x=\mathcal{O}_x-\{0\}$.

Lemma 8.7. *Let $D\in\mathcal{D}(\mathcal{G})$. Then there exists D^+,D^- such that D^+ is a positive divisor and*

$$D=D^+-D^-.$$

Furthermore, this decomposition is unique. In particular, D,D^+,D^- are principle. That is $D=(f)$ then for $f=u\frac{g}{h}$ where $u\in\mathcal{O}^*(\mathcal{G})$, we have ,

$$(f)=(g)-(h).$$

Definition 8.19. *For $0 \le r \le n$, consider, $\bigwedge^r \mathcal{O}$. Let $e_i \in \mathcal{O}^n$ denote*

$$e_i = (\delta_{ij})_{j=1,\cdots,n}.$$

The basis for $\bigwedge^r \mathcal{O}$ is

$$e_I = e_{i_1} \wedge \cdots \wedge c_{i_r},$$

where $1 \le i_1 < \cdots < i_r \le n$. We also have,

$$\bigwedge^1 \mathcal{O} = \mathcal{O}^n, \bigwedge^r \mathcal{O} = \mathcal{O}^{\binom{n}{r}}, \bigwedge^n \mathcal{O} = \mathcal{O}.$$

We then define,

$$\bigwedge^r \mathcal{O} \xrightarrow{d_r} \bigwedge^{r-1} \mathcal{O}, d_r e_{i_1} \wedge \cdots e_{i_r} = \sum_{j=1}^{r} (-1)^j z_{i_j} e_{i_1} \wedge \cdots e_{i_{j-1}} \wedge e_{i_{j+1}} \wedge \cdots e_{i_r}.$$

Remark 8.12. *By some calculations, we can show that $d_{r-1} d_r = 0$.*

Definition 8.20. *The following sequence is called a Kossul complex.*

$$0 \longrightarrow \bigwedge^n \mathcal{O}^n \xrightarrow{d_n} \bigwedge^{n-1} \mathcal{O}^n \xrightarrow{d_{n-1}} \cdots \xrightarrow{d_2} \bigwedge^1 \mathcal{O}^n \xrightarrow{d_1} \mathcal{M} \longrightarrow 0 \tag{K}$$

Lemma 8.8. *The sequence* (K) *is exact.*

Proof. See [9]. □

Theorem 8.18 (Gleason)**.** *Let $\mathcal{G}$ be pseudoconvex in $\mathbb{C}^n$ containing the origin. Let M be such that*

$$M := \{f \in \mathcal{O}(\mathcal{G}) \mid f(0) = 0\}.$$

Then we have,

$$M = (z_1, \cdots, z_n)\mathcal{O}(\mathcal{G}) = M.$$

That is for any $f \in M$, we have,

$$f = \sum_{i=1}^{n} f_i z_i,$$

for some $f_1, \cdots, f_n \in \mathcal{O}(\mathcal{G})$.

Note that $\mathcal{M}_x = \{f_x \mid f_x\}$.

Proof. $0 \longrightarrow \mathcal{K}_n \longrightarrow \mathcal{O}^n \xrightarrow{(f_1,\cdots,f_n)\mapsto\sum_{i=1}^n f_i} \mathcal{M} \longrightarrow 0$

Observe that $\mathcal{M}(\mathcal{G}) = M$. Now the question is

$$\cdots \longrightarrow \mathcal{O}^n(\mathcal{G}) \longrightarrow \mathcal{M}(\mathcal{G}) = M \longrightarrow \underbrace{H^1(\mathcal{G}, \mathcal{K}_n)}_{=0?} \longrightarrow \cdots$$

Using the Kossul complex of $\mathcal{O}^n$, we obtain,

$$0 \longrightarrow \mathcal{K}_1 \longrightarrow \bigwedge^1 \mathcal{O}^n \longrightarrow \mathcal{M} \longrightarrow 0$$

$$0 \longrightarrow \mathcal{K}_2 \longrightarrow \bigwedge^2 \mathcal{O}^n \longrightarrow \mathcal{K}_1 \longrightarrow 0$$

$$\vdots \qquad \vdots \qquad \vdots \qquad \vdots \qquad \vdots$$

$$0 \longrightarrow \mathcal{K}_{n-1} \longrightarrow \bigwedge^{n-1} \mathcal{O}^n \longrightarrow \mathcal{K}_{n-2} \longrightarrow 0$$

$$0 \longrightarrow \underbrace{\mathcal{K}_n}_{=0} \longrightarrow \bigwedge^n \mathcal{O}^n \longrightarrow \mathcal{K}_{n-1} \longrightarrow 0$$

□

Theorem 8.19. *Let $\mathcal{G}$ be a domain containing 0 and*

$$A(\mathcal{G}) := \{f \in \mathcal{C}^0(\overline{\mathcal{G}}) \mid f \text{ is holomorphic on } \mathcal{G}\}.$$

Consider $M := \{f \in A(\mathcal{G}) \mid f(0) = 0\}$. If $\mathcal{G}$ is strictly pseudoconvex, then $M = (z_1, \cdots, z_n)A(\mathcal{G})$ (i.e. M is an ideal generated by $z_1, \cdots, z_n$).

Remark 8.13. *$\bigwedge^r \mathcal{O}^n$ is acyclic and we have,*

$$H^1(\mathcal{G}, \mathcal{K}_1) \cong H^2(\mathcal{G}, \mathcal{K}_2) \cong H^3(\mathcal{G}, \mathcal{K}_3) \cong \cdots \cong \underbrace{H^n(\mathcal{G}, \mathcal{K}_n)}_{=0}.$$

We could also take an equivalent approach. This is done by directly using double complex and CR-equations. This approach can be found in the last chapter of [5].

Definition 8.21. *We define following subsets of holomorphic p-forms on a domain $\mathcal{G}$.*

1). $Z^p_{\mathrm{hol}}(\mathcal{G}) := \{f \in \mathcal{C}^\infty_{p,0}(\mathcal{G}) \mid f$ is holomorphic and $\partial f = 0\}$ (i.e. the set of holomorphic ∂-closed p-forms).

2). $B^p_{\mathrm{hol}}(\mathcal{G}) := \{f \in \mathcal{C}^\infty_{p,0}(\mathcal{G}) \mid f$ is holomorphic and $\exists u \in \mathcal{C}^\infty_{p-1,0}, f = \partial u\}$ (i.e. the coboundaries).

We then define,

$$H^p_{\mathrm{dR,hol}}(\mathcal{G}) := Z^p_{\mathrm{hol}}(\mathcal{G})/B^p_{\mathrm{hol}}(\mathcal{G}),$$

which is called the holomorphic de Rham group.

Corollary 8.1. *For a domain of holomorphy $\mathcal{G}$ in $\mathbb{C}^n$, we have,*

$$p > n \Rightarrow H^p(\mathcal{G}, \mathbb{C}) = 0.$$

Lemma 8.9. *Let $f \in \mathcal{C}^\infty_r(\mathcal{G})$ and suppose that $df \in \mathcal{C}^\infty_{r+1,0}(\mathcal{G})$. Then there is $\tilde{f} \in \mathcal{C}^\infty_{r,0}(\mathcal{G})$ such that,*

$$\exists g \in \mathcal{C}^\infty_{r-1,0}(\mathcal{G})\,\text{s. t.}\, f - \tilde{f} = dg.$$

Proof. Write,

$$f = \sum_{q=0}^{k} f_{r-q,q},$$

where $0 \leq k \leq r$. Induction with respect to k we have for $k = 0$, $f = f_{r,0}$. Thus take $\tilde{f} = f$, we have the statement. On the induction step,

$$\overline{\partial} f_{r-k,k} = 0,$$

thus there is $u \in \mathcal{C}^\infty_{r-k,k-1}(\mathcal{G})$ such that $\overline{\partial} u = f$. Then,

$$\begin{aligned} f - du &= f - \partial u - \overline{\partial} u, \\ &= \sum_{q=0}^{k-1} f_{r-q,q} - \partial u, \\ &= \tilde{f} + dv, \\ &\Rightarrow f = \tilde{f} + d(u+v). \end{aligned}$$

Here $\tilde{f} \in \mathcal{C}^\infty_{r,0}(\mathcal{G})$ is the one we get by the induction hypothesis. $\square$

Theorem 8.20. *If $\mathcal{G}$ is a pseudoconvex domain in $\mathbb{C}^n$, then*

$$H^p_{\text{dR,hol}}(\mathcal{G}) \cong \underbrace{H^p_{\text{dR}}(\mathcal{G}) \cong H^p(\mathcal{G}, \mathbb{C})}_{\textit{de Rham}}.$$

Proof. Let f be a d-closed r-form. Then by Lemma 8.9, we obtain $\tilde{f} \in \mathcal{C}^\infty_{r,0}(\mathcal{G})$ such that

$$f - \tilde{f} = dg.$$

By assumption $d\tilde{f} = 0$ so $\partial \tilde{f}, \overline{\partial} \tilde{f} = 0$. Therefore f is a holomorphic r-form.

We now show that f is exact from $\tilde{f}$ being ∂-exact. Clearly if $f = dg$, then $\tilde{f} = d\tilde{g}$. Using Lemma 8.9 to $\tilde{g}$, there is $\hat{g}_{r-1,0}$ with

$$\tilde{g} - \hat{g} = dv.$$

Thus,

$$d\hat{g} - d\tilde{g} = \tilde{f}, \overline{\partial}\hat{g} = 0, d\hat{g} = \partial \hat{g} = \tilde{f}.$$

Thus this yields a map,

$$H^r_{\text{dR}} \overset{f \mapsto \tilde{f}}{\to} H^r_{\text{hol,dR}}.$$

$\square$

8.7 Bounded, in Particular, strictly pseudoconvex, and pseudo convex domains

Definition 8.22. *Let $\mathcal{G}$ be a bounded domain in $\mathbb{R}^n$. Suppose U is a neighborhood of $\partial\mathcal{G}$. A function $r : U \to \mathbb{R}$ is said to define $\mathcal{G}$ or a defining function of $\mathcal{G}$ if*

$$\mathcal{G} = \{x \in U \mid r(x) < 0\} \cup (\mathcal{G}\backslash U),$$
$$\partial\mathcal{G} = \{x \in U \mid r(x) = 0\}.$$

Lemma 8.10. *Let $\mathcal{G} \subseteq \mathbb{R}^n$ be a domain given by the defining function $r : U \to \mathbb{R}$. Set*

$$r_j = \frac{1}{\frac{\partial r}{\partial x_j}}, \omega_j = (-1)^j dx_1 \wedge \cdots dx_{j-1} \wedge dx_{j+1} \wedge \cdots \wedge dx_n.$$

We have,

$$\frac{1}{r_j}\omega_j \circ \iota = \frac{1}{r_k}\omega_k \circ \iota.$$

Proof. Use $dr \circ \iota = 0$. □

Theorem 8.21 (General Stokes' Formula)**.** *Let $\mathcal{G}$ be a bounded domain whose boundary is piecewise smooth More precisely, there is a neighborhood U of $\partial\mathcal{G}$ and a functi onr $: U \to \mathbb{R}$ such that $dr \neq 0$ and we have,*

$$\mathcal{G} = \{x \in U \mid r(x) < 0\} \cup (\mathcal{G}\backslash U),$$
$$\partial\mathcal{G} = \{x \in U \mid r(x) = 0\}.$$

Such r is called the defining function.

Proof. We may assme that

$$|dr| = \left(\sum_{j=1}^{N} \left(\frac{\partial r}{\partial x_j}\right)^2\right)^{\frac{1}{2}},$$

and $|dr||_{\partial\mathcal{G}} \equiv 1$. Set

$$\omega_j = (-1)^j dx_1 \wedge \cdots dx_{j-1} \wedge dx_{j+1} \wedge \cdots \wedge dx_n.$$

Then

$$dx_j \wedge \omega_j = dV,$$

which is the volume form. Let $\iota : \partial\mathcal{G} \to \mathbb{R}^n$ be an embedding. By denoting $\frac{\partial r}{\partial x_j} = r_j$, we have, □

More general version of Stokes' theorem can be found in [10].

Definition 8.23. *Let $\mathcal{G} \subseteq \mathbb{R}^n$ be a bounded domain given by the defining function $r : U \to \mathbb{R}$. The surface element of $\partial\mathcal{G}$ is defined as,*

$$dS := \frac{1}{\frac{\partial r}{\partial x_j}}\omega.$$

Lemma 8.11. *Let f be a $n-1$-form (i.e. $f = \sum_{i=1}^n f_j\omega_j$). Then we have,*

$$\int_{\partial\mathcal{G}} f = \int_{\partial\mathcal{G}} \sum_{j=1}^{n} (f_j r_j) dS.$$

We pass these results to $\mathbb{C}^n = \mathbb{R}^{2n}$.

Definition 8.24. *Let $\mathcal{G}$ be a domain given by the defining function $r : U \to \mathbb{R}$. The surface element dS is a $2n-1$-form on $\partial\mathcal{G}$ such that*

$$\frac{\partial r}{\partial z_j} dS = \frac{1}{2}(\omega_{2j-1} - i\omega_{2j}), \frac{\partial r}{\partial \overline{z}_j} dS = \frac{1}{2}(\omega_{2j-1} + i\omega_{2j}).$$

Definition 8.25. *Suppose we have,*

$$f = \sum_{\substack{\alpha\in\mathbb{N}^n \\ |\alpha|=q+1}} f_\alpha d\overline{z}^\alpha,$$

then, we define,

$$\vartheta f := -\sum_{\substack{\alpha\in\mathbb{N}^n \\ |\alpha|=q}} \sum_{j=1}^{n} \frac{\partial f_{j\alpha}}{\partial z_j} d\overline{z}^\alpha.$$

Recall that the Hermitial product on $\mathcal{G} \subseteq \mathbb{C}^n$ is defined by,

$$(f, g) = \int_{\mathcal{G}} \langle f, g\rangle d\lambda,$$

where

$$\langle f, g\rangle = \sum_{\substack{\alpha\in\mathbb{N}^n \\ |\alpha|=q+1}} f_\alpha \overline{g}_\alpha.$$

Lemma 8.12 (Partial Integration)**.** *Let $r : U \to \mathbb{R}$ be a function that defines a bounded domain $\mathcal{G}$ and u, v be functions defined on $\overline{\mathcal{G}}$ and f, g differential forms. Then we have,*

$$\left(\frac{\partial u}{\partial \overline{z}_j}, v\right) + \left(u, \frac{\partial v}{\partial \overline{z}_j}\right) = \int_{\partial\mathcal{G}} u\overline{v}\frac{\partial v}{\partial \overline{z}_j} dS.$$
$$\left(\frac{\partial u}{\partial z_j}, v\right) + \left(u, \frac{\partial v}{\partial z_j}\right) = \int_{\partial\mathcal{G}} u\overline{v}\frac{\partial v}{\partial z_j} dS.$$
$$(\vartheta f, g) - (f, \overline{\partial} g) = \int_{\partial\mathcal{G}} \langle\vartheta(rf), g\rangle dS.$$

Corollary 8.2. *In the setting of Lemma 8.12, if f or g has a compact support then,*

$$(\vartheta f, g) = (f, \overline{\partial} g).$$

That is ϑ is a formal adjoint of $\overline{\partial}$.

Remark 8.14. *We have studied the following sequence,*

$$L^2_{0,q}(\mathcal{G}) \xrightarrow{T} L^2_{0,q+1}(\mathcal{G}) \xrightarrow{S} L^2_{0,q+2}(\mathcal{G})$$

We will denote the Hilbert adjoint of $\overline{\partial}$ as $\overline{\partial}^*$. The adjoint is defined as follows. For $f \in \operatorname{Dom} \overline{\partial}^*, g \in \operatorname{Dom} \partial$, we have

$$(\overline{\partial}^* f, g) = (f, \overline{\partial} g).$$

In order to have $\overline{\partial}^* = \vartheta$ on $\mathcal{C}^2(\overline{\mathcal{G}})$, we need,

$$\vartheta(rf) \equiv 0 \quad (\text{on } \partial\mathcal{G}).$$

Definition 8.26. *For $l \in \mathbb{Z}_{\geq 0}$, we define,*

$$\mathcal{D}^l_{\overline{\partial}^*} := \operatorname{Dom} \overline{\partial}^* \cap \mathcal{C}^l(\overline{\mathcal{G}}).$$

Definition 8.27. *The following equation is called the first Neumann condition.*
Let r be a function defining a bounded domain $\mathcal{G}$, we have,

$$\forall \alpha \in \mathbb{N}^n \,\text{s. t.}\, |\alpha| = q, \; \textit{we have} \; \sum_{j=1}^n f_{j\alpha} \frac{\partial r}{\partial z_j} = 0 \; \textit{on} \; \mathcal{G}. \tag{N1}$$

Lemma 8.13. *For $l \geq 1$ and $f \in L^2_{0,q+1}(\mathcal{G})$, we have, $f \in \mathcal{D}^l_{\overline{\partial}^*}$ if and only if $\vartheta(rf) = 0$ on $\partial\mathcal{G}$. That is f satisfies* (N1).

Lemma 8.14. *Let $\varphi \geq 0$ be a weight function and $\varphi \in \mathcal{C}^2(\mathcal{G})$. We note that $\varphi \equiv 0$ is also allowed. We define $L^2_{\cdot,\cdot}(\mathcal{G}, \varphi), \vartheta_\varphi$, accordingly. Then the first Neumann condition does not depends on φ.*

Definition 8.28. *Let $f \in \operatorname{Dom} \overline{\partial}^*_\varphi \cap \mathcal{C}^\infty(\mathcal{G})$. The graph norm of f is defined as,*

$$\|f\| + \|\overline{\partial} f\| + \|\overline{\partial}^* f\|.$$

Theorem 8.22 (Hörmander-Friedrich)**.** *Let $\mathcal{G}, \partial\mathcal{G}$ defined by $r : U \to \mathbb{R}$. We have, $\operatorname{Dom} \overline{\partial}^*_\varphi \cap \mathcal{C}^\infty(\mathcal{G})$ is dense in $\operatorname{Dom} \overline{\partial}^* \cap \operatorname{Dom} \overline{\partial}$. That is for any $f \in \operatorname{Dom} \overline{\partial}^* \cap \operatorname{Dom} \overline{\partial}$, there is a sequence $(f_j) \in \mathcal{C}^\infty(\mathcal{G})$ such that*

$$f_j \xrightarrow{L^2} f, \overline{\partial} f_j \xrightarrow{L^2} \overline{\partial} f, \vartheta f_j \xrightarrow{L^2} \overline{\partial}^* f,$$

and each f_j satisfies (N1).

Proof. The proof can be found in [5] chapter 5, section 4. $\square$

What we want to do now is to replicate the calculation we have done before to calculate adjoints but with only one weight function φ, but taking accout of boundary integrals. In general boundary integral $= 0$ because of (missing) (some integrals).

Theorem 8.23 (Kohn-Morrey)**.** *Let $\mathcal{G}, \partial\mathcal{G}$ be defined by r and $\varphi \in \mathcal{C}^\infty(\mathcal{G})$ be such that $\varphi \geq 0$. Then for*

$$f = \sum_{\substack{\alpha \in \mathbb{N}^n \\ |\alpha| = q+1}} f_\alpha d\overline{z}^\alpha,$$

which is smooth on $\overline{\mathcal{G}}$ and satisfies (N1)*, we have,*

$$\begin{aligned} \|\overline{\partial} f\|_\varphi^2 + \|\overline{\partial}_\varphi^* f\|_\varphi^2 = &\int_{\mathcal{G}} \sum_{\substack{\alpha \in \mathbb{N}^n \\ |\alpha| = q}} \sum_{j,k} \frac{\partial^2 \varphi}{\partial z_j \partial \overline{z}_k} f_{j\alpha} \overline{f}_{k\alpha} e^{-\varphi} d\lambda \\ &+ \int_{\mathcal{G}} \sum_{\substack{\alpha \in \mathbb{N}^n \\ |\alpha| = q+1}} \sum_j \left| \frac{\partial f_j}{\partial \overline{z}_j} \right|^2 e^{-\varphi} d\lambda \\ &+ \int_{\partial\mathcal{G}} \sum_{\substack{\alpha \in \mathbb{N}^n \\ |\alpha| = q}} \sum_{j,k} \frac{\partial^2 r}{\partial z_j \partial \overline{z}_k} f_{j\alpha} \overline{f}_{k\alpha} e^{-\varphi} dS. \end{aligned}$$

Corollary 8.3. *Take $\varphi \equiv 0$. Suppose $\mathcal{G}$ is a strictly pseudoconvex domain defined by a strictly plurisubharmonic function r. We then have,*

$$\|\overline{\partial} f\|^2 + \|\overline{\partial}^* f\|^2 \geq c \int_{\partial\mathcal{G}} |f|^2 dS.$$

This equation is called Kohn's basic estimate.

From now on, we will only use the first term of Theorem 8.23. That is

$$\|\overline{\partial} f\|_\varphi^2 + \|\overline{\partial}_\varphi^* f\|^2 \geq c\|f\|_\varphi^2.$$

Theorem 8.24. *Let $\mathcal{G} \subset\subset \mathbb{C}^n$ be a pseudoconvex domain. Then there is a constant $c > 0$, only depending on the diameter of $\mathcal{G}$, such that for any $f \in L^2_{0,q+1}(\mathcal{G})$, $\overline{\partial} f = 0$, there is $u \in L^2_{0,q}(\mathcal{G})$, with $\overline{\partial} u = f$ and $\|u\| \leq c\|f\|$.*

If f is smooth then u can be chosen as a smooth function.

In the equality of Theorem 8.23, instead of considering the whole, we may take only a single summation term on the right hand side, we aim to derive the following inequality.

Theorem 8.25 (Adjusted Kohn-Morrey)**.** *Let $\mathcal{G}$ be a strictly pseudoconvex domain and $\varphi \in \mathcal{C}^2(\overline{\mathcal{G}})$ be a weight function. Then for $f \in L^2_{0,q}(\mathcal{G}, \varphi)$, we have,*

$$\int \sum_K \sum_{j,k} \frac{\partial^2 \varphi}{\partial z_j \partial \overline{z}_k} f_{jK} \overline{f}_{kK} e^{-\varphi} d\lambda \leq \|\overline{\partial} f\|_\varphi^2 + \|\overline{\partial}_\varphi^* f\|_\varphi^2.$$

Proof. choose $\varphi(z) = t\|z\|^2$ and $t > 0$ will be chosen later.

$$t\|f\|_\varphi^2 \le \|\overline{\partial} f\|_\varphi^2 + \|\overline{\partial}_\varphi^*\|_\varphi^2.$$

In a favorable situation, we have,

$$L^2_{0,q+1}(\mathcal{G},\varphi) = \mathcal{N}_{\overline{\partial}_\varphi^*} \oplus \mathcal{R}_{\overline{\partial}}, L^2_{0,q}(\mathcal{G},\varphi) = \mathcal{N}_{\overline{\partial}} \oplus \mathcal{R}_{\overline{\partial}_\varphi^*}.$$

Therefore for $f \in L^2(\mathcal{G},\varphi)_{0,q+1}$ with $\overline{\partial} f = 0$, we have $f \in \mathcal{R}_{\overline{\partial}}$. That is $f = \overline{\partial} u$ for some $u \in \operatorname{Dom}\overline{\partial} \subseteq L^2_{0,q}(\mathcal{G},\varphi)$. We may assume that

$$u = \overline{\partial}_\varphi^* v, v \in \operatorname{Dom}\overline{\partial}_\varphi^*,$$

and $\overline{\partial} v = 0$. Apply this to v, we have,

$$\begin{aligned} t\|v\|_\varphi^2 &\le \|\overline{\partial}_\varphi^* v\|_\varphi^2, \\ \|u\|_\varphi^2 &= (u,u)_\varphi = (u, \overline{\partial}_\varphi^* v)_\varphi, \\ &= (\overline{\partial} u, v)_\varphi = (f,v)_\varphi, \\ &\le \|f\|_\varphi \|v\|_\varphi, \\ &\le \frac{1}{\sqrt{t}}\|f\|_\varphi \|\overline{\partial}_\varphi^* v\|_\varphi, \\ &= \frac{1}{\sqrt{t}}\|f\|_\varphi \|u\|_\varphi. \end{aligned}$$

Therefore, we have a solution u with,

$$\|u\|_\varphi \le \frac{1}{\sqrt{t}}\|f\|_\varphi.$$

Compute,

$$\begin{aligned} \int_{\mathcal{G}} |u|^2 e^{-t\|z\|^2} d\lambda &= \frac{1}{t}\int_{\mathcal{G}} |f|^2 e^{-t\|z\|^2} d\lambda, \\ &\le \frac{1}{t}\int_{\mathcal{G}} |f|^2 d\lambda. \end{aligned}$$

For $\delta > \operatorname{diam}\mathcal{G}$, we have,

$$e^{-t\delta^2}\int_{\mathcal{G}} |u|^2 d\lambda \le \int_{\mathcal{G}} |u|^2 e^{-t\|z\|^2} d\lambda.$$

This implies,

$$\int_{\mathcal{G}} |u|^2 d\lambda \le \frac{e^{t\delta^2}}{t}\int_{\mathcal{G}} |f|^2 d\lambda, \|u\|^2 \le e\delta^2\|f\|^2.$$

As $\mathcal{G}$ is bounded and pseudoconvex, take $\delta = \operatorname{diam}\mathcal{G}$, we may take a exhaustion by strictly pseudoconvex domains,

$$\mathcal{G}_1 \subset \mathcal{G}_2 \subset \cdots \subset \mathcal{G}.$$

For $f \in L^2_{0,q+1}(\mathcal{G})$, solve the Cauchy-Riemann equation on each domain $\mathcal{G}_i$ and denote the solution u_i, (i.e. $\overline{\partial} u_i = f$ on $\mathcal{G}_i$) with the inequality,

$$\|u_i\|^2 \leq e\delta^2 \|f\|^2.$$

Choose $t = \frac{1}{\delta^2}$. Now we have a sequence of bounded L^2 functions $(u_i)_{i\in\mathbb{N}}$. Without loss of genearlity, we assume

$$u_i \overset{\text{weakly in } L^2}{\longrightarrow} u \in L^2_{0,q}(\mathcal{G}), u_i \overset{\mathcal{D}'}{\to} u.$$

Of course we have,

$$\|u\|^2 \leq e\delta^2 \|f\|^2.$$

Furthermore, for $i \geq j$, we have,

$$\overline{\partial} u_i = \overline{\partial} u_j \text{ on } \mathcal{G}_j \Rightarrow \overline{\partial} u = f \text{ on } \mathcal{G}.$$

□

Theorem 8.26. *Let $\mathcal{G} \subset\subset \mathbb{C}^n$ be pseudoconvex and $f \in L^2_{0,q+1}(\mathcal{G}), \overline{\partial} f = 0$. Then there is $u \in L^2_{0,q}(\mathcal{G})$ such that*

$$\overline{\partial} u = f, \|u\|^2 \leq e\delta^2 \|f\|^2.$$

u carries any regularity that f has.

Notation 8.2. *$\|\cdot\|_{W^k}$ denotes the Sobolev norm of order k.*

Lemma 8.15. *Let $u \in W^s(\mathbb{R}^n)$ for $s \geq 0$ and let us denote its Fourier transform $\hat{u}$. We have,*

$$\|u\|_{W^s} = \|\hat{u}(1 + \|x\|^2)^{\frac{s}{2}}\|.$$

Definition 8.29. *Let $\mathcal{G}$ be a strictly pseudoconex domain. The Bergman projection is an orthogonal projection,*

$$P : L^2(\mathcal{G}) \to L^2(\mathcal{G}) \cap \mathcal{O}(\mathcal{G}).$$

Theorem 8.27 (Kohn)**.** *Let $\mathcal{G} \subset\subset \mathbb{C}^n$ be a strictly pseudoconvex domain and $f \in L^2_{0,q}(\mathcal{G})$ for $q \geq 1$.*

1). If $f \in \operatorname{Dom} \overline{\partial} \cap \operatorname{Dom} \overline{\partial}^*$, we have,

$$\|f\|_{W^{k+\frac{1}{2}}} \leq c(\|\overline{\partial} f\|_{W^k} + \|\overline{\partial}^* f\|_{W^k} + \|f\|).$$

2). There is $c > 0$ such that

$$\|u - Pu\|_{W^{k+\frac{1}{2}}} \leq c(\|\overline{\partial} u\|_{W^k} + \|u\|_{W^k})$$

Theorem 8.28 (Sobolev)**.** *Let $\mathcal{G} \subset\subset \mathbb{R}^N$ be a domain and $d = \left[\frac{N}{2}\right] + 1$. Then*

$$W^{s+d}(\overline{\mathcal{G}}) \subseteq \mathcal{C}^s(\overline{\mathcal{G}}),$$

where $\mathcal{C}^s(\overline{\mathcal{G}})$ denotes the space of functions whose derivative of order up to s are bounded.

Theorem 8.29. *Let $\mathcal{G}$ be strictly pseudoconevx with $\partial\mathcal{G}$ smooth and $f \in \mathcal{C}^\infty_{0,q}(\mathcal{G})$ with $\overline{\partial} f = 0, q \geq 1$. Then there is $u \in \mathcal{C}^\infty_{0,q+1}(\overline{\mathcal{G}})$ with $\overline{\partial} u = f$. Moreover, the canonical solution u satisfying either*

$$\begin{cases} \vartheta u = 0, \\ P u = 0, \end{cases}$$

has this property. (Recall ϑ is formal adjoint of $\overline{\partial}$).

For f defined on $\mathcal{G} \subset\subset \mathbb{C}$, set

$$u(z) = \int_{\mathcal{G}} C(\zeta, z) f(\zeta).$$

Now take f as a (p, q)-form on $\mathcal{G}$. We want to find a kernel $K(\zeta, z)$ such that

$$u(z) = \int_{\mathcal{G}} f(\zeta) K(\zeta, z), \overline{\partial} u(z) = f(z).$$

Theorem 8.30. *Let $\mathcal{G} \subseteq \mathbb{C}^n$ be a strictly pseudoconvex domain and $f_0 \in L^\infty_{0,1}(\mathcal{G}) \cap \mathcal{C}^\infty(\mathcal{G})$ with $\overline{\partial} f = 0$. Then there is a solution $u \in L^\infty(\mathcal{G}) \cap \mathcal{C}^\infty(\mathcal{G})$ with $\overline{\partial} u = f$. Furthermore, there is a constant $C \geq 0$, which is independent of f such that*

$$\|u\|_{L^\infty} \leq C \|f\|_{L^\infty}.$$

Remark 8.15 (Henkin-Ramírez)**.** *C is invariant under small pertubation of $\mathcal{G}$ but not under larger pertubations.*

Definition 8.30. *Let $f : \mathbb{R} \to \mathbb{R}$. The Hölder norm of f order k is*

$$\|f\|_{\mathcal{C}^k} := \inf\{C \geq 0 \mid \forall x, y \in \mathbb{R}, |f(x) - f(y)| \leq C|x - y|^{\frac{1}{2}}\}.$$

Theorem 8.31. *Let $\mathcal{G} \subseteq \mathbb{C}^n$ be a strictly pseudoconvex domain. There is a linear continuous operator $S : L^1_{0,q+1} \to L^1_{0,q}$ with*

$$\overline{\partial} f = 0 \Rightarrow \overline{\partial} S f = f.$$

Sf is smooth in $\mathcal{G}$ if f is smooth (i.e. S is the solution operator).

$$\|Sf\|_{\mathcal{C}^{k+\frac{1}{2}}} \leq \text{const}\, \|f\|_{\mathcal{C}^k}.$$

This constant is invariant under small pertubations.

Proof. The proof can be found in [5]. □

Theorem 8.32. *Let $\mathcal{G} \subseteq \mathbb{C}^n$ be a strictly pseudoconex domain, then there is a Hermitian metric on $\mathbb{C}^n$ such that*

i). for $f \in L^2_{0,q+1}(\mathcal{G}), f \in \operatorname{Dom}\overline{\partial} \cap \operatorname{Dom}\overline{\partial}^*$,

$$\|f\|_{\mathcal{C}^{k+\frac{1}{2}}} \leq \text{const}(\|\overline{\partial} f\|_{\mathcal{C}^k} + \|\overline{\partial}^* f\|_{\mathcal{C}^k} + \|f\|_{L^2}).$$

ii). for $f \in L^2_{0,0}(\mathcal{G})$, we have,

$$\|f - Pf\|_{\mathcal{C}^{k+\frac{1}{2}}} \leq \text{const}(\|\overline{\partial} f\|_{\mathcal{C}^k} + \|f\|_{L^2}).$$

Note that the metric is used to define L^2-space, the projection P, and the adjoint $\overline{\partial}^*$.

Remark 8.16 (Beals-Greiner-Stanton)**.**
The Theorem 8.32 holds for arbitrary metrics.

9 What now?

9.1 Stein Manifolds

We will examine how the results we have seen can be studied further. One canonical generalization of these results would be to consider complex manifolds instead of $\mathbb{C}^n$.

Definition 9.1. *Let X be a n-dimensional complex manifold. X is said to be Stein if*

i). it has countable topology (i.e. admits a countable basis of the topology),

ii). it is holomorphically convex,

iii). it is holomorphically separable (i.e. for any $x \neq y, \exists f \in \mathcal{O}(X)$ such that $f(x) \neq f(y)$),

iv). for all $x \in X$, there is a global holomorphic coordinates (i.e. there is $f_1, \cdots, f_m \in \mathcal{O}(X)$ that give local coordinate at x).

Remark 9.1. *$\mathcal{G} \subseteq \mathbb{C}^n$ is a Stein manifold if and only if it is a domain of holomorphy.*

Our motivation is to show natural domains of existence for holomorphic functions (in $\mathbb{C}^n$) are complex manifold $X \xrightarrow{P} \mathbb{C}^n$ where X is an unbranced Riemann domain.

Definition 9.2. *X is pseudoconvex if there is smooth, strictly plurisubharmonic exhaustive functions.*

Theorem 9.1. *X is Stein if and only if it is pseudoconvex.*

Proof. We will not give the full proof, however we will note some tools which we need to utilise to show the statement. Namely, $\overline{\partial}$-complex, (p, q)-forms, instead of $(0, q)$-forms, L^2 spaces with respect to a hermitian metrix. $\square$

Theorem 9.2 (Cartan-Thullen-Oka)**.** *X is a domain of holomorphy of $\mathbb{C}^n$ then X is Stein.*

Theorem 9.3 (Behnke-Stein)**.** *Open Riemann surfaces are Stein.*

Theorem 9.4 (Remmert)**.** *X is n-dimensional Stein manifold if and only if $\phi : X \to \mathbb{C}^{2n+1}$ biholomorphic embedding.*

Theorem 9.5. *The second and second Cousin problems are solvable in $\mathbb{C}^n$.*

Theorem 9.6 (Cartan-Serre)**.** *If S is a coherent analytic sheaf on a Stein manifold X, then $H^q(X, S) = 0$ for all $q \geq 1$.*

Remark 9.2. *In the first and the last condition of Stein manifolds are redundant.*

Theorem 9.7 (Grauert)**.** *X is Stein if and only if it is holomorphically convex and for all $x \in X$, there are holormophic functions $f_1, \cdots, f_k \in \mathcal{O}(X)$ such that x is isolated in $V(f_1, \cdots, f_k)$.*

Theorem 9.8 (Radó)**.** *All connected Riemann surfaces have countable topology.*

9.2 Complex Spaces

Definition 9.3. *A topological space X is Stein if it is holomorphically convex and for all $x \in X$, there are holormophic functions $f_1, \cdots, f_k \in \mathcal{O}(X)$ such that x is isolated in $V(f_1, \cdots, f_k)$ (as per Theorem 9.7).*

Theorem 9.9. *A topological space X is Stein if and only if it is pseudoconvex.*

Theorem 9.10 (Cartan-Serre)**.** *If S is a coherent analytic sheaf on a Stein space X, then $H^q(X, S) = 0$ for all $q \geq 1$.*

Let X be a topological space and $S \subset X$ be a subset. Suppose $X \backslash S$ is a connected n-dimensional manifold. One can use the arsenal of differential forms to study this and consider S on a boundary which can be totally pathological.

Theorem 9.11 (Milestones of Complex Analysis)**.**

I). *(Hironaka) Let X be a complex space, then there is $X' \xrightarrow{\pi} X$ where X' is a complex manifold and π is holomorphic and biholomorphic over the regular points. That is to say, we can desingularize complex spaces.*

II). *(Grauert) Let $p : X \to Y$ be a holomorphic map between complex spaces which is proper (i.e. each compact set in Y has its preimage compact in X under p). Then $R^i p\mathcal{S}$ where $\mathcal{S}$ is coherent sheaf on X are coherent.*

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